Iterative Queries in a Low-Rank Memory Adapter: Composition, Writer Dependence, and Budget Limits

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Abstract

Can a low-rank adapter use extra computation to extract compositions from a fixed context memory? We study a reader that updates a low-dimensional query several times over a fixed delta-rule memory, and we compare it with a single read at the same rank. With an oracle writer, more reads help. In a symbolic task, repeated reads improve the reachable hops and beat fixed-query and memory-free controls. In a frozen language model at rank 128, one shared module read four times beats a single read of equal parameter count on hops 2 to 3 at two backbone sizes of one family (Qwen3-0.6B-Base and Qwen3-1.7B-Base), and the four-unshared-reads reader fails its registered gate at both; a copy of it started from the shared solution keeps that solution and leaves the verdict unchanged. At its trained depth the shared reader reaches hop 3; run deeper, the same checkpoint reaches hops 4 to 8 at 9.9 to 16.6% (0.6B) and 9.8 to 20.9% (1.7B) with the depth chosen by the label, while it loses hop 1. A label-free per-question stop raises hops 4 to 8 over the trained depth by +3.90 pp (0.6B) and +5.68 pp (1.7B), at the cost of hop 2 (-10.83 and -12.27 pp) and of extra reads, and no efficiency claim is made. Learned writing is the weak point. In the symbolic task a learned writer gives a smaller, seed-dependent benefit, and a factorized writer keeps a walk-specific depth effect but fails to preserve single-hop accuracy. In the language model, the completed validation screens, including the registered DPFP, DPFP-X, ridge-storage (SQ2) and trained least-squares (LSQT) screens, select no learned writer, despite informative input probes and a usable supplied-answer-code output path; the best cells reach 72.3% (SQ2) and 46.1% (LSQT) against the 0.80 gate. A parser into a fixed table of the oracle's own codes drives the frozen reader at the oracle's level (validation, one seed), but it is not a learned writer and is not compared with the DPFP results. Per-step readouts are decoding only and bridge patching gives no causal claim; at rank 32 the readers did not learn to use the oracle memory. The comparison is against a single read at the same rank, extra latency is reported as a cost, and no efficiency advantage over a larger-rank single read is claimed. All results use a closed 1,024-name vocabulary, depths within those trained, one oracle seed and three training seeds. The contribution is a bounded account of when repeated memory access helps, with negative results that keep capability gains from being mistaken for efficiency or end-to-end reasoning gains.

Reading guide. Sections 1 to 8 give the question, the setting, the main results and the limits. Every principal claim, with its comparator, writer, split and gate, is in Appendix C. The appendices hold the method and theory in full, every symbolic and language-model study, and the full related work. Numerical and claim authority is the pair of evidence ledgers (`evidence_symbolic.md`, `evidence_llm.md`); the ledger wins over any shortened wording here. Section numbers inside Appendices D to H follow the numbering of the record from which they were taken. Appendix I holds the full tables of the three follow-up studies (stop rules, B8 warm start, classify-then-code writer).

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1 Introduction

A context-dependent adapter has to decide what to store and how to query it. A larger memory can keep more information. Spending more computation on the stored representation is a different option: an extra query may reveal a composition that the smaller state already holds. Neither option follows from counting parameters. We ask whether the query can improve over several updates while the input, the context and the frozen base projection stay fixed.

The test case is a chain of associations. A query names a starting entity, and the target is the terminal of its chain. A reader with a suitable memory map can read a successor, form a new query from it, and continue. High accuracy on the terminal task does not by itself show that this happens. A learned writer can expose information about the terminal set directly, so a model can answer some questions without following the path. Proposition 10 (Appendix D) gives a conditional construction of such a statistic, and factorizing the writer inputs removes one direct term of it without removing every global statistic.

We study a low-rank adapter whose reader updates a low-dimensional query several times over a fixed delta-rule memory (Section 2). The writer makes one decay-first delta update per eligible context token. Three questions organize the paper. First, does a larger evaluation depth improve one trained reader while its input and memory stay the same? Second, do separately trained evolving-query systems beat single-read, fixed-query and memory-free controls at the same rank? Third, what do these comparisons cost in parameters and measured latency? Rank is held fixed so that extra reader computation is not mixed up with extra representation width, and extra latency is reported as a cost.

What we find.

- With an oracle writer, more reads help. In a symbolic task with orthogonal oracle codes, repeated reads improve the reachable hops and beat fixed-query and memory-free controls. In a frozen language model at rank 128, one shared module read four times beats a single read with the same parameter count on hops 2 to 3, at two backbone sizes of one family: by +21.550 pp (Qwen3-0.6B-Base) and +25.783 pp (Qwen3-1.7B-Base), in every seed. The four-unshared-reads reader B8 fails its registered gate at both sizes; a copy of B8 started from the shared solution keeps that solution, and the verdict stays.
- Learned writing is the weak point. In the symbolic task a learned writer gives a smaller, seed-dependent benefit, and a factorized writer gives a stronger walk-specific effect but loses single-hop accuracy. In the language model, no learned writer reaches the registered 0.80 gate, including the DPFP, DPFP-X, ridge-storage (SQ2) and trained least-squares (LSQT) screens; each registered decision keeps the learned-writer line stopped, and the trained least-squares line stays open in a band that neither closes nor triggers it. A parser into a fixed table of the oracle's own codes drives the frozen reader at the oracle's level, but it is not a learned writer. We therefore make no learned-writer and no end-to-end claim.
- The oracle result is bounded. At its trained depth the reader reaches hop 3; run deeper, the same checkpoint reaches hops 4 to 8 at low accuracy while it loses hop 1. A label-free per-question stop raises hops 4 to 8 over the trained depth by 3.90 and 5.68 pp, but hop 2 falls by 10.83 and 12.27 pp and no efficiency claim is made. Per-step readouts are decoding only, and bridge patching gives no causal claim.
- Stopping rules, memory load and budget controls are reported as costs and limits. Same-rank latency is a cost, and no efficiency claim is made over larger-rank single reads.

Limits that apply to every claim. All entities come from a closed 1,024-name vocabulary, and every evaluation set draws on its seen pool. The one exception is a descriptive scope row on 256 unseen-pool names of the same vocabulary, which supports only the statement that the depth gain is also present there. Nothing here concerns names outside the vocabulary or new entity codes. Effects hold at the trained hops and depths, and longer hops appear only as measured cells. Oracle writers and learned writers answer different questions and are never pooled, and the language-model oracle differs from the symbolic oracle. Every language-model composition number rests on one random oracle code table (one oracle seed) and

three training seeds. Appendix C lists each principal claim with its comparator, writer, split, gate and evidence rows.

2 Setting: one write schedule, repeated low-dimensional reads

The reader. A frozen linear projection maps an input x to $Wx + b$. The adapter first forms $u = Ax$, where A has output dimension r. A memory matrix M has shape $r_{\text{mem}} \times r_{\text{mem}}$. With column vectors, the shared reader is

$$z^{(0)} = u, \qquad z^{(s+1)} = u + \rho\, \phi(Rz^{(s)} + UMQz^{(s)}), \qquad y = Wx + b + \alpha Bz^{(L)}.$$

The same M is used in every inner step. Q maps the current state to a memory query, U maps the read back into the reader coordinates, and B is the output bottleneck. The input is reinjected at every update, so a larger L changes the reader computation and leaves the base projection unchanged. Setting $L = 0$ gives the ordinary low-rank branch.

The writer. The writer uses a normalized key k and a value v derived from the designated writer inputs. For an eligible write the decay-first update is

$$\bar{M} = \lambda M, \qquad M^{+} = \bar{M} + \beta\,(v - \bar{M}k)k^{\mathsf{T}}.$$

Reading precedes the current token's write. Padding neither reads nor writes, reset clears the state before use, and a skipped write does not trigger decay. In the memory question-answering protocol, support facts fill the state and questions read without writing. Separate questions and evaluation depths fork from the same completed context snapshot. Oracle conditions specify their codes and gate values directly; learned writers train them. Appendix D gives the full method and the propositions.

Readers compared. Table 1 lists the readers. M1 is the shared reader with ReLU, and M1-lin has identity activation. B8 uses step-specific matrices and is trained for a fixed unrolled depth. It is a separate architecture, not a reader that can be extended by reusing a next shared step, and it is not parameter-matched to the shared readers: its extra step matrices add $3(r^2 + 2r\, r_{\text{mem}})$ parameters. That is 9,216 at $r = r_{\text{mem}} = 32$ and 147,456 at $r = r_{\text{mem}} = 128$, or +8.9% (rank 32) and +31.0% (rank 128) of the language-model adapter, and +69.2% of the symbolic unfactorized joint model.

Table 1: The readers compared. Rank is the same for every reader in a comparison unless the row says otherwise.

Reader	Definition
B3	Memory path removed (memory-free control).
B4	Single read at the same rank ($L = 1$).
B5	Initial memory query held fixed; the read is cached while internal iterations continue (fixed-query control).
B6	Larger rank in a single step (historical control).
B7	Larger memory dimension with the smaller reader dimension (historical control).
B8	Step-specific R/U/Q matrices, fixed unrolled depth; not parameter-matched.
M1, M1-lin	Shared reader with ReLU, and with identity activation.

Which comparisons hold the memory fixed. Oracle writing uses specified entity codes. A T1-pretrained frozen writer is shared across the reader variants for its seed. Under joint writing each reader variant trains its own writer, so a contrast with B5 or B3 compares learned systems and does not isolate query evolution under an identical learned writer. Factorized conditions restrict keys and the initial query to head and relation information and values to the tail input, which removes the direct head-subtraction statistic of Proposition 10.

Tasks. The symbolic primary layout has 12 edges in chains of lengths 8 and 4, 64-dimensional entity representations and $\rho = 0.25$; reader and memory dimensions are both 32. Readers train on hops 1 to 3 with mixed depths $\{1, 2, 4\}$ unless a fixed-depth comparison is named. In the language model the adapter sits on the layer-14 attention output projection of a frozen Qwen3 Base model (2048→1024 for Qwen3-0.6B-Base and 2048→2048 for Qwen3-1.7B-Base). The registered oracle writer builds the memory directly from the fact list with frozen random unit key codes for (entity, relation) and fixed entity value codes, so every read after the first needs a learned value-to-key lookup over the 768 seen names; this is not the symbolic oracle. These task conditions are part of every claim below.

Metrics and intervals. The depth effect D is L4 minus L1 on hops 2 to 3, and walk specificity W subtracts the corresponding effect on far hops 6 to 8, so that a generic all-hop shift is not described as selective improvement. Symbolic accuracies are candidate-restricted; language-model results use full-vocabulary exact match (EM), and the two are not exchanged. Confirmation intervals are 97.5% paired episode-bootstrap intervals that condition on the fixed training seeds. Percentage points are written pp.

3 With an oracle memory, more reads help

3.1 Symbolic oracle

With orthogonal oracle writing, the rank-32 shared reader M1 has $D = +92.883$ pp [92.233, 93.550] and $W = +92.872$ pp [92.208, 93.564] on confirmation seed 777001, and it also improves hop 1. The separately trained evolving-query reader exceeds B5 by +92.892 pp [92.150, 93.542] and B3 by +85.233 pp [84.233, 86.183] on hops 2 to 3 at L4, with the same primary oracle codebook. These contrasts support repeated, query-dependent memory use for the tested task, rather than an explanation in terms of more memory-free iterations alone.

Exact one-hop-per-read timing is not an empirical result. For the trained orthogonal M1-lin reader, hop 4 accuracy at L4 is 2.483%, and hop 8 at L8 is zero, although some intermediate longer hops are accurate at L8. Its hop 1 accuracy falls from 99.567% at L1 to 96.550% at L4 and 26.300% at L8. These cells reject the automatic identification of a trained reader with the aligned construction of Proposition 3; they do not refute its sufficient conditions. The benefit of depth has to be described by hop and evaluation depth, including the short-hop losses.

Random oracle codes give a different balance. At rank 32, M1 improves the reachable hops by +33.258 pp [32.016, 34.484] but loses 18.933 pp on hop 1, and M1-lin likewise has a positive depth effect and a large hop 1 loss. Both conditions fail only the hop-preservation criterion. Random oracle codes are therefore not interference-free storage, and their positive reachable-hop effect is not a full gate pass.

3.2 Language-model oracle at two backbone sizes

The language-model oracle is the registered P1-L study. It puts the oracle memory into a frozen Qwen3 Base model and trains the question encoding and the reader and output path. The confirmation at Qwen3-0.6B-Base used split `test`, episode indices [100000, 102000), 2,000 episodes, training seeds 17, 29 and 43, and 97.5% paired episode-bootstrap intervals (2,000 draws) in the memory-only protocol. One oracle memory (oracle seed 17) is shared by every reader, and the intervals condition on the three seeds. Only the `oracle_random` condition ran, because the writer screens selected no learned writer. A second confirmation (P5L) repeated the rank-128 condition at Qwen3-1.7B-Base on a range no earlier study used, `test` [104000, 106000), with the rank-128 learning rates carried over without a pilot.

Memory-use check. Before any gate is interpreted, each reader's hop 1 EM must exceed the memory-free B3-L4@4 with a 97.5% interval above 0 (Amendment B7). Table 2 gives the result at Qwen3-0.6B-Base. It fails at rank 32: the readers did not learn to use the oracle memory, and no claim about multiple reads in the language model is made at rank 32. The gate outcomes there (G1-L and G2-L fail; B8@4−B4@1 on hops 2 to 3 is −0.008 pp [−0.108, 0.100]; M1-L4@4−B4@1 is +0.058 pp [−0.050, 0.167]) are reported and not interpreted. The check passes at rank 128 in every seed, at both backbones. It is a floor at hop 1 and does not show that composition uses the memory.

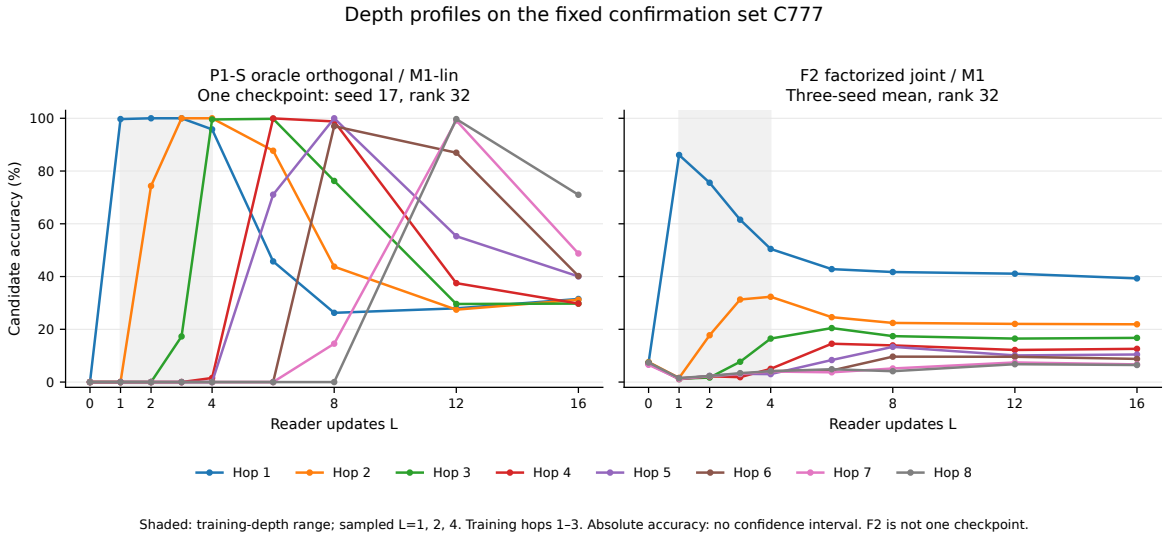


Figure 1: Quality versus evaluation depth by hop. The P1-S oracle_orth/M1-lin curves use the individual seed-17 checkpoint; the F2 joint_fact/M1 curves use the three-seed mean of fixed-checkpoint sweeps, because individual-checkpoint F2 cells are not committed. Candidate-restricted accuracy, confirmation seed 777001. The curves have no confidence bands and do not imply exact one-hop timing or hop preservation.

Table 2: Memory-use check at Qwen3-0.6B-Base: hop 1 full-vocabulary EM, and each reader minus the memory-free B3-L4@4 in percentage points with 97.5% intervals.

Rank	B4@1	B8@4	M1-L4@4	B3-L4@4	Check
32	0.067% (+0.033 [0.000, 0.100])	0.067% (+0.033 [−0.050, 0.133])	0.150% (+0.117 [0.017, 0.233])	0.033%	fails
128	19.633% (+19.633 [18.417, 20.933])	19.033% (+19.033 [17.867, 20.367])	47.850% (+47.850 [46.016, 49.667])	0.000%	passes, every seed

The shared reader composes. On hops 2 to 3 at rank 128, the shared reader M1-L4@4 exceeds B4@1 by +21.550 pp [20.383, 22.758] (26.908% against 5.358%; seeds +24.450, +15.100, +25.100) with equal adapter parameter counts (475,138 each). This meets the Amendment-B6 criterion and is the only equal-parameter claim. It also exceeds B5-L4@4 by +21.400 pp [20.217, 22.550] and B3-L4@4 by +26.908 pp [25.758, 27.975]. The claim is bounded by Amendment B7: the reader learns the value-to-key lookup over the 768 seen names and composes reads inside the language model for this oracle memory; it is not equivalent to the symbolic oracle, and nothing is claimed for unseen names or entity codes, a learned writer or end-to-end use.

The unshared reader B8 does not pass. The gated reader B8 (four unshared reads) exceeds B4@1 by only +0.983 pp [0.267, 1.717] (6.342% against 5.358%; seeds +0.925, +0.050, +1.975), below the registered 3 pp bar, so G1-L fails, although its hop 1 condition (−0.600 pp [−1.984, 0.767]) and its typed-replication condition (+1.550 pp [0.767, 2.367]) hold. B8 adds 147,456 parameters (+31.0%) over B4. G2-L passes (+0.833 pp [0.058, 1.550] over the fixed-query B5-L4@4 and +6.342 pp [5.758, 6.933] over the memory-free B3-L4@4), but a G2-L pass without G1-L is not a multi-read claim for B8. B8@4 is 20.567 pp [19.358, 21.783] below M1-L4@4 on hops 2 to 3, a secondary and ungated contrast. This is a difference between two architectures on one oracle memory in 17,000 steps; it is not a statement about unshared reads in general.

The second backbone. At Qwen3-1.7B-Base the memory-use check passes in every seed: hop 1 EM exceeds the memory-free reader (0.033%) by +30.617 pp [29.167, 32.234] for B4@1, +28.900 pp [27.517, 30.433] for B8@4 and +58.667 pp [56.983, 60.400] for M1-L4@4. The Amendment-B6 contrast M1-L4@4−B4@1 on hops 2 to 3 is +25.783 pp [24.533, 27.026] (31.958% against 6.175%; seeds +24.925, +26.500, +25.925), with 606,210 adapter parameters in each, so the equal-parameter result holds at a second scale of the same family. It has the limits of the 0.6B result, and in addition one oracle seed, three training seeds and learning rates carried without a pilot; it covers the two sizes tested and no other. B8@4−B4@1 is +1.033 pp [0.292, 1.800] (seeds +0.200, +1.725, +1.175), below the 3 pp bar, so G1-L fails again, and its hop 1 condition also fails here (−1.717 pp [−3.317, −0.016]) where it held at 0.6B; G2-L passes (+1.675 pp [0.908, 2.383] over B5-L4@4 and +7.167 pp [6.583, 7.758] over B3-L4@4). Two sizes support no scaling trend, no test across scales was registered, and the values are placed side by side without one.

A copy of B8 started from the shared solution keeps it. G3 (B8W) tests whether a B8 reader started at the M1-L4 solution keeps that solution when its step matrices are untied. Each M1-L4 checkpoint (Qwen3-0.6B-Base, rank 128) was copied into B8 and gave identical logits to the original in every seed (tolerance 0) before any test episode. Then the untied copy and a continued shared M1-L4 were trained for 2,000 more steps and compared at L=4 on hops 2 to 3 of a fresh range (test [108000, 110000)). The registered decision is that untying keeps the solution. The untied arm minus the shared arm is +0.90 pp [0.09, 1.71] (32.40% against 31.50%; seeds +1.05, +0.90, +0.75), so no seed loses more than 2 pp and the pooled lower bound is above −2 pp. The shared arm moved up from its start by +5.09 pp [4.18, 6.01], so the starting reader had not converged. The registered wording stays: B8 has no equal-parameter status (622,594 adapter parameters against 475,138), G1-L stays failed, and nothing is said about why B8 fails from a random start. The untied arm is nominally above the shared arm, but the 97.5% interval [−0.04, 1.81] touches 0 and no superiority test was registered, so that difference is exploratory and is not a finding. The result rests on one learning rate, one training length (2,000 steps), three seeds and one range, and it is sensitive to the length because the shared reference was still improving.

Same-rank table. Table 3 gives hop-wise EM at rank 128 for the single read B4@1 and for the two four-read readers at their trained depth. Table 4 gives their costs. M1-L4 and B4 have equal parameter counts, B8 adds 147,456 parameters, and the state bytes are the persistent FP32 delta memory ($r_{\text{mem}} \times r_{\text{mem}} \times 4$ bytes per episode, derived from the rank and identical for the three readers, which read the same memory). Time per question is a measured cost, compared only within one backbone; no efficiency claim is made.

Three kinds of evidence. Table 5 separates the evidence by the question it answers. The three kinds are never pooled.

Table 3: Exact-match accuracy (%) by hop at rank 128, primary setting, seed means of three training seeds, one oracle seed.

Backbone	Reader	hop 1	hop 2	hop 3	hop 4	hop 5	hop 6	hop 7	hop 8
0.6B-Base	B4@1	19.63	5.35	5.37	5.82	5.32	5.00	5.40	5.17
0.6B-Base	M1-L4@4	47.85	35.35	18.47	7.25	3.90	4.50	4.72	4.97
0.6B-Base	B8@4	19.03	6.65	6.03	6.10	6.17	5.38	6.25	6.37
1.7B-Base	B4@1	30.65	6.17	6.18	5.53	5.53	5.62	5.55	5.72
1.7B-Base	M1-L4@4	58.70	41.45	22.47	6.23	3.10	3.78	3.63	3.92
1.7B-Base	B8@4	28.93	8.17	6.25	6.28	5.97	5.73	5.57	6.20

Table 4: Costs of the same readers: parameters, persistent state, and measured time per question (memory-only protocol, batch of 16 episodes).

Backbone	Reader	Adapter parameters	State bytes per episode	Time per question (ms)	Time relative to B4@1
0.6B-Base	B4@1	475,138	65,536	0.392	1.00
0.6B-Base	M1-L4@4	475,138	65,536	0.556	1.42
0.6B-Base	B8@4	622,594	65,536	0.617	1.57
1.7B-Base	B4@1	606,210	65,536	0.486	1.00
1.7B-Base	M1-L4@4	606,210	65,536	0.625	1.29
1.7B-Base	B8@4	753,666	65,536	0.625	1.29

Table 5: Three kinds of evidence.

Kind	Question	Result	Rows
Oracle composition	Does a trained reader compose reads over a privileged memory?	M1-L4@4 exceeds B4@1 at equal parameters on hops 2 to 3 by +21.550 pp (0.6B) and +25.783 pp (1.7B); B8 fails G1-L at both backbones	LP7 to LP20
Learned-writer access	Can a learned writer build that memory?	No writer reaches the 0.80 gate in any screen (best final WP-L4 arm 33.5%; DPFP 60.8% and 73.0% at E=8; DPFP-X 66.2% and 59.3%; SQ2 best cell 72.3%; LSQT best arm 46.1%); each registered decision keeps the learned-writer line stopped	LL9, DP1, DP4, PD1, SQ1, LQ1
Mechanism evidence	Where and how does the walk happen?	Per-step readouts are decoding only; bridge patching at the final question position shifts the answer by +0.391 pp against a 10 pp bar; final-position-only reads lower M1-L4's hops 1 to 3 EM by 29.906 pp (a distribution shift)	LP5, PC1 to PC3

3.3 Depth by hop

The M1-L4 checkpoint is trained at L=4 on hops 1 to 3. Tables 6 and 7 evaluate that same checkpoint at depths L=1 to 8 on hops 1 to 8. The best depth of each hop is in bold. The last row, the oracle stop, takes the best depth for each hop using the label; it is an upper bound and a maximum over eight noisy cells, not a stopping rule.

Table 6: Qwen3-0.6B-Base: seed-mean EM (%) of the M1-L4 checkpoint by evaluation depth L and hop (primary setting, rank 128). The last row gives the best depth in parentheses.

L	hop 1	hop 2	hop 3	hop 4	hop 5	hop 6	hop 7	hop 8
1	37.8	1.1	0.9	1.1	0.8	0.9	1.1	1.1
2	**64.2**	10.9	1.4	1.8	1.6	1.7	1.7	1.8
3	61.3	30.0	7.5	2.9	2.6	2.5	2.9	3.2
4 (trained)	47.9	**35.4**	18.5	7.3	3.9	4.5	4.7	5.0
5	37.2	29.3	**24.2**	14.1	7.2	6.1	6.7	6.7
6	28.1	22.8	20.6	**16.6**	11.6	8.8	8.5	8.9
7	22.6	18.8	17.7	15.4	13.5	10.9	9.5	9.7
8	19.0	16.4	15.0	13.8	**13.5**	**12.4**	**10.6**	**9.9**
Oracle stop (L)	64.2 (2)	35.4 (4)	24.2 (5)	16.6 (6)	13.5 (8)	12.4 (8)	10.6 (8)	9.9 (8)

Table 7: Qwen3-1.7B-Base: the same matrix.

L	hop 1	hop 2	hop 3	hop 4	hop 5	hop 6	hop 7	hop 8
1	34.1	0.5	0.5	0.5	0.5	0.5	0.5	0.5
2	**77.6**	10.9	1.2	1.3	1.1	1.2	1.2	1.1
3	70.6	34.6	6.6	1.8	2.0	2.3	2.2	1.9
4 (trained)	58.7	**41.5**	22.5	6.2	3.1	3.8	3.6	3.9
5	44.9	36.5	**29.3**	15.3	5.9	5.2	5.8	6.0
6	34.9	30.2	27.0	**20.9**	12.0	7.5	6.9	7.7
7	29.0	24.6	24.3	20.6	16.9	11.2	8.7	8.8
8	24.1	20.9	21.4	18.4	**17.8**	**14.7**	**10.7**	**9.8**
Oracle stop (L)	77.6 (2)	41.5 (4)	29.3 (5)	20.9 (6)	17.8 (8)	14.7 (8)	10.7 (8)	9.8 (8)

At the trained depth the reader reaches hop 3. At L=4 its EM on hops 1 to 3 is 47.9/35.4/18.5% at 0.6B and 58.7/41.5/22.5% at 1.7B. On hops 5 to 8 it is 3.9 to 5.0% and 3.1 to 3.9%, not above the single read B4@1 (5.0 to 5.8% and 5.5 to 5.7% on hops 4 to 8). The same checkpoint reaches hops 4 to 8 at 9.9 to 16.6% (0.6B) and 9.8 to 20.9% (1.7B) when run deeper. Hop 4 peaks at L=6, and hops 5 to 8 peak at L=8, the largest depth evaluated, so their best depth is not located. Over the same range hop 1 is lost: at L=8 it is 19.0% and 24.1%, against 47.9% and 58.7% at L=4 and 64.2% and 77.6% at L=2. These cells are at hops and depths that were never trained, they carry no interval, and they rest on one oracle seed and three training seeds. The paper does not say that the reader cannot go beyond hop 3, and it does not say that the reader composes up to hop 8. A label-free stop is evaluated in Section 3.4.

3.4 A label-free stop at inference

The matrix suggests that the depth that suits a hop grows with the hop, and that the deployed depth L=4 is a compromise. LSR (G2) asks whether a per-question stop that uses no label can raise hops 4 to 8 over fixed L=4 without a large loss on hop 1. It evaluates the six rank-128 M1-L4 checkpoints (three seeds at each backbone) and trains nothing. Three rules were registered: R1 stops when the relative change of the state falls below a threshold, R3 when the norm of the read, relative to the first read, falls below a threshold, and R2 when the model's top-1 answer has not changed over two consecutive depths (no parameter). The thresholds of R1 and R3 were chosen on validation episodes under a hop-1 constraint, and the registered comparisons use the test range [106000, 108000) with 99.17% intervals for six comparisons.

R1 and R3 had no feasible threshold. At both backbones no value of the 25-value validation grid kept the hop-1 loss against fixed L=4 within the registered margin (8 pp pooled, 10 pp per seed), so no test

statistic was computed. The traces show why. At 0.6B the read norm relative to the first read never falls below 1 (median 1.0 to 1.1), so a threshold below 1 never fires and one above 1 stops at the first step. At 1.7B it grows to about 10 and the relative change of the state stays at 0.5 to 0.7, so both rules stop at the first step or run to depth 8. This does not show that no other label-free signal works.

R2 raises hops 4 to 8 over fixed L=4. Against fixed L=4, which is also the best fixed depth under the validation constraint, the R2 stop raises hops 4 to 8 by +3.90 pp [3.35, 4.43] at 0.6B and +5.68 pp [5.09, 6.25] at 1.7B, in every seed (+4.58, +3.25, +3.87 and +5.46, +5.71, +5.86), and it raises hop 1 by +6.28 and +4.70 pp. So a label-free per-question stop raised hops 4 to 8 over fixed L=4 at both scales, in the oracle-memory condition, evaluation only. The result comes with its costs. Hop 2 falls by 10.83 pp [−11.85, −9.87] at 0.6B and by 12.27 pp [−13.32, −11.23] at 1.7B, and hop 3 by 3.28 and 3.65 pp, so R2 does not improve every hop. On hops 4 to 8 it stays below fixed L=8 (−3.54 and −5.05 pp), and it recovers about half of the headroom between L=4 and the best fixed depth for each hop. It reads 5.60 and 5.66 times per question against 4, and each stopping decision needs a pass through the frozen upper layers, so a question that stops after s^* reads needs $s^* + 1$ such passes (6.60 and 6.66 on average). R2 was not timed, and no efficiency or latency claim is made. The stop depends only weakly on the hop: the mean number of reads rises from 4.97 at hop 1 to 5.86 at hop 8 at 0.6B, and from 4.87 to 5.93 at 1.7B. Table 8 puts the R2 row next to the fixed-depth rows. The LSR range differs from the range of the matrices above, so its L=4 cells differ slightly from them (hop 1 is 48.0% here and 47.9% there), and the two are not pooled. The intervals condition on three training seeds per backbone and one oracle code table.

Table 8: Exact match (%) of the L=4-trained M1-L4 checkpoints on hops 1 to 8 at rank 128 on the LSR test range (episodes [106000, 108000), seeds 17, 29, 43 pooled, oracle memory). R2 is the label-free per-question stop; O1 and O2 choose a depth per hop (O1 on validation, O2 on the test cell with the label), so they are upper references and not stopping rules. The LSR range differs from the range of the depth matrix of Tables 6 and 7, so its rows are compared only with each other.

Row	1	2	3	4	5	6	7	8	4 to 8
Qwen3-0.6B-Base									
L = 2	64.6	9.6	1.5	1.9	1.8	2.3	2.1	1.8	2.0
L = 4 (trained)	48.0	33.8	18.1	6.1	4.4	4.9	5.1	5.5	5.2
L = 6	28.9	23.0	20.8	15.8	12.1	9.6	8.9	9.4	11.2
L = 8	20.2	16.7	15.5	13.8	14.4	13.2	11.5	10.3	12.6
R2 (label-free stop)	**54.3**	**22.9**	**14.8**	**10.6**	**9.4**	**9.1**	**8.2**	**8.2**	**9.1**
O1 (validation depth per hop)	64.6	33.8	23.3	15.8	14.4	13.2	11.5	10.3	13.0
O2 (best test cell, labels)	64.6	33.8	23.3	15.8	14.4	13.2	11.5	10.4	13.0
Qwen3-1.7B-Base									
L = 2	77.2	10.9	0.8	1.3	1.2	1.2	1.3	1.5	1.3
L = 4 (trained)	59.2	42.4	23.0	6.5	3.4	4.7	4.1	4.0	4.5
L = 6	36.8	29.9	28.2	22.5	13.0	8.5	7.6	8.2	11.9
L = 8	24.6	21.1	21.2	19.6	18.8	15.5	12.2	10.2	15.3
R2 (label-free stop)	**63.9**	**30.1**	**19.3**	**13.9**	**11.2**	**10.0**	**8.4**	**7.5**	**10.2**
O1 (validation depth per hop)	77.2	42.4	30.6	22.5	18.8	15.5	12.2	10.2	15.8
O2 (best test cell, labels)	77.2	42.4	30.6	22.5	18.8	15.5	12.2	10.2	15.8

3.5 Seed-level statistics

The registered intervals resample episodes and hold the three fitted seeds fixed. Tables 9 and 10 print the three seed effects of each headline contrast beside a seed-level interval. The seed-level interval is the mean of the three seed effects plus or minus $t(0.9875; 2) = 6.205$ times their sample standard error, the two-sided 97.5% level of the registered intervals. Both kinds of interval are conditional on three fitted seeds and one fixed oracle code table (oracle seed 17, one layout). Neither covers future training seeds or other code tables, and the t-interval treats the three seed effects as exchangeable draws, which three values cannot test.

The seed-level interval excludes 0 for the B6 contrast (M1-L4@4−B4@1) and for G2-L against the memory-free reader, at both backbones. It includes 0 for the G1-L contrasts and for G2-L against the fixed-query

Table 9: Qwen3-0.6B-Base: seed effects and intervals for the headline contrasts at rank 128 (pp).

Contrast (pp)	Seed 17	Seed 29	Seed 43	Mean	Seed-level 97.5%	Episode bootstrap 97.5%
B6 contrast M1-L4@4 − B4@1, hops 2 to 3	+24.450	+15.100	+25.100	+21.550	[+1.50, +41.60]	[+20.383, +22.758]
G1-L reach B8@4 − B4@1, hops 2 to 3	+0.925	+0.050	+1.975	+0.983	[−2.47, +4.44]	[+0.267, +1.717]
G1-L hop 1 B8@4 − B4@1	−2.300	+1.200	−0.700	−0.600	[−6.88, +5.68]	[−1.984, +0.767]
G1-L typed replication B8@4 − B4@1, hops 2 to 3	+1.575	+0.600	+2.475	+1.550	[−1.81, +4.91]	[+0.767, +2.367]
G2-L B8@4 − B5-L4@4 (fixed query), hops 2 to 3	+0.875	+0.100	+1.525	+0.833	[−1.72, +3.39]	[+0.058, +1.550]
G2-L B8@4 − B3-L4@4 (memory-free), hops 2 to 3	+6.525	+5.675	+6.825	+6.342	[+4.20, +8.48]	[+5.758, +6.933]

Table 10: Qwen3-1.7B-Base: the same contrasts.

Contrast (pp)	Seed 17	Seed 29	Seed 43	Mean	Seed-level 97.5%	Episode bootstrap 97.5%
B6 contrast M1-L4@4 − B4@1, hops 2 to 3	+24.925	+26.500	+25.925	+25.783	[+22.93, +28.64]	[+24.533, +27.026]
G1-L reach B8@4 − B4@1, hops 2 to 3	+0.200	+1.725	+1.175	+1.033	[−1.73, +3.80]	[+0.292, +1.800]
G1-L hop 1 B8@4 − B4@1	−1.800	−0.300	−3.050	−1.717	[−6.65, +3.22]	[−3.317, −0.016]
G1-L typed replication B8@4 − B4@1, hops 2 to 3	+0.375	+3.300	−0.300	+1.125	[−5.73, +7.98]	[+0.275, +1.942]
G2-L B8@4 − B5-L4@4 (fixed query), hops 2 to 3	+0.550	+3.600	+0.875	+1.675	[−4.33, +7.68]	[+0.908, +2.383]
G2-L B8@4 − B3-L4@4 (memory-free), hops 2 to 3	+6.225	+8.825	+6.450	+7.167	[+2.01, +12.33]	[+6.583, +7.758]

reader at both backbones, so the G2-L pass over B5-L4@4 is a statement about the fitted seeds. The registered decisions use the episode intervals and are unchanged.

4 Learned writers are the weak point

4.1 Symbolic learned writers

The unfactorized joint rank-32 reader M1 passes the registered gate with a much smaller effect than the oracle: $D = +8.292$ pp [7.383, 9.292] and $W = +2.492$ pp [1.244, 3.733]. Its mean advantage over B5 is +7.767 pp [6.516, 9.050], but the three seed effects are +9.300, −17.700 and +31.700 pp. The pooled interval is therefore compatible with a failed training seed, and it does not establish improvement for a population of new seeds. The joint linear reader fails the primary depth gate, with specificity −0.408 pp [−1.475, 0.753]. Unfactorized frozen T1 writers fail the replication-specificity criterion for both readers, and the M1 replication interval includes zero. This does not show that joint training is necessary for every composable writer: factorized frozen writers give positive depth and specificity effects under the tested protocol. Even matched factorized T1 sources with validation accuracy above 91% do not certify the full gate after freezing; the resulting readers keep positive depth effects but fail hop 1 preservation.

Factorization separates composition evidence from overall utility. On the reused confirmation set C777, joint_fact M1 has $D = +22.975$ pp [22.017, 23.975] and $W = +20.158$ pp [19.058, 21.192], but $H = -35.617$ pp [−37.217, −33.933] on hop 1. All four factorized writer and reader conditions fail the complete depth gate because of the hop 1 loss, although their reach, specificity and control contrasts pass. At L4, joint_fact M1's reachable-hop contrast against the unfactorized joint system is −19.108 pp [−20.534, −17.600], despite a larger depth contrast. Improving a contrast from a weaker L1 baseline is not the same as improving the deployed L4 system. The larger-rank follow-up gives stronger joint-code effects in its tested regimes: rank-64 M1 has $D = +30.533$ pp [29.225, 31.900] and a pooled B5 contrast of +24.017 pp [22.575, 25.575] (seeds +22.575, +25.075, +24.400). Rank and training steps are confounded there (20,000 steps against 15,000), it has no replication, and it reuses the C777 set, so it is not an independent confirmation.

4.2 Language-model writer screens

In one frozen language-model module, validation screens (training seed 17, validation seed 4242) tested learned writers against the WP-L4 gate: validation EM of at least 0.80. The screens select no learned writer. Input information is recoverable at the adapter's input positions (tail-token probe 97.6%, question-final start 99.4%, relation 100.0%), and a supplied-answer-code control reaches 98.7% EM, so the frozen output path is usable. The deployed read is the problem. For the audited W5 arm, the clean-value head reads 85.4% while the read-value head reads 10.5%, and zero- or shuffled-memory interventions show sensitivity to the memory channel without localizing the failure. The final rank-128 arms reached at best 33.5% (R), 29.9% (W5) and, for the ineligible value/output ceiling OA, 84.0%. A read-failure factorial (RFX) on frozen checkpoints found that only an exact store together with the true key rescues the read, and the true key is a label, not deployable; the exact store with the deployed or unit query does not. The verdict does not separate store from query; the trained least-squares study LSQT, registered and run afterwards, is reported below.

DPFP. A registered screen replaced the keys and queries of the WP-L4 learned writer by DPFP features: three arms (D32-3 at rank 32, D128-1 and D128-3 at rank 128), one seed, 1,000 validation episodes each. No arm reaches the 80% gate: D32-3 reaches 39.0%, D128-1 58.0% and D128-3 60.8%. The best arm is above the 50% trigger, so D128-3 was retrained once at E=8 under the registered rule and reached 73.0%, again below 0.80. The registered decision is stop: no writer is selected, no further writer screen follows from that rule (the later, separately registered screens are reported below), and P1L-D (the P1-L learned condition for a selected writer) was not registered and was not run. DPFP does raise EM (by 27.0 pp at rank 32 and by 24.5 and 27.3 pp at rank 128 over the WP-L4 references), lowers the normalized read error (0.39, 0.43 and 0.36 against 0.67 and 0.63) and raises the read-value head accuracy (30.2%, 46.0% and 51.0% against 8.8% and 19.7%), but the clean-to-read gap narrows without closing. DPFP changes the feature geometry and the query normalization together (Amendment A1), so neither these gains nor the 12.2 pp gain of the E=8 retrain, which also halves the memory load, is attributed to decorrelation alone. A zero- or shuffled-memory audit leaves at most 0.2% EM for every arm, which shows that the answers depend on the episode's own memory, not that the read is accurate. The result is one seed, one learning rate and fixed budgets, and it is not a proof that learned writing in a language model is impossible.

DPFP-X. A follow-up registered after this stop trained the best DPFP design at rank 256 on Qwen3-0.6B-Base (X256) and at rank 128 on Qwen3-1.7B-Base (X17), everything else as D128-3. Neither arm reaches 0.80 (X256 66.2%, X17 59.3%), so the learned-writer line stays stopped and both arms are descriptive rows next to DPFP. X17 differs from D128-3 (60.8%) by −1.5 pp, which the registered rule calls not distinguished, and it is a diagnostic and not a scaling test. Both curves were still rising at the final checkpoint (+3.5 and +2.3 pp over the last interval), but the study has no longer run and nothing is extrapolated. X256's higher EM is not accompanied by a lower read error or a better read-value head than D128-3's, and the paper gives no explanation for it. The zero- and shuffled-memory audit again leaves at most 0.4% EM.

SQ2: ridge storage on the DPFP features. SQ2 continued the final D128-3 checkpoint for 6,000 steps in four arms that cross the store (the delta store as trained, or a ridge store on the DPFP features) with a coefficient loss (off or on); it used the same validation episodes and one seed. The registered primary contrast is S-RC (ridge with the coefficient loss) against S-DE (the continued delta control), and it does not advance: the EM gain is +4.1 pp where +10 is needed, and the read error is 0.554 of S-DE's where at most 0.5 is needed; only the stability condition is met. No cell reaches 0.80, so no writer candidate is flagged and the learned-writer line stays stopped. The registered prediction, that S-RC would gain 10 pp and halve the read error, did not hold. Table 11 gives the four cells and the parent. They are descriptive only, and each reading carries its limit.

- Ridge storage on the DPFP features roughly halves the read error (0.196 for S-RE against 0.359 for S-DE) and is +6.9 pp over the continued delta control, which still misses both registered conditions. This is one seed, and the validation episodes selected D128-3.
- The delta continuation alone gains 4.6 pp (60.8 to 65.4%), so the parent had not converged and part of every continuation gain is the continuation.
- The coefficient loss did not help: S-DC is 2.4 pp below S-DE, S-RC is 2.8 pp below S-RE, the interaction is −0.4 pp, and the read error is unchanged by it.

- Swapping only the store at step 0, with no retraining, gives 68.7% (read error 0.312); the ridge arms first dip below that value and then recover, which is consistent with starting from AdamW moments of another objective (registered deviation 2) but was not tested.
- The gradient clip (norm 1.0) is active at every logged step in all four arms, as it was in the training of D128-3; the paper makes no statement about what it does to the effect of the coefficient loss.

Zeroing or shuffling the memory leaves at most 2 of 1,000 answers correct in every arm, so the answers come from the stored episode, but the read error stays between 0.20 and 0.36.

Table 11: SQ2: four continuations of the DPFP learned writer D128-3 for 6,000 steps (validation episodes [0, 1000), one seed). Exact match (EM), normalized read error and clean-value head accuracy at the last checkpoint. No cell reaches 0.80, and all cells are descriptive; the registered primary contrast S-RC against S-DE does not advance.

Arm	Storage and coefficient loss	EM (%)	Read error	Clean-value head (%)	Step 0 EM (%)
S-DE	delta, no coefficient loss	65.4	0.359	93.7	60.8
S-DC	delta, coefficient loss	63.0	0.353	92.9	60.8
S-RE	ridge, no coefficient loss	72.3	0.196	93.6	68.7
S-RC	ridge, coefficient loss	69.5	0.199	92.9	68.7
D128-3 (parent)	delta, 22,370 steps	60.8	0.363	90.6	

LSQT: trained least-squares stores. LSQT trains three arms from scratch at rank 128 with the unit query, with the delta store replaced by a least-squares store: LT-K4 (four Kaczmarz sweeps) and LT-RG001 and LT-RG1 (ridge with $\eta = 0.01$ and $\eta = 1$). The registered decision is that no arm reaches 0.80. The best ridge arm, LT-RG001, reaches 46.1%, inside the band from 40% to below 48.5%, which neither falsifies the line nor triggers W3 (a separate ridge-plus-DPFP study that would need its own registration), so the trained least-squares line stays open and the learned-writer line stays stopped. All three arms are above R-r128 (33.5%), by +4.8, +12.6 and +9.1 pp, and well below D128-3 (60.8%), by 22.5, 14.7 and 18.2 pp (Table 12; the standard error of a difference is about 2.2 pp, one seed). The read error is not removed (0.554, 0.512 and 0.499, against 0.631 for R-r128 and 0.363 for D128-3), and the read-value head stays at 22 to 27% against a clean-value head of 92 to 96%. The registered prediction, that the best ridge arm would reach 50% with a read error of at most 0.35, did not hold. The unit query is in every LSQT arm and in none of R-r128, so the gain over R-r128 cannot be split between the store and the unit query; that split is untested. The arms differ in step count, two of the curves were still rising at the end, and nothing is extrapolated.

Table 12: LSQT: trained least-squares stores at rank 128, trained from scratch with the unit query (validation episodes [0, 1000), one seed), against the delta-store references. The registered decision is that no arm reaches 0.80, the best ridge arm lies in the open band 40% to below 48.5%, and the trained-LSQ line stays open. The unit query is in every LSQT arm and not in R-r128. Clean and read head: accuracy of the value decoded from the stored and from the read value.

Arm	Store	EM (%)	Read error	Clean head (%)	Read head (%)	vs R-r128 (pp)
LT-K4	four Kaczmarz sweeps	38.3	0.554	92.3	27.4	+4.8
LT-RG001	ridge, $\eta = 0.01$	46.1	0.512	93.9	26.5	+12.6
LT-RG1	ridge, $\eta = 1$	42.6	0.499	95.9	22.3	+9.1
R-r128 (WP-L4)	delta	33.5	0.631	93.3	19.7	
D128-3 (DPFP)	delta	60.8	0.363	90.6	51.0	

4.3 Negative interventions and the gate each did not meet

Table 13 lists each repair or test that failed its registered criterion. Each row is one study under its own conditions (one training seed or three, one optimizer, a fixed budget). The rows record which criterion was not met. They do not show that learned writing, an exact store, unshared reads or composition at rank 32 is impossible. Not triggered: the LSQ arms of the RFX rule (the separately registered LSQT is reported above). Not run: P1L-D and P1L-S. A warm start of B8 from M1-L4 (Section 3.2) kept the shared solution, so it is not a failed repair and has no row here.

Table 13: Negative interventions and the registered criterion each did not meet.

Intervention	Conditions	Result	Criterion not met	Rows
Learned writers of the WP-L4 screen (R, W5, read-supervision variants)	Qwen3-0.6B-Base, ranks 32 and 128, one seed, 1,000 validation episodes	Best eligible arm 33.5% (R, rank 128)	Validation EM ≥ 0.80 (and the 0.50 trigger of the E=8 retraining)	LL9
Read-failure factorial (RFX)	Frozen R/r128, R/r32 and W5/r32 checkpoints, evaluation only	Only an exact store with the true key (a label) rescues the read; the exact store with the deployed or unit query gives 38.8% and 38.75% against 33.8% at R/r128	Rescue: normalized read error ≤ 0.5 of the baseline and EM gain ≥ 10 pp	RX1
Oracle codes at rank 32 (WP-L6)	M1-L4, $r = r_{\text{mem}} = 32$, one seed, 5,666 steps	Hop 1 EM 0.1% (O-rot) and 11.8% (O-shared)	Hop 1 full-vocabulary EM $\geq 30\%$ at L=4	WL1
DPFP writer	Ranks 32 and 128, E=16, E=8 retrain of the best arm, one seed	39.0%, 58.0% and 60.8% at E=16; 73.0% at E=8	Validation EM ≥ 0.80	DP1
DPFP-X writer	X256: Qwen3-0.6B-Base, $r = 256$; X17: Qwen3-1.7B-Base, $r = 128$; one seed	66.2% and 59.3%	Validation EM ≥ 0.80	DP4
Oracle readers at rank 32	B4, B8 and M1-L4 on the oracle memory, three seeds	Hop 1 EM at most 0.15%; the memory-use check fails	Amendment B7 rule 1: each reader's hop 1 EM above B3-L4@4 with a 97.5% interval above 0 (lower bounds 0.000 for B4, -0.050 for B8)	LP7
B8 (four unshared reads) at 0.6B	Rank 128, three seeds	B8@4$-$B4@1 on hops 2 to 3: +0.983 pp [0.267, 1.717]	G1-L point estimate ≥ 3 pp (its hop 1 and typed conditions held)	LP9
B8 at 1.7B	Rank 128, three seeds	+1.033 pp [0.292, 1.800]; hop 1 -1.717 pp	G1-L point estimate ≥ 3 pp; hop 1 point estimate ≥ -1 pp	LP16
Ridge storage and coefficient loss on the DPFP features (SQ2, primary contrast S-RC against S-DE)	Continuation of D128-3 for 6,000 steps, rank 128, one seed, 1,000 validation episodes	EM gain +4.1 pp, read error ratio 0.554; best cell S-RE 72.3%	EM gain $\geq +10$ pp and read error ratio ≤ 0.5 (stable decoding met); no cell reaches 0.80	SQ1
Trained least-squares stores (LSQT: Kaczmarz, ridge $\eta = 0.01$ and $\eta = 1$)	Rank 128 from scratch with the unit query, one seed, 1,000 validation episodes	38.3%, 46.1% and 42.6%	Validation EM ≥ 0.80; the best ridge arm lies in the open band 40% to below 48.5% (line open, W3 not triggered)	LQ1
Bridge patching	Rank 128, 0.6B, L=4, state after the first read at the final question position	+0.391 pp [-0.130, 0.781] (M1-L4), 0.000 pp (B8)	Shift ≥ 10 pp, 95% interval above 0, positive in every seed	PC1

4.4 A learned parser with a fixed code table (CTC)

This study is not a learned writer, and its numbers are not comparable with the DPFP rows above. A classify-then-code writer reads the support text with frozen layer-14 states, classifies the head, relation and tail of each fact with trained heads, and writes the oracle's fixed codes for the predicted names into the memory. The frozen P1-L rank-128 readers then run unchanged. The writer has 268,292 parameters; no parameter match is claimed. The registered decision flags a candidate (validation episodes $[0, 1000)$, one seed, arm CE selected because CE and CE+ans have the same hop-1 EM). The wording registered for it is: a learned parser plus a fixed code table drives the frozen composing reader. The per-fact classification accuracy is p=1 (12,000 of 12,000 facts correct), and M1-L4@4 reaches 50.7/36.5/21.6% on hops 1 to 3 against 50.6/36.6/21.5% with the oracle memory, so the EM equals the oracle reference within 0.2 pp. The gain over B4@1 on hops 2 to 3 is +24.05 pp [21.80, 26.30] (+23.90 pp with the oracle memory), and zero or shuffled memory leaves 0.0% and 0.2% hop-1 EM. The accuracy is 1 because both entities are read at their own tokens and the relation is constant in this task (it is 0 for every fact), so the key is always the code of the head entity. These are properties of the task and of the closed 768-name vocabulary. The codes are the oracle's privileged codes. The study shows that the reader transfers to near-oracle codes and that it uses its memory; it does not show that a model can learn its own codes. No confirmation was run, and a three-seed confirmation would need its own registration and the user's decision. The learned-writer line stays stopped (Section 4).

5 What the mechanism evidence shows, and what it does not

Per-step readouts are decoding only. For M1-L4 and B8 the state after each read $s = 1, \dots, 4$ was decoded at the answer position on the confirmation episodes (three seeds pooled), through the trained output path and the frozen upper layers (logit lens), by projection onto name-gradient directions, and by a random direction. At rank 128 the logit lens recovers earlier-hop entities from M1-L4's early states, and best at a step at or above the entity's hop rather than at $s = e$. On hop-1 questions the hop-1 entity is the top-1 name in 38.6%, 64.3%, 61.1% and 47.9% of episodes at $s = 1, \dots, 4$, and on hop-2 questions the hop-2 entity in 1.1%, 11.0%, 30.2% and 35.4%. The rates at $s = 4$ equal the deployed EM within 0.1 pp, and the hop-1 rates at $s = 2$ and $s = 3$ are close to the EM of the same checkpoint at depth 2 and 3 (64.15% and 61.28%), which the corrected record reads as the answer the reader would give if it stopped at that step. B8's lens decodes names only at $s = 4$ (the hop-1 entity in 19.1% of hop-1 questions) and at most 0.23% at $s = 1, \dots, 3$. The name-gradient projections and the random direction are at or near chance for both readers (maximum cell 0.28%), and at rank 32 every method is at chance, which adds nothing because the rank-32 readers did not use the memory. These are decoding results. They are not gate criteria, not an ordered entity trace, not path following, and not evidence for the composition claim; bridge patching gives no causal claim. Liang and Pan warn that decodable intermediate states do not establish a complete serial algorithm; the readouts respect that boundary.

P1L-C: no causal bridge. P1L-C evaluated the nine rank-128 checkpoints (M1-L4, B8 and B4, three seeds; nothing retrained) at Qwen3-0.6B-Base on a second range, `test` $[102000, 104000)$, with 95% intervals. Keeping the memory only at the final question position lowers the hops 1 to 3 EM of M1-L4@4 from 33.64% to 3.74% (-29.906 pp $[-30.756, -29.094]$), of B8@4 from 10.48% to 0.99% (-9.489 pp $[-10.078, -8.911]$) and of B4@1 from 10.39% to 1.26% (-9.133 pp $[-9.711, -8.589]$), in every seed and hop. The readers were trained with reads at every question position, so the drop is a distribution shift, not evidence of leakage, and it does not locate the bridge. Bridge patching replaces the state after the first read at the final question position by the state from a world whose first edge points to another bridge. It shifts the answer by +0.391 pp $[-0.130, 0.781]$ for M1-L4 and 0.000 pp $[-0.521, 0.000]$ for B8, against a registered bar of 10 pp with an interval above 0 and a positive shift in every seed. There is no causal bridge claim for either reader. The M1-L4 answer does follow the other chain when the memory itself carries the other bridge (37.24% against 7.03%), but not when the state after the first read alone is replaced at the final or at every question position (+0.911 pp $[-0.651, 2.344]$). The P1-L contrasts replicate on this range (M1-L4@4$-$B4@1 +20.808 pp [19.725, 21.850], B8@4$-$B4@1 +0.925 pp [0.283, 1.533]) without changing any gate.

Mechanism wording. The adapter acts at every question position. Each position performs its own vertical walk: its branch output enters that position's residual after the layer-14 attention and is not an input

of any later position's adapter, and the frozen upper layers aggregate across positions. Which position's walk carries the answer is untested. "Composes reads" in this paper means the iterated reads inside one position's loop.

What the theory does and does not give. The propositions of Appendix D are conditional. With a fixed memory and depth the branch output lies in a fixed subspace, so a depth comparison measures different computation inside one bottleneck (Proposition 1). An existence construction shows that a linear reader with an aligned successor map, orthonormal codes and exact storage can follow a chain (Proposition 3); the trained M1-lin reader is not identified with it (Section 3). Non-orthogonal delta writes can alter earlier associations, and a one-read reader sees the context only through one observation (Propositions 4 and 5). A fixed-point error bound needs a valid contraction constant and says nothing about a label without decoder-margin assumptions (Proposition 7). These are conditional explanations and diagnostic requirements, not fitted causes of the symbolic or language-model outcomes.

6 Costs, memory load, stopping, and the other registered studies

Costs. The primary comparison is a single read at the same rank, with added latency reported as a cost, and no efficiency advantage over a larger-rank single read is claimed. Earlier P2-S and P2-S-F1 comparisons with larger single reads remain historical evidence, including the original practical-gate failure. They supply no efficiency claim for this paper. The joint M1 point has 13,314 trained parameters and 4,096 persistent FP32 bytes per episode, against 114,690 parameters and 65,536 bytes for B6 at rank 128. A reader may therefore be attractive under a strict state allocation even when it does not win the measured latency comparison. That is a conditional design choice, not a practical advantage established for the full request path, because the reported timings exclude the context writer and other preparation.

Memory load. At rank 32 the reachable-hop L4−L1 benefit shrinks from E12 to E36 under the registered load and layout changes, while the multi-read capability criterion still holds at the tested training loads. The largest tested load is not an identified failure boundary.

Stopping and overshoot. Post-hoc learned halting fails hop 1 preservation in the six primary frozen-reader families. A hop-1-constrained threshold gives one registered pass, for random-oracle writing, after about two executed updates and with no latency gain. Stay training and a sticky gate do not remove the overshoot, and fixed-depth training remains the remedy found. Training-time terminal-Jacobian regularization reduces overshoot without meeting the registered removal criterion, and it decreases longer-walk accuracy.

Table 14 gives these studies with the registered criterion each rests on; the full results are in Appendix E.

Table 14: Other registered results and their criteria (wording of the claim contract, Appendix C).

Result	Gate or criterion	Rows
Same-rank policy: several reads against a single read at the same rank, with latency reported as a cost and no efficiency claim over larger-rank single reads	None: a policy and a static count, not an accuracy result	FR1
Memory load: the depth benefit shrinks from E12 to E36 for fixed checkpoints, while M1@4 and B8@4 keep exceeding B4@1 at the trained loads; updated-path diagnostics are partial; key interference is identified for oracle_random only (IVD)	IS-E1 (a negative DeltaD with CI<0 is consistent with Prop.4), IS-E2 (C(E)≥3 pp, CI>0); IS3 and IS4 descriptive; IVD (IV1, IV2) is a separate study with its own gate	IS1-IS4, IV1, IV2

Result	Gate or criterion	Rows
Historical larger single reads: under joint writing B6/r128 and both B7 reads score above shared M1 (claim B1, that a larger single read could not replace it, is not supported); under random-oracle and factorized writing M1 outperforms them; the P2-S practical criterion fails; the factorized latency frontier is unmeasured	P2-S Gate 3: the capability claim needs E1/E2 positive at every tested size and writer; the practical claim also needs a latency win (E4), which fails. P2-S-F1 F-E3: the joint_fact capability claim needs every contrast with CI>0	B1, B2, B5, B7, B8 (open)
Sharing: step-specific B8 exceeds shared M1-L4 at matched training depth, but not on every hop; fixed-depth training improves shared M1	P2-S-F1 decision rule: B8 is carried into P1-L if F-E1>0 (CI) for joint and joint_fact, and fixed-depth training if F-E2>0 (CI); no gate on hop-wise superiority	B3, B4, B6, B9
Post-hoc stopping pilots: label-free rules trade long-hop gain for hop1 loss and give no speedup	Screening only; no gate	H1-H6
Learned post-hoc halting does not repair hop1 preservation in six primary frozen-reader families	Combined criterion: H≥−1 pp (point), D≥3 pp with CI>0, W with CI>0; every per-seed H also fails	HL1-HL5
Terminal-Jacobian regularization reduces overshoot, does not meet the removal criterion, and costs longer-walk accuracy	Removal needs DeltaS>0 with CI>0 and S(RS+J)≥−1 pp; fails for both writers	JS1-JS6
A halting head with the correctness label and a hop-1-constrained validation threshold repairs overshoot for oracle_random on frozen readers; no other CVH arm passes; no CVH arm has a latency claim	Gain ≥3 pp with CI>0, hop-1 change ≥−1 pp, mean executed updates ≤8; a latency claim only where wall time is below fixed L16 and the arm passes (it does not hold for the passing arm)	CV1-CV4
Stay training (T1) and a content-gated sticky update (T2) do not remove the overshoot under this protocol; fixed-depth training remains the remedy found	An arm passes only for both writers: absolute L16−L4 difference ≤1 pp on hops1-3, hops4-8 at L8 at least −1 pp against T0, and gain ≥3 pp with CI>0 over B4; no arm passes	TS1-TS2
The depth gain on hops2-3 is also present on unseen-pool entities (256 further names of the same vocabulary); no claim beyond the closed vocabulary or for new entity codes	Descriptive wording rule: unseen-pool gain ≥3 pp with CI>0 (all six results meet it); Part A also reproduces the 18 committed M1 depth-contrast rows	UN1-UN4

7 Claim contract, limitations, and reproducibility

What the paper claims.

- The equal-parameter oracle contrast: at rank 128 with a clean oracle memory, the shared reader M1-L4 read four times beats a single read of equal parameter count on hops 2 to 3 at two backbone sizes of one family.
- A negative result for B8, the four-unshared-reads reader, under one optimizer; a copy started from the M1-L4 solution keeps it, which leaves G1-L failed and gives B8 no equal-parameter status.
- A label-free per-question stop (R2) raises hops 4 to 8 over fixed L=4 at both backbones, in the oracle-memory condition and by evaluation only, at the stated cost of hop 2, of extra reads and upper-layer passes, with no efficiency or latency claim; R1 and R3 had no feasible threshold.
- A learned parser plus a fixed code table drives the frozen composing reader at p=1 on validation episodes of one seed; this is not a learned-writer result.
- A negative result for learned writers: the symbolic learned writers give a smaller or hop-1-damaging benefit, and no language-model writer reaches the 0.80 gate (DPFP, DPFP-X, SQ2 and LSQT included); every registered stop stands.
- Per-step readouts as decoding only, and no causal bridge at the final question position.
- No scale trend: two sizes, carried learning rates and no registered test across scales.

What the paper does not claim. The language-model oracle is not the symbolic oracle: it needs a learned value-to-key lookup for every read after the first, so symbolic oracle results do not transfer to it. No conclusion about multiple reads in the language model is drawn at rank 32. No claim covers unseen entities, names beyond the closed 1,024-name vocabulary, or new entity codes. OA is the value and output ceiling, not an addressing control. B8 is not parameter-matched to the single read or to M1: it adds 9,216 parameters (+8.9%) at rank 32, 147,456 (+31.0%) at rank 128 and +69.2% in the symbolic unfactorized joint model. The paper also does not say, and no evidence here supports, any of the following.

- That the reader cannot go beyond hop 3, or that it composes up to hop 8 (Section 3.3).
- That B8 cannot compose, or that sharing is necessary: B8 contains M1-L4, it was trained with one optimizer and no learning-rate pilot of its own, and a warm start from M1-L4 keeps the shared solution without changing the G1-L verdict (BW1), so it says nothing about why B8 fails from a random start.
- That an exact store would not help, or that trained least-squares storage works or is falsified: RFX is evaluation only, and the trained least-squares study LSQT left its line open (best arm 46.1%, read error 0.50 to 0.55).
- That the recurrence composes across question positions, or the converse: each position performs its own walk, and which position's walk carries the answer is untested.
- That rank 256 or the 1.7B backbone fixes the learned writer, that DPFP-X selects a writer, any extrapolation of its rising curves, a scaling claim from X17, an explanation of X256's higher EM, or any learned-writer or end-to-end claim at either rank or backbone.
- That the intervals cover future training seeds, other oracle code tables or layouts.
- That adaptive stopping is efficient, that R2 improves every hop, that label-free stopping closes the gap to the best depth, or that no label-free stop can work (R1 and R3 had no feasible threshold; R2 lowers hops 2 and 3 and was not timed).
- That untying improves B8, that B8 passes G1-L after the warm start, or that untied is better than shared (the difference is exploratory), or any reason why B8 fails from a random start.
- That ridge storage solves or removes the read distortion, or that the coefficient loss helps: on the DPFP features ridge storage halves the read error and is +6.9 pp over the continued delta control (one seed, on the validation set that selected D128-3), the primary SQ2 contrast does not advance, and the coefficient loss did not help.
- That any SQ2 cell or LSQT arm is a writer candidate or a selected method, or that the LSQT gain over R-r128 is due to the store: the unit query is in every LSQT arm and in none of R-r128, and that split is untested.
- That CTC is a learned writer, that the model learns its own codes, or that a CTC number is comparable with a DPFP or DPFP-X number.

Uncertainty. The bootstrap conditions on the fixed training seeds, and the follow-ups reuse confirmation samples rather than supplying independent confirmation of every successive claim. Training-seed failures, longer rank-follow-up training and the disclosed F2 decision deviation remain relevant even when a pooled interval is positive. The P1-L confirmation (test [100000, 102000), 97.5%, $k = 2$) and P1L-C (test [102000, 104000), 95%) condition on three training seeds and one oracle seed and layout. Section 3.5 adds seed-level intervals; with three seeds they are wide, and they include 0 for the G1-L contrasts and for G2-L against the fixed-query reader at both backbones. The depth matrix evaluates one L4-trained checkpoint at depths and hops that were never trained, without intervals, and its oracle-stop row uses the label. The language-model oracle results rest on one random oracle code table. The stop-rule, warm-start and CTC results (Sections 3.4 and 4.4) share that limit; CTC is validation episodes of one seed, B8W is one learning rate, one training length and three seeds, and SQ2 and LSQT are one seed on the validation set that selected D128-3.

Stops and open items. The registered stops are kept. No language-model writer was selected, so P1L-D and P1L-S did not run, and the LSQ screen triggered by RFX ran no arm; the separately registered LSQT left the trained least-squares line open. The rank-32 oracle-code pilot stopped on one seed, one learning rate and 5,666 steps, which is not a proof of impossibility. Every study reported here has its

registered decision; studies registered later are not part of this version. The symbolic and language-model correspondence figure was not produced.

Reproducibility. The symbolic figures are produced by `scripts/paper_figures.py` from committed CSV and JSON exports, and the tables of Sections 3.2 to 3.5 by `scripts/paper_tables_g1.py` from the committed per-seed accuracy, timing and endpoint files. No checkpoint is re-evaluated and no bootstrap or timing measurement is rerun for this manuscript. The evidence ledgers (`docs/paper/evidence_symbolic.md` and `docs/paper/evidence_llm.md`) hold the exact stored values and artifact commits and take priority over any shortened wording here. Appendix B records the sets, seeds and comparisons, and Appendix C the claim contract.

8 Related work and boundaries

This section gives the closest comparisons and what each does and does not bear on; Appendix H has the full text. The dated literature audit (`docs/reports/lit_check.md`) is the scope record, and we keep its access limitations rather than turn an incomplete search into a priority claim. Associative memory, key-value separation, multi-hop memory reading and recursive sharing are established ingredients, and we claim none of them as new.

Low-rank adaptation and fast weights. LoRA supplies the parameterization: a frozen base projection plus a trained low-rank branch. Ba et al. already read fixed fast weights repeatedly with an evolving inner state, and Schlag et al. connect linear attention to fast-weight programming and delta-style updates. δ-mem combines associative online state with low-rank corrections to a frozen language model; the distinction is the read schedule and query evolution, not associative memory itself.

Looped and recurrent-depth models. MixerLoop repeats a delta-network mixer, and MemReasoner is a close memory-and-composition neighbor with an end-to-end trained memory-augmented model. Huginn, Ouro, Relaxed Recursive Transformers and Mixture-of-Recursions place and train repeated computation differently. The loss of short-hop accuracy at larger L is a known failure of recurrent depth (Bansal et al., Think-at-Hard, Kohli et al.), so we read it as an instance in a new setting and do not test the remedies those works propose. STARS and ReLIT narrow any claim about reinjection, frozen-backbone recurrence or stabilization.

Latent multi-hop composition in language models. Yang et al. (2024), Yang et al. (2025), Balesni et al., Biran et al., Yu et al. and Liu et al. study bridge entities and layer order in two-hop queries; Zhang et al., Lin et al. and Wang et al. study out-of-distribution composition. In our setting facts reach the reader only through a fixed-size adapter memory and hops are spent in a reader loop, so these studies bound what we may claim without predicting our results. In particular they make it unsafe to read B8's in-distribution advantage over the shared reader as evidence for the unshared design, and the paper reports only a descriptive scope row on unseen-pool names.

Regression memories, writers and retrieval in weights. A delta-rule write is one gradient step on a recall loss (Wang et al.), and Titans, ATLAS and TTT layers train the memory itself at test time. Our writer is the one-step delta update; whether that gap contributes to the deployed-read distortion is untested, and we study none of these alternative writers. Slot and lookup memories, parameter edits and episodic memories for frozen models (Appendix H) give each item its own address or write into weights; no slot-memory baseline is run. Two further reviews added works on writers as online regression or filtering, on retention and update rules, and on looped supervision and distillation (Appendix H). We cite their abstract pages as the reviewers recorded them, did not re-open the pages, claim no priority, and study none of the writers, objectives or architectures named there. Zahn et al. report that semantic keys overlap and recall falls near chance after a few related facts, a setting close to the key-overlap questions of the writer screens. Sanford et al. show that logarithmic depth suffices for k-hop problems, so one hop per read is a design choice of this adapter and not a bound. Liang and Pan warn that decodable intermediate states do not establish a complete serial algorithm, which is the boundary the per-step readouts respect.

Verbalizable directions and the readout boundary. The J-lens reads intermediate residuals through averaged transport to the final layer. It motivates asking whether reader states expose the expected entities while separating readability from causal use; our readouts are a proxy that differs from it, and they support no new priority claim.

9 Conclusion

Repeated low-dimensional queries are useful under particular writing and access conditions. The clearest evidence is the orthogonal-oracle and factorized capability comparisons in the symbolic task and the equal-parameter oracle contrast in the language model at two backbone sizes. The claim compares several reads with a single read at the same rank and reports their costs. Historical larger-rank controls and the original practical-gate failure stay in the record, and neither is rewritten as a system-efficiency success. The shared reader reaches hop 3 at its trained depth and walks to later hops at low accuracy when run deeper, losing hop 1 on the way. A label-free stop recovers part of the long-hop accuracy at the price of hop 2 and of extra reads, and it is not an efficiency result. The learned-writer screens, DPFP, DPFP-X, SQ2 and LSQT included, select no writer, a parser into a fixed table of oracle codes is not a learned writer, and the oracle result is not an end-to-end result. The appropriate conclusion is a bounded mechanism result and a set of reproducible failures, with no efficiency claim over increasing rank.

A Proof sketches and non-implications

The full arguments, counterexamples, and constructed checks are in theory.md, Props.1-11. These sketches specify the scope needed to read the experimental sections; they are not new empirical results.

A.1 Subspace and polynomial. Induction gives a fixed-memory reader state f(Ax); multiplication by B places every branch output in col(B), and the chain rule factors its Jacobian through A and B. For an identity activation, substituting $z_{s+1} = u + Tz_s$ repeatedly yields $\sum_{j=0}^{L} T^j u$. This may increase memory-composition degree while preserving a linear input map at fixed M. It does not imply a larger branch-output subspace or a lower bound against snapshot-specific precomputation (Props.1-2).

A.2 Ideal chain construction. For disjoint directed chains, writing each edge once with orthogonal entity codes, unit gates, and zero initial memory gives $M = \sum_{i \to j} c_j c_i^T$, independently of fact order. With an aligned linear transition of gain a>1 that maps the terminal to zero without competing-code leakage, the state is $\sum_{j=0}^{\min(L,h)} a^j c_{i_j}$. The matched decoder selects the deepest visited node. Nilpotence kills further transition terms after the terminal. Nonorthogonal codes, imperfect gates, misaligned projections, or a different decoder invalidate this particular sufficient argument without invalidating the theorem (Prop.3).

A.3 Error amplification and one-read observations. The identity $X^j - Y^j = \sum_{k=0}^{j-1} X^{j-1-k}(X - Y)Y^k$ gives an operator-norm error bound after summing reader powers; it is an upper bound, not a lower bound on damage. A cached-read reader is a deterministic function of its input, one MQu, and allowed side information. Identical observations imply identical predictions, and hence an ambiguity limit only for distributions actually containing such observations. An arbitrary learned Q need not satisfy an aligned-first-hop restriction (Props.4-5).

A.4 ReLU and elimination. The output of a width-r bias-free ReLU through-map lies in the cone generated by r columns of its output matrix. It cannot equal identity on a set positively spanning the entire r-dimensional space; the residual reader need not implement that through-map. Separately, if every high-dimensional update is B times a reproducible low-dimensional function, substituting h=x+Bz gives an exact reduced recurrence. Extra operations that leave col(B) require a different proof. Shared linear powers satisfy $(BA)^L = B(AB)^{L-1}A$; nonlinearities cannot generally be moved through the factors (Props.6,8-9).

For the pure double-BA branch, local Lipschitz continuity and $\phi(0) = 0$ imply $\|BA\phi(BAx)\| = O(\|B\|^2)$ for bounded A near initialization. Its joint first derivative at B=0 therefore vanishes, even if a ReLU Hessian does not exist. A skip path changes that first-order statement; the same argument does not apply to V2's single output factor (Prop.9).

A.5 Stopping certificate. If F is a contraction with constant c<1, the triangle inequality gives $\|z - z^*\| \le \|z - F(z)\| + c\|z - z^*\|$. Rearrangement proves the residual state-error bound. Multiplying by a decoder norm bounds output perturbation; an additional positive class margin is needed for label stability. No

assumption connects that stable label to the target, so correctness is not a consequence of contraction alone (Prop.7).

A.6 Writer shortcut and exact delta read. Common unit keys, zero initial memory, and constant scalar gates give an exact weighted value recursion with decay factor $\lambda(1-\beta)$. Difference values $v_i = D(e_{tail} - e_{head})$ telescope to endpoint differences only in an equal-weight reference sum; using the weighted recursion requires controlling its error relative to decoder margins and the relevant code geometry. Common keys or generic head information alone do not ensure a usable terminal statistic. Tail-only values remove the direct head-subtraction term. More generally, for unit keys, zero initial memory, constant scalar gates, and E actual writes without an intervening reset, unrolling the decay-first writer yields $M_E q = \sum_i \beta\lambda^{E-i} v_i k_i^T \prod_{j>i}(I - \beta k_j k_j^T) q$, where the matrix product follows write order. The resulting coefficients are signed, later writes affect earlier contributions, and interference power contains off-diagonal value-Gram terms. An empty product is the identity. Neither auxiliary angular alignment nor isolated value decoding guarantees the actual downstream answer (Props.10-11).

B Experimental protocol and evidence map

All empirical numbers in this draft are rounded from the authoritative symbolic and LLM evidence ledgers; figures use the symbolic ledger's committed exports. The ledgers' status and conditions take priority over shortened manuscript wording.

Stage	Evaluation set	Training and comparison conditions	Ledger IDs
P1-S	C777: confirmation seed777001, 2,000 episodes	r32; 15,000 steps; train hops1-3, mixed L={1,2,4}; B4 uses L1. Depth sweeps hold x/u/M fixed within checkpoint.	S1-S7
Rank follow-up F1	C777 reused	r64/128; 20,000 steps; mixed depths; no replication. Rank and training duration both change relative to P1-S and are confounded.	S8
Factorized F2	C777 reused	r32; 15,000 steps; joint_fact and t1_frozen_fact; mixed depths; no replication. No factorized condition passes hop1 preservation.	S9-S12
P2-S	C888: confirmation seed888001, 2,000 episodes	Primary endpoint models use 15,000 steps. Mixed-depth M1/r32 at L4 versus larger one-read B6/B7 or fixed-L4 B8; descriptive longer-trained references excluded from the primary gate.	B1/B2/B7
P2-S-F1	C888 explicitly reused	Fixed-L4 M1 versus B8; fixed versus mixed M1; factorized mixed M1 versus larger one-read systems. No factorized timing benchmark.	B3-B6/ B8/B9
IS	C999: new confirmation seed999001, 2,000 episodes per load	r32; fixed E12 checkpoints across E6/12/18/24/36; E24/E36 training at inherited LR/15,000 steps; update check uses 14 physical records, base_E12, hops1-4.	IS1-IS4
A5 swaps	S4242 screening	Stratum-specific original/new-terminal diagnostics; no CIs; P1-S fitting-path and F2 new-path-length-one strata differ.	S11
Stopping pilots	V4242 validation and S4242 screening, 2,000 episodes per checkpoint/split	Thresholds selected on validation, fixed before screening. Reuse mixed-depth checkpoints; no retraining or confirmation, no stopping CIs.	H1-H6
P4-S-L	C999002: new confirmation seed999002, 2,000 episodes	Six primary frozen-reader families, r32; 3,393-parameter supervised heads, thresholds locked on V4242. Descriptive joint families stay outside primary qualification.	HL1-HL5
JS	C999003: new confirmation seed999003, 2,000 episodes	r32 M1; RS/RS+J train 15,000 steps with one uniform depth4-8 draw per batch; unchanged fixed-L4/mixed-depth references freshly evaluated. Stopped policies use locked V4242 thresholds and cap16.	JS1-JS6
IVD	C999004: new confirmation seed999004, 2,000 episodes per checkpoint	12 frozen r32 checkpoints (oracle_random and joint_fact × M1 and B4); up to 24 extra facts at fixed load with low, natural and high key overlap; IC=G(low)−G(high) at load 24 on hops2-3, 97.5% intervals; the joint_fact overlap is selected, not constructed.	IV1-IV3
CVH	C999005: new confirmation seed999005, 2,000 episodes per reader	Nine frozen P4-S-L M1 readers; A0 reference, A1 hop-1-constrained threshold, A2 continuation-value head (0.25-nat margin); 99.166667% intervals (six comparisons); wall-time latency on prepared batches of 64.	CV1-CV4

Stage	Evaluation set	Training and comparison conditions	Ledger IDs
TSM	C999007: new confirmation seed999007, 2,000 episodes	18 newly trained r32 M1 arms (T0 M1-L4, T1 stay loss at depth 8, T2 sticky gate) at the released learning rate 0.003, no search; B4 at L1 as the same-rank reference; 98.75% intervals.	TS1-TS2
UNS	Part A: S4242 screening (`test`, `test_unseen`); Part B: C999006, seed999006, 2,000 episodes per pool	joint, joint_fact and oracle_random (control); released mixed-depth M1 (Part A) or M1 trained at L4 (Part B) and B4; wording rule on the unseen-pool depth gain; descriptive 95% intervals; the parts are never pooled.	UN1-UN4
NCR	S4242 and C777 saved predictions; no new evaluation	L1 errors of released M1 and of B4 on hops2-3, six writers of the existing r32 P1-S and F2 layouts; first-hop (neighbour) category; descriptive 95% intervals.	PM1, NC2
Same-rank design	Architectural accounting; no accuracy sample or CI	Multi-read versus single-read at the same rank; registered adapter parameters and extra latency reported. LLM B8 overhead is dimension-specific architectural accounting, not a measured accuracy result: +9,216 parameters at r=r_mem=32 and +147,456 at r=r_mem=128, that is +8.9% and +31.0% of the P1-L adapter and +69.2% of the symbolic unfactorized joint model.	FR1
LLM Pilot-1 and diagnosis	Validation seed4242; training seed17	T1; initial 1,500-step capacity screen, separate input probes, supplied-answer-code output control, 6,400-step unassisted writer/input-site fits; no confirmation CIs.	LL1-LL4
LLM writer fixes	Same validation set; training seed17	Seven 4,800-step arms; fresh 32,900-step OA (value/output ceiling) and W5 reruns; stronger auxiliary supervision, no multihop confirmation.	LL5/LL6
LLM read fixes and audit	Same validation set; training seed17	Audited historical W5 and OA (value/output ceiling); seven r32/r128 final arms with unequal timing-sized step budgets; final-checkpoint selection, no learned writer/no E8 retry.	LL7-LL10
RFX	`val`, seed4242, episode indices [200000,204000), one hop-1 query per episode; fitting set of 4,096 `train` episodes	Frozen R/r128, R/r32 and W5/r32 (training seed17): the final-position read replaced by store (deployed delta, exact ridge) × query (raw, unit, true) cells; rescue rule and first-applying verdict; no training, no CIs.	RX1-RX3
WP-L6	`val`, seed4242, 1,000 episodes	Rank-32 M1-L4 with O-rot and O-shared oracle codes, lr 0.003, seed17, 5,666 steps; hop-1 full-vocabulary EM at L=4 against a 30% threshold; pilot, no CIs; P1L-S not run.	WL1-WL2
DPFP	`val`, seed4242, 1,000 episodes per arm	Learned LLM writers with DPFP key and query features (D32-3, D128-1, D128-3; T1, E=16, training seed17) against the WP-L4 references R at the same rank; registered E=8 retrain of D128-3; zero- and shuffled-memory audit; no CIs. P1L-D not run.	DP1-DP3, PD1
DPFP-X	`val`, seed4242, 1,000 episodes per arm	The best DPFP design (ν=3, E=16, training seed17) at rank256 on Qwen3-0.6B-Base (X256) and at rank128 on Qwen3-1.7B-Base (X17) against D128-3; five-checkpoint curves, read diagnostics, zero- and shuffled-memory audit; no E=8 retrain; no CIs.	DP4-DP7
P1-L registered oracle readers	test episode indices[100000,102000), 2,000 episodes, eval seed4242	oracle_random at r32/r128; seeds17/29/43, primary fixed-L4 B8 (not parameter-matched: +8.9% at r32, +31.0% at r128), same-rank gates, k2 and 97.5% intervals; Amendments B6 (equal-parameter rule) and B7 (per-rank memory-use check; no multi-read claim at r32). Rank32 fails the check, rank128 passes; at rank128 G1-L fails, G2-L passes and M1-L4@4 against B4@1 meets the B6 criterion. LP6 is training/validation evidence; the per-step readouts (LP5) are descriptive.	LP1-LP13
P1L-C	test episode indices[102000,104000), 2,000 episodes (256 hop-2 episodes per seed for patching)	The nine rank128 checkpoints (M1-L4, B8, B4 × seeds17/29/43), nothing retrained; read-position analysis (descriptive), bridge patching with a registered causal bar of 10 pp, replication of the P1-L contrasts; 95% intervals.	PC1-PC4
P5L	test episode indices[104000,106000), 2,000 episodes, eval seed4242	Qwen3-1.7B-Base (revision ea980cb0...), oracle_random at rank128, readers B3-L4/B4/M1-L4/B8/B5-L4 × seeds17/29/43, S=17,000, learning rates carried from the 0.6B rank128 lock without a pilot, 97.5% intervals; B7 check then the B6 contrast decide; G1-L/G2-L reported only; no B1/B2, P1L-C or readouts.	LP14-LP20

Stage	Evaluation set	Training and comparison conditions	Ledger IDs
Revision tables (Section 8.6)	P1-L and P5L confirmation ranges (no new evaluation)	Seed means, per-seed effects and a seed-level t-interval from the committed per-seed `accuracy.csv`, `inference.csv` and `endpoints.json` of both ranges; M1-L4 scanned at L=1…8 on hops1-8; state bytes derived as $4{\cdot}r^2$	LP21-LP23
LSR	test episode indices[106000,108000), 2,000 episodes × 8 hops per checkpoint, eval seed4242 (thresholds on `val` [1000,3000))	The six rank128 M1-L4 checkpoints (P1-L at Qwen3-0.6B-Base, P5L at Qwen3-1.7B-Base; seeds17/29/43), nothing retrained; label-free stop rules R1 (relative state change), R2 (top-1 answer unchanged over two depths) and R3 (read norm) against fixed L=4 (= L_c) and L=8; R1 and R3 thresholds chosen on validation under a hop-1 constraint; 99.17% intervals for six comparisons; evaluation only.	LS1-LS4
B8W	test episode indices[108000,110000), 2,000 episodes, eval seed4242	B8 copies of the P1-L M1-L4 checkpoints (Qwen3-0.6B-Base, rank128; seeds17/29/43) trained 2,000 steps with untied matrices against the continued shared M1-L4; equality gate with tolerance 0; non-inferiority within 2 pp on hops2-3; 95% intervals.	BW1-BW3
CTC	`val`, episodes [0,1000), seed4242, one seed	Classify-then-code writer (arms CE and CE+ans) into the frozen P1-L rank128 readers B3-L4, B4 and M1-L4 of seed17; hard and soft lookup, zero- and shuffled-memory controls; candidate rule hop-1 EM ≥40% and gain over B4@1 ≥+10 pp; no test split read.	CT1-CT3
SQ2	`val`, episodes [0,1000), seed4242, one seed	Four continuations (6,000 steps, one fresh stream, lr 0.003) of the final D128-3 checkpoint, crossing the store (delta, or ridge with $\eta = 10^{-4}$ on the DPFP features) with the coefficient loss (off, or $w_c = 1$); primary contrast S-RC against S-DE (EM gain ≥ +10 pp, read error ratio ≤ 0.5, stable decoding); step-0 check; zero- and shuffled-memory audit; no CIs.	SQ1-SQ3
LSQT	`val`, episodes [0,1000), seed4242, one seed	Three arms trained from scratch at rank128 with the unit query: LT-K4 (four Kaczmarz sweeps), LT-RG001 and LT-RG1 (ridge, $\eta = 0.01$ and 1), 17,970, 28,005 and 28,245 steps; gate EM ≥ 0.80; best-ridge-arm bands (at least 48.5% triggers W3, 40% to below 48.5% leaves the line open, below 40% closes it); R-r128 and D128-3 as references, not rerun; audit; no CIs.	LQ1-LQ3

The symbolic studies use training seeds 17/29/43. P1-S/P2-S and their follow-ups use 2,000 paired episode-bootstrap draws with seed2718 and confidence 97.5%, conditional on those seeds; episodes are the pairing units, and the same episode is not independent merely because several queries or model seeds are scored on it. IS uses the same fixed-seed episode-bootstrap convention within a load, but its E36−E12 change uses independent load-specific streams derived from base seed2718. IS4 additionally exports absolute-rate episode-bootstrap intervals; they are not current−stale or affected−untouched contrast intervals. These intervals are endpoint-specific, not simultaneous guarantees for every sentence or model comparison. Absolute accuracy and timing points do not acquire intervals from a neighboring contrast. P4-S-L instead uses confidence 1−0.05/6=99.166667% for its six primary families, with 2,000 paired episode draws conditional on the same three fixed seeds. JS uses 97.5% paired episode intervals; DeltaS jointly resamples all four accuracy terms. Neither study's uncertainty includes retraining, head fitting, or threshold reselection on new training seeds. The LLM writer screens have no such intervals: their reused validation episodes, single seed, diagnostic supervision, and sequential selections limit all cross-arm and cross-budget interpretations. P1-L's 97.5% intervals belong to its confirmation (Section 8.5); they must not be attached to these writer-screen values.

The learned-writer P1-S replication uses typed chains(8,1), E=12, with distractors. Both primary oracle rows use a random-code replication at E=18 with chains(8,8,2); the orthogonal primary pass is not a second orthogonal replication. Within each oracle or frozen-writer condition, reader variants share the designated writer/codebook; joint variants train separate writers. The fixed-versus-mixed M1 selected learning rates all equal 0.003 despite separate selection procedures; the manuscript does not invent a numerical learning-rate difference as an observed confound.

P2-S and P4-S prepared-query timings use FP32 on GH200 with five warmups and 20 repeats. P2-S excludes context/writer preparation and input projection; P4-S includes candidate decoding and stopping bookkeeping but excludes writer/backbone preparation. State bytes are per episode, and trained versus

merely registered parameters must be distinguished for frozen oracle components. P4-S stopping caps at L16 and begins after the first update; read-norm uses L+1 reads, while relative-change and output-stability use L. Exported tail latency summaries are kept descriptive; they are not a pooled end-to-end request percentile. P4-S-L times a prepared 64-query batch per seed, separately from complete-set accuracy/execution counters; selected depth can be less than executed depth on fallback, and post-state features require executed updates+1 reads. JS radius summaries pool reached states over seeds and hops at L4/8/16; no radius CI, global contraction bound, or fixed-point certificate is exported.

Reproduction command and compute-job provenance are recorded in WP-T5, WP-T6, and WP-T7. Figure generation is a rendering job only: it neither opens model checkpoints nor changes the scientific estimands or registered decisions. Pending figure items are marked in Sections 8-9 rather than populated from unmerged or provisional results.

Manuscript figure	Plan §9.3 item	Committed figure pair
Figure 1	1: quality-L by hop	quality_depth.pdf / quality_depth.png
Figure 2	3: hop×L heatmap	hop_depth_heatmap.pdf / hop_depth_heatmap.png
Figure 3	5: fixed-query and memory-free paired contrasts	paired_controls.pdf / paired_controls.png
Figure 4	2: quality-latency, larger single reads marked	budget_latency.pdf / budget_latency.png
Figure 5	7: stopping quality versus mean and tail latency	stopping_latency.pdf / stopping_latency.png
Figure 6, Section 6	4: memory load×L	memory_load_depth.pdf / memory_load_depth.png
Figure 7, Section 7	JS extension: overshoot and longer walks	js_overshoot.pdf / js_overshoot.png
TODO, Section 8	6: single-module / end-to-end correspondence	Not produced; the P1-L confirmation is reported in tables (Section 8.5), and the per-step readouts (LP5) are decoding only.

C Claim contract

Each row states one principal claim of the manuscript with the comparator it rests on, the writer, the evaluation split, the gate or criterion that applies, and the ledger rows that carry its numbers. Exact values and statuses stay in the symbolic and LLM ledgers; where this table is shorter than a ledger row, the ledger governs.

Claim	Comparator and parameter status	Writer	Split	Gate or criterion	Evidence rows
More reads of one checkpoint help under orthogonal oracle writing, and the evolving query beats fixed-query and memory-free controls	The same checkpoint at L1 (identical parameters, x, u and M); B5 fixed query and B3 memory-free are separately trained at the same rank r32, and no parameter-equality claim is made for them	oracle_orth, shared across reader variants	C777: test, seed777001, 2,000 episodes; hops2-3; registered random-code replication at E=18	P1-S G1 (D≥3 pp, CI>0, every seed, walk specificity, hop1 preservation, replication) and G2 (97.5% CI>0 against B5 and B3)	S1, S2; S7 limits exact one-hop timing
Random oracle codes give a positive depth effect but do not preserve hop1	The same checkpoint at L1	oracle_random	C777	P1-S G1: fails the hop1-preservation criterion only	S6
Unfactorized learned writing gives a smaller, seed-dependent benefit; M1-lin and frozen unfactorized writers do not pass	The same checkpoint at L1; B5 and B3 are separately trained systems with their own learned writers, so these are not fixed-writer contrasts; same rank r32	joint (a writer trained with each reader); t1_frozen (T1-pretrained, frozen, shared across readers)	C777; typed-chain replication (8,1) for learned writers	P1-S G1 and G2: joint/M1 passes with B5 contrasts by seed of +9.300/ −17.700/+31.700 pp; joint/M1-lin fails the depth gate; t1_frozen fails replication specificity	S3, S4, S5
Factorized writing gives positive, walk-specific depth effects without hop1 preservation; T1 validation alone is not retrieval; the depth contrast improves while final L4 accuracy does not	The same checkpoint at L1; B5 and B3 at the same rank; joint_fact against joint compares separately trained systems	joint_fact (a writer trained with each reader); t1_frozen_fact (frozen)	C777, reused after P1-S, so not independent confirmation; no replication	P1-S G1 fails on hop1 for all four writer/reader conditions; G2 passes	S9, S10, S12; S11 is a screening diagnostic (S4242, no CIs)
Larger-rank follow-up: larger learned-code depth effects	The same checkpoint at L1; B5 at the same rank (r64/r128); rank and training steps (20,000 versus 15,000) are confounded	joint	C777, reused after P1-S; no replication	The follow-up's unreplicated gate only; B5 contrast by seed +22.575/+25.075/ +24.400 pp at r64	S8
Same-rank policy: several reads against a single read at the same rank, with latency reported as a cost and no efficiency claim over larger-rank single reads	B4, the single read at the same rank. Equal architectural parameter counts only for shared M1-L4 versus B4. B8 is not parameter-matched: +9,216 parameters, +8.9% at r32 and +31.0% at r128 of the P1-L adapter, +69.2% of the symbolic unfactorized joint model	Not applicable	Not applicable (deterministic accounting)	None: a policy and a static count, not an accuracy result	FR1

Claim	Comparator and parameter status	Writer	Split	Gate or criterion	Evidence rows
Historical larger single reads: under joint writing B6/r128 and both B7 reads score above shared M1 (claim B1, that a larger single read could not replace it, is not supported); under random-oracle and factorized writing M1 outperforms them; the P2-S practical criterion fails; the factorized latency frontier is unmeasured	B6 larger-rank and B7 larger-memory single reads at L1 against r32 M1 at L4: different parameter and state budgets, so neither same-rank nor equal-parameter comparators	joint; oracle_random; joint_fact	C888: test, seed888001, 2,000 episodes (P2-S); reused by P2-S-F1	P2-S Gate 3: the capability claim needs E1/E2 positive at every tested size and writer; the practical claim also needs a latency win (E4), which fails. P2-S-F1 F-E3: the joint_fact capability claim needs every contrast with CI>0	B1, B2, B5, B7, B8 (open)
Sharing: step-specific B8 exceeds shared M1-L4 at matched training depth, but not on every hop; fixed-depth training improves shared M1	B8@4 against shared M1-L4@4 at the same rank r32 and training depth; B8 adds 9,216 parameters (+69.2% in the joint model) and is not parameter-matched. M1-L4 against mixed-depth M1 compares separately trained protocols	joint; joint_fact; oracle_random	C888, reused by P2-S-F1; not a fresh confirmation sample	P2-S-F1 decision rule: B8 is carried into P1-L if F-E1>0 (CI) for joint and joint_fact, and fixed-depth training if F-E2>0 (CI); no gate on hop-wise superiority	B3, B4, B6, B9
Memory load: the depth benefit shrinks from E12 to E36 for fixed checkpoints, while M1@4 and B8@4 keep exceeding B4@1 at the trained loads; updated-path diagnostics are partial; key interference is identified for oracle_random only (IVD)	Same checkpoint at L1 (IS1); M1@4 and B8@4 against B4@1 at the same rank r32 and each load (IS2), where B8 adds 9,216 parameters and is not parameter-matched; IS3 compares load-trained with E12-trained systems	oracle_random and joint_fact (joint is descriptive)	C999: test, seed999001, 2,000 episodes per load (fresh)	IS-E1 (a negative DeltaD with CI<0 is consistent with Prop.4), IS-E2 (C(E)≥3 pp, CI>0); IS3 and IS4 descriptive; IVD (IV1, IV2) is a separate study with its own gate	IS1-IS4, IV1, IV2
Post-hoc stopping pilots: label-free rules trade long-hop gain for hop1 loss and give no speedup	Fixed-depth evaluation of the same checkpoints (same parameters)	oracle_random at r128 from the 20,000-step F1 checkpoints (rank and steps confounded); joint_fact; joint at r32	V4242 selection, S4242 screening, 2,000 episodes per checkpoint; no CIs	Screening only; no gate	H1-H6
Learned post-hoc halting does not repair hop1 preservation in six primary frozen-reader families	Learned head against L1 of the same frozen reader; label-free reference; any-step oracle (label-informed, an upper bound only among the tested states)	joint_fact, oracle_random, t1_frozen_fact (primary); joint (descriptive)	C999002: test, seed999002, 2,000 episodes (fresh); 99.166667% adjusted CIs	Combined criterion: H≥−1 pp (point), D≥3 pp with CI>0, W with CI>0; every per-seed H also fails	HL1-HL5
Terminal-Jacobian regularization reduces overshoot, does not meet the removal criterion, and costs longer-walk accuracy	RS+J against RS: the same architecture, rank and parameters, shared initialization and RNG streams per seed, only the penalty differs; M1-L4 and M1-mixed references are descriptive	oracle_random; joint_fact	C999003: test, seed999003, 2,000 episodes (fresh); 97.5% CIs	Removal needs DeltaS>0 with CI>0 and S(RS+J)≥−1 pp; fails for both writers	JS1-JS6

Claim	Comparator and parameter status	Writer	Split	Gate or criterion	Evidence rows
Key interference explains the IS shrinkage of the depth benefit for oracle_random, and is not offered as the identified explanation for joint_fact, where load, layout and candidate distribution changed together	Same frozen checkpoint, M1@4 against M1@1 (identical parameters), low- against high-overlap extra facts at equal load; B4@1 descriptive	oracle_random (constructed key codes); joint_fact (overlap selected among candidate heads, associational)	C999004: test, seed999004, 2,000 episodes per checkpoint (fresh); 97.5% CIs	IVD: IC=G(low)−G(high)≥5 pp at load 24 on hops2-3 with 97.5% lower bound>0, per writer; passes for oracle_random (+13.975 pp), fails for joint_fact (+3.208 pp)	IV1-IV3
A halting head with the correctness label and a hop-1-constrained validation threshold repairs overshoot for oracle_random on frozen readers; no other CVH arm passes; no CVH arm has a latency claim	Arm against L1 of the same frozen reader (identical parameters); fixed L4 and L16 and the P4-S-L head (A0) as references; latency against fixed L16 on prepared batches of 64	oracle_random (pass); joint_fact and t1_frozen_fact (no pass)	C999005: test, seed999005, 2,000 episodes per reader (fresh); 99.166667% CIs	Gain ≥3 pp with CI>0, hop-1 change ≥−1 pp, mean executed updates ≤8; a latency claim only where wall time is below fixed L16 and the arm passes (it does not hold for the passing arm)	CV1-CV4
Stay training (T1) and a content-gated sticky update (T2) do not remove the overshoot under this protocol; fixed-depth training remains the remedy found	T1 and T2 against T0 (M1-L4 retrained under the same code) and the released B4 at L1 (same rank); T2 adds two parameters over T1	oracle_random; joint_fact	C999007: test, seed999007, 2,000 episodes (fresh); 98.75% CIs	An arm passes only for both writers: absolute L16−L4 difference ≤1 pp on hops1-3, hops4-8 at L8 at least −1 pp against T0, and gain ≥3 pp with CI>0 over B4; no arm passes	TS1-TS2
The depth gain on hops2-3 is also present on unseen-pool entities (256 further names of the same vocabulary); no claim beyond the closed vocabulary or for new entity codes	M1 at L4 against L1 of the same checkpoint (Part A released mixed-depth, Part B trained at L4); B4@1 descriptive; parts never pooled	joint; joint_fact; oracle_random (control)	Part A: S4242 screening (`test_unseen`); Part B: C999006 (fresh); descriptive 95% CIs, no multiplicity correction	Descriptive wording rule: unseen-pool gain ≥3 pp with CI>0 (all six results meet it); Part A also reproduces the 18 committed M1 depth-contrast rows	UN1-UN4
In the groups where the hop-2 single read is below the chance line, its errors are mostly the one-hop neighbour; this qualifies the reading of D	L1 of released M1 and of B4 at the same rank r32; no multi-read comparator	t1_frozen_fact, joint_fact, oracle_random, oracle_orth (below chance); joint and t1_frozen read above chance	S4242 and C777 saved predictions, 2,000 episodes; descriptive 95% CIs	None (descriptive); a first-hop share above its uniform value does not identify the mechanism, and no fraction of D is estimated	PM1, NC2
LLM writer screens, including the registered DPFP and DPFP-X screens, select no learned writer; P1L-D is not run	Writer arms and read-supervision variants at unequal fixed budgets; OA and OAρ (the value/output ceiling) and the D2 supplied-answer control are privileged and ineligible; DPFP arms against the WP-L4 references R at the same rank (DPFP changes the feature geometry and the query normalization together, Amendment A1); no parameter matching beyond equal inference parameters at each rank	Learned writers on the T1 task; no oracle	val: validation seed4242, 1,000 episodes; training seed17; no CIs	Selection EM≥0.80 (partial-learning threshold 0.30; E8 trigger 0.50); DPFP: no E=16 arm reaches 0.80 (39.0%, 58.0%, 60.8%) and the E=8 retrain of D128-3 reaches 73.0%, so the decision is stop; DPFP-X: X256 66.2% and X17 59.3%, neither reaches 0.80, stop (X17 against D128-3 −1.5 pp, not distinguished; descriptive rows only)	LL1-LL10, DP1-DP7, PD1
At the frozen R/r128 and R/r32 checkpoints only an exact store together with the true key rescues the deployed read (verdict store+query); W5/r32 is undecided; no LSQ arm is triggered	Cells that replace the final-position read of the same frozen checkpoint; no parameter comparison	Learned writers R (WP-L4) and W5 (WP-L3 rerun); no oracle	val: seed4242, episode indices [200000,204000), 4,000 queries; training seed17; no CIs	Rescue: normalized error ≤0.5 of the baseline and EM gain ≥10 pp; the first rule that applies gives the verdict	RX1-RX3

Claim	Comparator and parameter status	Writer	Split	Gate or criterion	Evidence rows
At rank 32 the LM path did not learn the O-rot and O-shared oracle codes within the pilot budget; P1L-S is not run	None: a pilot against a 30% hop-1 threshold; no B4 or B8 comparison, so no equal-parameter or multi-read claim	oracle_rot and oracle_shared (LLM oracle designs, not the symbolic oracle)	val: seed4242, 1,000 episodes; training seed17; no CIs	Hop-1 full-vocabulary EM ≥30% at L=4, final checkpoint, one arm per design; neither passes (0.1%, 11.8%)	WL1-WL2
At rank128, with a clean oracle memory, the shared M1-L4 reader exceeds a single read of equal parameter count on hops2-3, and the gated B8 reader fails G1-L (G2-L passes); at rank32 the readers did not learn to use the oracle memory, and no multi-read claim is made	Per rank, B8@4 and M1-L4@4 against B4@1, B5-L4@4 and B3-L4@4 at the same rank. The equal-parameter claim only from M1-L4@4−B4@1 at rank128 (475,138 parameters each; Amendment B6); B8 is not parameter-matched (+8.9% at r32, +31.0% at r128); latency (M1-L4 about 1.42 and B8 about 1.57 times B4 at rank128) is a cost	oracle_random (the LLM oracle, with independent random pair-key and value tables, is not the symbolic oracle); no learned writer	test, episode indices [100000,102000), 2,000 episodes; seeds17/29/43; 97.5% CIs (k=2)	G1-L (B8@4−B4@1 ≥3 pp, CI>0, every seed, hop1 ≥−1 pp, typed CI>0): fails at r128 (+0.983 pp), outcome at r32 not interpreted; G2-L passes at r128; Amendment B6 criterion met by M1-L4@4−B4@1 (+21.550 pp); Amendment B7 memory-use check: fails at r32, passes at r128	LP1-LP13; B9
The equal-parameter result holds at a second scale of the same family: at Qwen3-1.7B-Base, rank128, M1-L4@4 exceeds B4@1 on hops2-3 (the registered decision); B8 fails G1-L again (reported only); no scaling trend and no other size	B4@1 at the same rank; equal adapter parameters for M1-L4 and B4 (606,210); B8 is not parameter-matched (+24.3%); latency (M1-L4 and B8 about 1.29 times B4) is a cost and is not compared across scales	oracle_random (the LLM oracle, not the symbolic oracle); no learned writer	test, episode indices [104000,106000), 2,000 episodes; seeds17/29/43; 97.5% CIs	Memory-use check passes; B6 contrast ≥3 pp, CI>0, positive in every seed (+25.783 pp: met); G1-L (+1.033 pp, hop 1 −1.717 pp) and G2-L reported only; cross-scale values descriptive, no test	LP14-LP20
At the trained depth the shared reader reaches hop 3; run deeper, the same checkpoint reaches hops4-8 at 9.9-16.6% (0.6B) and 9.8-20.9% (1.7B) with the best depth chosen by the label, while it loses hop 1	None: the M1-L4 checkpoint at its other depths; B4@1 and the memory-free B3-L4@4 on hops4-8 are shown beside it	oracle_random	P1-L [100000,102000) (0.6B) and P5L [104000,106000) (1.7B); seed means of three training seeds, no interval	None (descriptive; the oracle stop uses labels and is an upper bound; hops and depths beyond training)	LP21
Headline contrasts carry the three seed effects and a seed-level interval beside the registered episode interval; both are conditional on three fitted seeds and one oracle code table	As the contrasts they accompany (M1-L4@4−B4@1, G1-L, G2-L at rank128)	oracle_random	P1-L and P5L ranges	None (descriptive; the registered decisions use the episode intervals)	LP22
P1L-C: no causal bridge claim at rank128 for M1-L4 or B8; the read-position and replication results are descriptive	The same rank128 checkpoints (nothing retrained); no new parameter comparison	oracle_random	test, episode indices [102000,104000); 256 hop-2 episodes per seed for patching; 95% CIs	Causal bridge claim: pooled shift ≥10 pp, CI>0 and positive in every seed (M1-L4 +0.391 pp, B8 0.000 pp: not met); read position and replication gate nothing	PC1-PC4
A label-free per-question stop (R2) raises hops4-8 over fixed L=4 at both backbones, with hop 2 falling by 10.83 and 12.27 pp, extra reads and no efficiency claim; R1 and R3 had no feasible threshold	Fixed L=4 (= L_c) and L=8 of the same checkpoints; R2 has no parameter, so no parameter comparison arises	oracle_random; no learned writer	test, episode indices [106000,108000), 2,000 episodes per checkpoint; seeds17/29/43; 99.17% intervals	Hops4-8 gain ≥3 pp with interval >0, hop 1 ≥−10 pp, every seed (met: +3.90 and +5.68 pp); R1 and R3: no feasible validation threshold (0 of 25)	LS1-LS4

Claim	Comparator and parameter status	Writer	Split	Gate or criterion	Evidence rows
A copy of B8 started from the M1-L4 solution keeps it after 2,000 untied steps (untying keeps the solution); B8 has no equal-parameter status and G1-L stays failed	untied@4 against the continued shared@4 from the same start; adapter parameters 622,594 (B8) and 475,138 (M1), not matched	oracle_random; no learned writer	test, episode indices [108000,110000), 2,000 episodes; seeds17/29/43; 95% intervals	Every seed ≥ -2 pp and pooled lower bound > -2 pp (met: +0.90 pp [0.09, 1.71]); equality gate with tolerance 0	BW1-BW3
A learned parser plus a fixed code table drives the frozen composing reader at p=1 on validation episodes; this is not a learned writer and is not comparable with DPFP or DPFP-X	B4@1 and the oracle-memory reference on the same readers; the writer has 268,292 parameters, no parameter match	Classify-then-code parser of entities read at their own tokens (relation constant), with the oracle's privileged codes	`val`, episodes [0,1000), seed4242, one seed; 95% interval of the paired gain	Candidate rule hop-1 EM $\geq$40% and gain $\geq$+10 pp (met: 50.7%, +24.05 pp); falsifiers did not fire; no confirmation run	CT1-CT3
Ridge storage with the coefficient loss on the DPFP features does not advance the learned-writer line; no SQ2 cell reaches 0.80 (descriptive: the ridge store alone halves the read error and is +6.9 pp over the continued delta control, one seed)	S-DE, the continued delta control, and the parent D128-3 (60.8%)	Learned DPFP writer D128-3, continued 6,000 steps	`val`, episodes [0,1000), seed4242, one seed; no CIs	EM gain $\geq$+10 pp and read error ratio $\leq$0.5 (not met: +4.1 pp, 0.554); stable decoding met; no cell $\geq$0.80	SQ1-SQ3
Trained least-squares stores at rank128 reach 38.3%, 46.1% and 42.6%, below the 0.80 gate; the trained-LSQ line stays open (neither falsified nor W3 triggered), and no store-versus-query split is made	R-r128 (33.5%) and D128-3 (60.8%), not rerun; the unit query is in every arm and in none of R-r128	Learned writers with least-squares stores (Kaczmarz sweeps, ridge), trained from scratch	`val`, episodes [0,1000), seed4242, one seed; no CIs	EM $\geq$0.80 (not met); best ridge arm 46.1%, inside the band 40% to below 48.5%	LQ1-LQ3
Conditional theory (Props.1-11)	Not applicable	Not applicable	Not applicable	None: conditional results that experiments are compatible with, or contradict only in their unconditional readings	Theory-to-measurement ledger in the symbolic ledger

Reading rules.

- **Comparators.** A comparator is at the same rank unless its cell says otherwise. The larger-rank and larger-memory single reads of P2-S and P2-S-F1 are historical controls with different budgets, not same-rank comparators.
- **Parameters.** "Equal parameters" is claimed only for shared M1-L4 against B4, whose architectures have identical parameter counts under matched dimensions. B8 is never parameter-matched: its step-specific R/U/Q add 9,216 parameters at r=r_mem=32 and 147,456 at r=r_mem=128, which is +8.9% (rank32) and +31.0% (rank128) of the P1-L adapter and +69.2% of the symbolic unfactorized joint model; each percentage belongs to its own model. No equal-parameter

claim is made for B5, B3, B6 or B7, which are compared at the stated rank or budget.

- **Writers.** Oracle and learned writers are separate claims. Under joint writing every reader variant trains its own writer, so a contrast between variants compares trained systems. The language-model oracle needs a learned value-to-key lookup for every read after the first (Section 8.4), so no symbolic oracle result is carried over to it.
- **Splits and reuse.** C777 is reused by the rank and factorized follow-ups and C888 by P2-S-F1, so their intervals there are not independent confirmation; C999, C999002 and C999003 and the follow-up sets C999004-C999007 are fresh; the P1-L confirmation range [100000,102000), the P1L-C range [102000,104000) and the P5L range [104000,106000) are separate. NCR re-reads saved S4242 and C777 predictions and UNS Part A audits S4242; neither adds an evaluation. Every interval conditions on the fixed training seeds 17/29/43 and is specific to its endpoint; Section 8.6 prints a seed-level interval beside the registered one for the headline contrasts (LP22). Per-seed effects appear next to the pooled ones where the ledgers record them (S3, S8, B1, IS1, HL1, JS1-JS4).
- **Scope.** Every set draws on the seen pool of a closed 1,024-name vocabulary, except the descriptive UNS scope row on the 256-name unseen pool (UN1-UN3), so no claim covers names outside the vocabulary, new entity codes, or composition "for unseen entities". Effects hold at the trained hops (1-3) and trained depths; longer hops are reported as measured cells (S7, B6, H1-H6, JS3), not as extrapolation claims. No claim is made about multiple reads in the language model at rank32. The language-model results cover Qwen3-0.6B-Base and, for P5L only, Qwen3-1.7B-Base; no claim is made for another size or as a trend across the two (LP20).
- **Names.** Ledger IDs name claim rows. The reader names B3-B8 (for example B3 memory-free and B8 step-specific) and the amendments B1-B7 of the P1-L preregistration (written "Amendment B6" and "Amendment B7" here) are separate namespaces: an HTML comment such as in this manuscript always cites claim row B7 of the symbolic ledger, whereas "B7" in running text names the larger-memory reader.
- **Descriptive rows.** Rows marked descriptive in the ledgers (IV3, CV4, TS2, UN2-UN4, PM1, NC2, RX2, RX3, WL2, LP5, LP8, LP12, LP13, LP18-LP23, PC2-PC4, DP2, DP3 and DP5-DP7) have no gate in their preregistrations. They support only the wording that the preregistration attaches (for UNS, "the depth gain is also present on unseen-pool entities"; for NCR, the qualification of D).
- **Pending.** Nothing registered is pending. The registered results of IVD, CVH, TSM, UNS, NCR, RFX, WP-L6, the P1-L confirmation with the memory-use check, P1L-C with the per-step readouts, its second-scale repeat (P5L), and DPFP with DPFP-X (P1L-D not run) are reported.

D Method and theory in full

Record sections 3 and 4 of the manuscript; section numbers below follow the record numbering.

3. Method: one write schedule, repeated low-dimensional reads

Let a frozen linear projection map an input x to Wx+b. Its adapter first forms u=Ax, where A has output dimension r. A memory matrix M has shape r_mem×r_mem. With column-vector notation, the shared reader is

$$z^{(0)} = u, \qquad z^{(s+1)} = u + \rho\,\phi(Rz^{(s)} + UMQz^{(s)}), \qquad y = Wx + b + \alpha Bz^{(L)}.$$

The same M is used throughout the inner loop. Q maps the current state to a memory query, U maps its read back into the reader coordinates, and B supplies the output bottleneck. The input is reinjected at every update. Increasing L therefore changes the reader computation while leaving the base-projection schedule unchanged. Setting L to zero gives the ordinary low-rank branch; this identity specifies a limiting case, not an empirical claim that different trained readers coincide.

The memory writer uses a normalized key k and value v derived from the designated writer inputs. The unfactorized learned writer takes $k = Ku/\max(\|Ku\|_2, \epsilon)$ and $v = Vu$, with a positive norm clamp and learned scalar sigmoid gates; factorization instead separates its head/relation and tail projections. Oracle conditions specify their codes and gate values directly. For an eligible write, the decay-first update is

$$\bar{M} = \lambda M, \qquad M^{+} = \bar{M} + \beta(v - \bar{M}k)k^{T}.$$

Reading precedes the current token's write. Padding neither reads nor writes, reset clears the state before use, and a skipped write does not trigger decay. In the memory-QA protocol, support facts populate the state and questions read without writing. Separate questions and evaluation depths fork from the same completed context snapshot. These semantics define which information remains fixed when a reader is evaluated at a different depth; the binding specification is notation.md.

We use M1 for ReLU and M1-lin for identity activation. B5 holds its initial memory query fixed and caches the resulting read while allowing internal iterations; B3 removes the memory path. B6 spends budget on larger rank in a single step, and B7 enlarges the memory dimension while retaining the smaller reader dimension. B8 replaces shared R/U/Q with step-specific matrices and is trained for a fixed unrolled depth. It is a separate architecture comparison, not a reader that can be extended by reusing a uniquely defined next shared step. It is also not parameter-matched to the shared readers: the extra step matrices add $3(r^2+2r\cdot r_mem)$ parameters, that is 9,216 at r=r_mem=32 and 147,456 at r=r_mem=128, or +8.9% (rank32) and +31.0% (rank128) of the language-model adapter and +69.2% of the symbolic unfactorized joint model.

Writer conditions determine which cross-reader comparisons hold memory representations fixed. Oracle writing uses specified entity codes. A T1-pretrained frozen writer is shared across the reader variants for its seed. Under joint writing, however, each reader variant trains its own writer, so a contrast with B5 or B3 compares learned systems rather than isolating query evolution under an identical learned writer. The factorized learned conditions restrict keys and the initial query/input projection to head/relation information and values to the tail input. Later reader queries can incorporate previously read values. This removes the direct head-subtraction statistic analyzed in Proposition 10. This is a structural restriction on an identified shortcut, not a universal guarantee about learned algorithms.

The symbolic primary layout contains 12 edges in chains of lengths 8 and 4, with 64-dimensional entity representations and rho=0.25; the primary reader and memory dimensions are both 32. Readers train on hops1-3, with mixed depths {1,2,4} unless a fixed-depth comparison is explicitly named. These task conditions are part of every claim below. We separately account for trained parameters, persistent state, prepared-query latency, and training protocol, because matching one does not match the others.

4. Theory: useful computation without unconditional guarantees

Fixed subspace, variable computation (Proposition 1). At fixed memory and depth, the branch has the form alpha B f(Ax), so its output lies in col(B), and its local input-Jacobian rank is bounded by r wherever

differentiable. Equal Ax implies equal branch outputs only when the other inputs, including M and L, are fixed. Nothing in this statement prevents f from changing with reader depth. Nor is it a global rank bound on a stacked sequence map through history-dependent memories. Thus a depth comparison can measure different computation inside the same bottleneck without claiming an expanded output subspace.

Composition order (Proposition 2). For an identity activation, shared matrices, constant reinjection, and a fixed snapshot, define T=rho(R+UMQ). The exact state is $z^{(L)} = \sum_{j=0}^{L} T^j u$. Additional steps permit higher-order products involving memory. When the recurrent and memory terms do not commute, these are ordered matrix products, not a scalar binomial expansion. At fixed M and L the map is still linear and may be precomputed at a preparation/storage cost; the identity therefore supplies neither an irreducible serial-computation lower bound nor an accuracy guarantee.

An existence construction for chain following (Proposition 3). Suppose all episode entity codes in disjoint directed chains are orthonormal across the entire episode (so their number does not exceed r), memory starts at zero, and each edge is written once with exact unit gates. Suppose the linear reader implements an aligned successor map with effective gain greater than one, maps terminals to zero, and has no leakage onto competing codes. With a matching code decoder, successive visited nodes receive increasing weights and the terminal remains the prediction after enough steps. This aligned chain transition is nilpotent on the entity-code span, so finite stabilization is possible even without Euclidean contraction; acyclic data alone do not make an arbitrary learned transition nilpotent. Exact storage alone does not ensure the required gain, query/value alignment, or decoder; the theorem does not state that optimization must find this construction.

Interference and information access (Propositions 4 and 5). Nonorthogonal delta writes can alter previously stored associations. A telescoping bound on differences of operator powers permits memory error to be amplified with depth, while allowing special cases with no growth. Separately, a one-read or fixed-query reader observes the context only through MQu, given its fixed input and side information. Episodes with identical observations must receive identical predictions. A chain-specific Bayes ceiling needs the additional aligned first-hop and side-information assumptions; it is not an impossibility theorem for every learned query or larger one-read model.

Fixed-point error versus answers (Proposition 7). For $F(z) = u + \rho\phi((R + UMQ)z)$, a valid global Lipschitz constant c<1 gives a unique fixed point and $\|z - z^*\| \le \|F(z) - z\|/(1 - c)$. One sufficient upper bound is $\bar{c} = |\rho|L_\phi(\|R\| + \|U\|\|M\|\|Q\|)$. A bound above one is inconclusive about actual contraction. Even a valid state-error bound needs decoder-margin and task assumptions before it can say anything about a label, and label stability alone is not correctness. Residual thresholds without a certificate are empirical stopping policies.

The JS penalty is motivated by a different sufficient condition: for a smooth map at an actual fixed point, transition spectral radius below one implies local asymptotic stability. A finite last-read state need not be fixed, pointwise radii do not control all transient products, and ReLU boundaries need additional care. Thus the preregistered stability study tests a regularizer rather than certifying a fixed point. Its reached-state radii and reduced overshoot do not verify Proposition 3's exact terminal-annihilation construction or establish a contraction or correctness certificate. The failed sufficient bounds from P4 establish no state-divergence claim.

Writer information and the final decoder (Propositions 10 and 11). Under specified common unit keys, constant gates, and zero initial memory, the writer admits an exact exponentially weighted value expression. Difference values of the form $D(e_{tail} - e_{head})$ can expose endpoint statistics when the weighting error is controlled relative to the decoder's endpoint margins; common keys alone are insufficient. Tail-only factorization removes the direct head-subtraction term but leaves other global statistics possible. More generally, an exact delta read is a signed combination of written values whose coefficients depend on later-write key erasure. Its interference depends on the value Gram matrix as well as key alignment. Normalized addressing objectives do not set raw query magnitude, and an auxiliary value decoder need not agree with the actual reader or downstream model. These are conditional explanations and diagnostic requirements, not fitted causes of the symbolic or language-model outcomes.

The remaining propositions set additional boundaries. The bias-free ReLU transmission obstruction concerns an exact through-map on a positively spanning code set, not universal classification failure of the residual reader (Prop.6). Exact elimination of a high-dimensional recurrence requires its entire update to remain in a fixed output subspace (Prop.8). The pure zero-output-factor double-BA dead start concerns that branch's first derivative, not every V2 initialization (Prop.9). We use none of these identities as evidence of a measured elimination speedup or a universal ranking between linear and nonlinear readers.

Full assumptions, proofs, and counterexamples are in theory.md; Appendix A gives proof sketches.

E Symbolic studies in full

Record sections 5 to 7: the oracle and learned-writer mechanism experiments, the same-rank and historical budget controls, and the stopping studies.

5. Mechanism experiments: depth is useful under specific writing conditions

We report candidate-restricted accuracy and distinguish percentage points (pp) from absolute percentages. Confirmation intervals are 97.5% paired episode-bootstrap intervals conditional on the three fixed training seeds; absolute point profiles have no standalone confidence intervals. The depth effect D is L4 minus L1 on hops2-3; walk specificity W subtracts the corresponding effect on far hops6-8. This subtraction prevents a generic all-hop shift from being described as selective improvement on the trained reachable hops. How far D measures composition rather than escape from a neighbour error is qualified in Section 6.

The primary protocol treats reach, specificity, preservation of the single-hop task, and replication as separate requirements; an improvement on only the reachable-hop average is insufficient for its full gate. The fixed-query comparison addresses a different alternative explanation: internal iterations may help even when the context observation is cached. The memory-free comparison asks whether those iterations need any stored context at all. Neither control is replaced by a larger-rank comparison, which changes the representation budget. These distinctions explain why we retain both the registered composite outcome and its component contrasts instead of selecting whichever metric is most favorable for a particular writer.

The primary stage and its follow-ups also differ in evidential strength. The original stage includes the registered changed-layout replication, whereas the rank and factorized follow-ups report their own unreplicated results on the reused confirmation set. For learned writers, a within-checkpoint depth contrast fixes that checkpoint's writer and memory; a comparison of reader variants generally does not fix the jointly trained writer. We consequently interpret depth effects, control effects, and cross-writer differences separately. A figure showing one checkpoint helps expose the depth trajectory, but it cannot replace the pooled endpoint or its training-seed breakdown.

5.1 Oracle writing and controlled reader contrasts

With orthogonal oracle writing, r32 M1 has D=+92.883 pp [92.233, 93.550] and W=+92.872 pp [92.208, 93.564] on confirmation seed777001, while also improving hop1. The separately trained evolving-query reader exceeds B5 by +92.892 pp [92.150, 93.542] and B3 by +85.233 pp [84.233, 86.183] on hops2-3 at L4, using the same primary oracle codebook. These contrasts support repeated, query-dependent memory use for the tested task, rather than an explanation solely in terms of more memory-free iterations.

Exact one-hop-per-read timing is nevertheless not an empirical result. For the trained orthogonal M1-lin reader, hop4 accuracy at L4 is 2.483%, and hop8 at L8 is zero, despite high accuracies for some intermediate longer hops at L8. Its hop1 accuracy falls from 99.567% at L1 to 96.550% at L4 and 26.300% at L8. These cells reject the automatic identification of a trained reader with Proposition 3's aligned construction; they do not refute its sufficient conditions. The apparent benefit of depth must be described by hop and evaluation depth, including the short-hop losses.

Random oracle codes give a different balance. At r32, M1 improves reachable hops by +33.258 pp [32.016, 34.484] but loses 18.933 pp on hop1; M1-lin likewise has a positive depth effect and a substantial hop1 loss. Both conditions fail only the hop-preservation criterion. Random oracle codes therefore should not be treated as interference-free storage, and their positive reachable-hop effect should not be relabeled a full gate pass.

5.2 Learned writers, factorization, and seed dependence

The unfactorized joint r32 M1 condition passes the registered gate with a much smaller effect: D=+8.292 pp [7.383, 9.292] and W=+2.492 pp [1.244, 3.733]. Its mean advantage over B5 is +7.767 pp [6.516, 9.050],

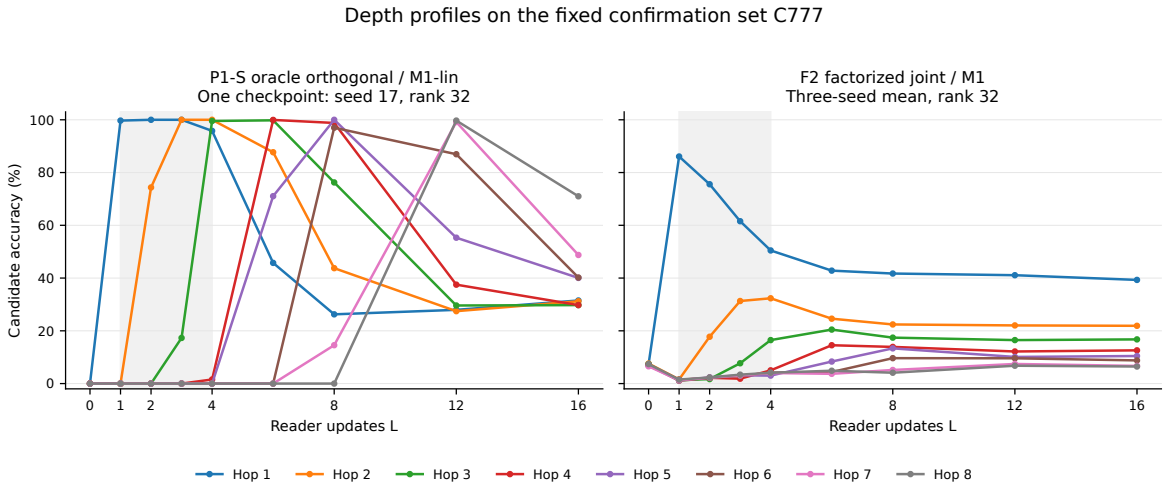


Figure 2: Quality versus evaluation depth by hop: P1-S oracle_orth/M1-lin uses the individual seed17 checkpoint; F2 joint_fact/M1 uses the three-seed mean of fixed-checkpoint sweeps because individual-checkpoint F2 cells are not committed. Candidate-restricted accuracy, confirmation seed777001; source selectors are recorded with the figure. Curves have no confidence bands and do not imply exact one-hop timing or hop preservation.

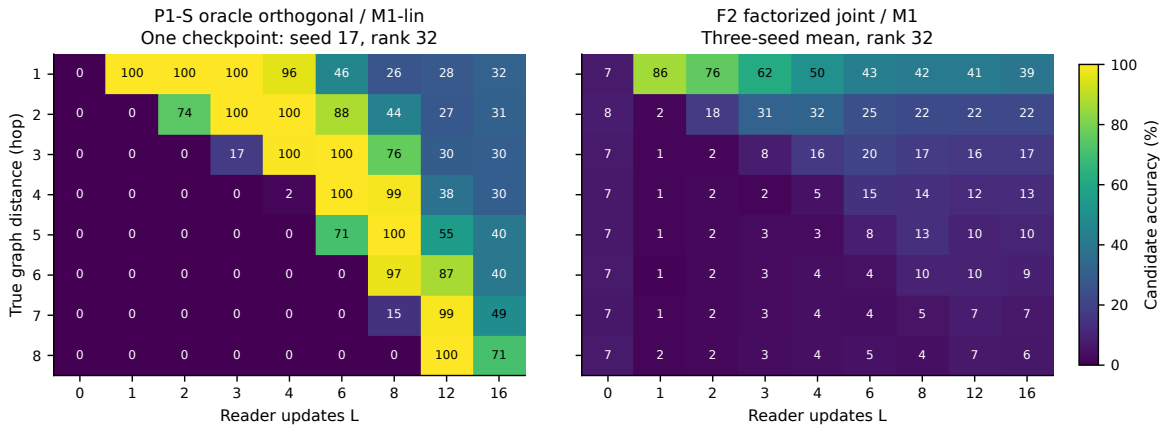


Figure 3: Hop×L views of the same committed conditions: the left panel is seed17, and the right panel is the three-seed mean, as in Figure 1. Training hop lengths and evaluation depths must be distinguished when reading extrapolation cells; successful individual cells do not establish uniform longer-hop success.

but the three seed effects are +9.300, −17.700, and +31.700 pp. The pooled interval is consequently compatible with a failed training seed, and it does not establish improvement for a population of new seeds. The joint linear reader fails the primary depth gate, with specificity −0.408 pp [−1.475, 0.753].

Unfactorized frozen T1 writers fail the replication-specificity criterion for both readers; the M1 replication interval includes zero. This result does not show that joint training is necessary for every composable writer, because factorized frozen writers produce positive depth and specificity effects under the tested F2 protocol. Even the matched factorized T1 sources' validation accuracies above 91% do not certify the full T2 gate after freezing: the resulting readers retain positive depth effects but fail hop1 preservation.

Factorization makes the distinction between composition evidence and overall utility particularly clear. On the reused confirmation set C777, joint_fact M1 yields D=+22.975 pp [22.017, 23.975] and W=+20.158 pp [19.058, 21.192], but H=−35.617 pp [−37.217, −33.933]. All four factorized writer/reader conditions fail the complete depth gate because of hop1 loss, even though their reach, specificity, and control contrasts pass the reported criteria. Moreover, joint_fact M1's final L4 reachable-hop contrast against the unfactorized joint system is −19.108 pp [−20.534, −17.600], despite a larger depth contrast. Improving a contrast from a weaker L1 baseline is therefore not equivalent to improving the deployed L4 system.

The larger-rank follow-up gives stronger joint-code effects in its tested regimes: r64 M1 has D=+30.533 pp [29.225, 31.900], and its B5 contrast is +24.017 pp [22.575, 25.575] pooled, with seed effects +22.575, +25.075, and +24.400 pp. However, rank and training steps are confounded: it uses 20,000 training steps instead of the primary study's 15,000, and has no replication, so the comparison cannot isolate rank or establish new-seed robustness. Its evaluation also reuses the C777 confirmation set after P1-S, so its intervals are not an independent confirmation.

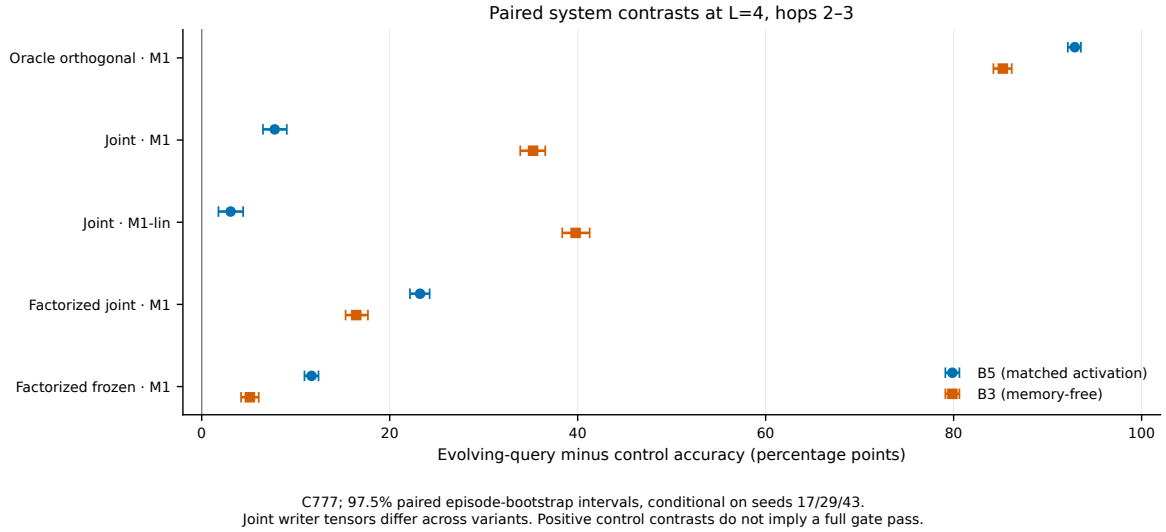


Figure 4: Paired hops2-3 contrasts at L4 against fixed-query B5 and memory-free B3; displayed intervals are the released 97.5% episode-bootstrap intervals. Oracle and learned writer conditions must retain their different sharing controls, and positive pooled effects do not require every training seed to improve.

5.3 Counterfactual specificity and decision history

Successor swaps provide screening-only diagnostics without released confidence intervals. For joint M1 at L4, among the specified P1-S rows whose new path fits, the new-terminal follow rate is 44.220% versus 35.293% for the unswapped query; joint M1-lin is nearly unchanged at 48.336% versus 47.913%. The factorized M1 new-path-length-one stratum has 47.985% versus 4.426%, but that different stratum prevents interpreting the cross-study rate difference as a controlled factorization effect. These responses are consistent with path-sensitive behavior without uniquely identifying pointer chasing or excluding global terminal statistics.

The continuation decision also needs disclosure. The evidence ledger's decision-history correction, item 4 records an explicit deviation from F2's original no-go rule: the orchestrator continued toward P1-L using the earlier joint/M1 pass, while using F2 to inform writer form. This decision did not turn the factorized failures into passes. It belongs in the interpretation of the study, because a preregistered gate and a later

justified continuation decision are different records.

6. Same-rank comparisons, reported costs, and historical budget controls

The primary baseline is the single read at the same rank, so the shared-reader comparison changes computation depth without increasing parameter count. With an otherwise identical architecture, shared M1-L4 and B4 have exactly the same parameter count, and "equal parameters" is reserved for that pair. B8 is not parameter-matched to either: its step-specific R/U/Q add $3(r^2+2r\cdot r_mem)$ parameters, 9,216 at r=r_mem=32 and 147,456 at r=r_mem=128. The registered P1-L adapter counts are 103,426 becoming 112,642 at rank32 (+8.9%) and 475,138 becoming 622,594 at rank128 (+31.0%); in the symbolic unfactorized joint model the counts are 13,314 becoming 22,530 (+69.2%). Each percentage belongs to its own model; none is the proportion of trainable oracle parameters or a whole-backbone figure. Extra reader latency is reported as a cost of the additional computation.

Historical rank, state, and sharing controls

P2-S compares r32 shared M1 at L4 with larger single reads and an unshared reader on confirmation set C888; the reference M1 uses mixed-depth training, while B6/B7 train at L1 and B8 (+9,216 parameters, not parameter-matched) at fixed L4. With unfactorized joint writing, M1 beats B6/r64 on reachable hops by +14.875 pp [13.658, 16.184] pooled, but the three seed effects are +33.125, −20.675, and +32.175 pp, so seed29 is negative. The M1−B6/r128 contrast is −3.233 pp [−4.675, −1.833], and M1 also scores below both larger-state B7 configurations. Thus the claim that these larger single reads cannot replace the tested joint system is not supported. With random oracle codes, by contrast, M1 exceeds B6/r64 and r128 by +34.750 and +34.775 pp, respectively, with positive intervals and positive effects in every tested seed.

The factorized follow-up, which reuses the P2-S confirmation set C888, also supports a bounded capability claim. Mixed-depth joint_fact M1/r32 at L4 exceeds B6/r64, B6/r128, B7/r_mem64, and B7/r_mem128 by +24.017, +24.242, +23.875, and +23.550 pp, respectively; every reported interval and tested-seed effect is positive. These are comparisons against the specified trained architectures, not an information-theoretic ceiling for every single-read function.

On the primary E12 layout the uniform-candidate chance level is $1/14\approx7.1\%$. The single reads in these comparisons score far below it on reachable hops: 0.025% and 0.000% for B6/r64 and r128 under random oracle codes, and 0.242-0.933% for the larger single reads under factorized joint writing. The frozen-writer factorized M1 at L1 shows the same pattern, at 1.983% on hop2 and 1.033% on hop3. Accuracy far below chance indicates systematic errors rather than uninformed guessing. One candidate explanation is that a single read tends to return the one-hop neighbour of the queried entity, in which case part of a depth effect would be escape from that error rather than composition. **Neighbour-confusion rate.** NCR categorises the wrong hop2 answers of the single read (L1) of released M1 and of B4, from the saved screening (S4242) and confirmation (C777) predictions of the existing r32 P1-S and F2 runs, with descriptive 95% intervals. In the 16 groups whose hop2 accuracy is below the 7.143% chance line (t1_frozen_fact, joint_fact, oracle_random and oracle_orth, each with M1 and B4 on both sets), the share of errors that name the one-hop neighbour is 58.667-100.000%, against 7.692% for a uniform wrong guess, and every interval lower bound exceeds the uniform value. The share is 89.739-100.000% for B4 and 58.667-87.159% for M1, while the unfactorized joint and t1_frozen writers read above chance at hop2 with first-hop shares of 5.3-23.0%. A share above its uniform value says that the single read returns the neighbour more often than a uniform error would; it does not identify the mechanism. The tables qualify the reading of D only: in the groups below chance, part of the L4−L1 gain on hops2-3 may measure escape from the neighbour error rather than composition, and because the errors at L4 are not categorised, no fraction of D is estimated.

The two kinds of positive result should not be combined into an unrestricted resource claim. A fixed-rank depth sweep spends more reader operations at the same persistent state size. A larger-rank single read changes both its representational width and state storage, while a larger-memory comparison changes the state while retaining the same reader width. Each comparison rules out a different explanation for the observed accuracy; none licenses replacing measured latency by an analytic operation count. The current factorized result establishes only the tested capability contrast, with its accuracy-latency frontier still unmeasured.

Matching training depth narrows the sharing question. On the reused set C888, B8 minus shared M1-L4 has reachable-hop effects of +13.217 pp [11.983, 14.450] for joint writing, +5.458 [4.242, 6.750] for

joint_fact, and +3.200 [1.958, 4.517] for random oracle writing, each positive in every tested seed. This comparison also changes parameter count and, for learned joint writing, writer parameters; it is not a parameter-neutral causal estimate of sharing. B8 is not parameter-matched to M1-L4: the joint model has 22,530 versus 13,314 parameters (+9,216, +69.2%), and oracle_random has 16,386 versus 7,170 registered and 13,312 versus 4,096 trained parameters (+128.5% and +225.0%); joint_fact totals are not recorded. Fixed-L4 shared training improves over the mixed-depth reference, reducing the joint B8 gap from 33.733 to 13.217 pp, but the selected protocol comparison does not identify a unique causal fraction due to depth. B8's pooled long-hop accuracy at evaluation L4 is lower for all three writers, so its trained-hop advantage is not evidence of superior depth extrapolation.

Under P2-S's original registered practical criterion, the result is negative. Joint M1/r32 at L4 attains 48.017% hops1-3 accuracy at 0.848 ms per prepared batch64, whereas joint B6/r128 at L1 attains 57.244% at 0.352 ms; these are descriptive CUDA-event means without timing intervals. Every eligible joint M1 depth point is matched by an equal-or-faster single-step point for both tested batch sizes, and Gate 3 fails. Those models spend different parameter and state budgets, so the negative practical result does not erase the fixed-budget composition contrast; it rejects a broader efficiency claim. P2-S-F1 supplies no factorized accuracy-latency benchmark. This historical measurement gap is not converted into an efficiency claim, and the same-rank design does not relabel the earlier gate.

The parameter and storage differences make the intended tradeoff concrete: the joint M1 point has 13,314 trained parameters and 4,096 persistent FP32 bytes per episode, versus 114,690 parameters and 65,536 bytes for B6/r128. A reader may therefore be attractive under a strict state allocation even when it does not win the measured latency comparison. That possible application is a conditional design choice, not a practical advantage established for the full request path. The reported timings exclude the context writer and other preparation, and the follow-up's aggregate resource ledger cannot supply missing comparative training costs for individual models.

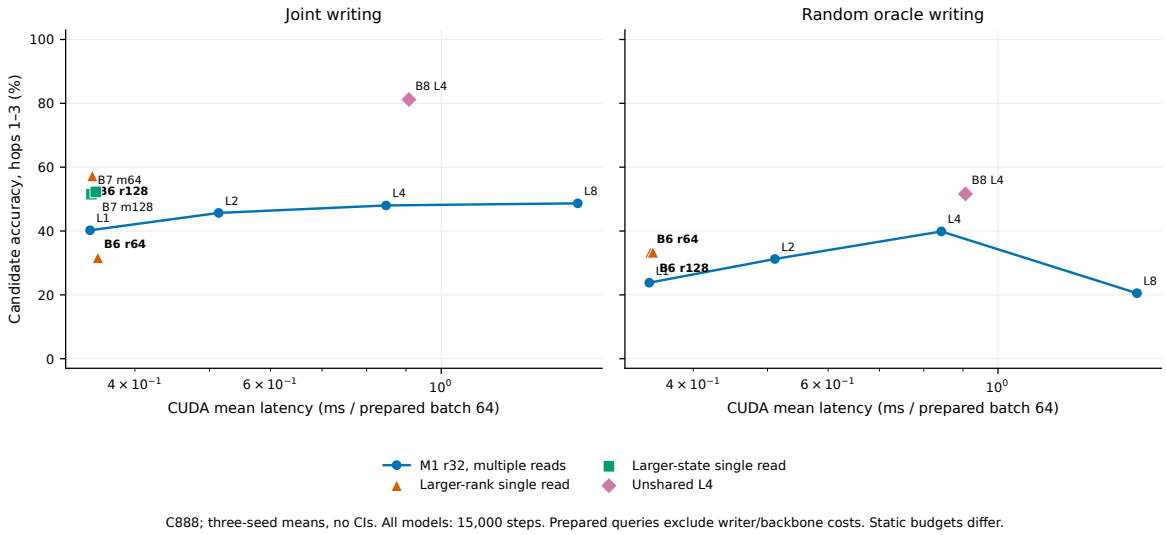


Figure 5: P2-S accuracy-latency points, with larger-rank single reads marked. Accuracy averages hops1-3, rather than only the reachable-hop endpoint; timings exclude writer/context preparation and are descriptive means. Different ranks trade different parameters and persistent-state budgets.

Memory load, adaptation, and updated facts

Study IS uses a new confirmation set, C999 (evaluation seed999001, 2,000 episodes per load), at r=r_mem=32, with oracle_random and joint_fact as primary writers. Its fixed E12-trained M1 checkpoints are evaluated at E6/12/18/24/36; increasing load also changes chain layout and candidate count. D(E) compares L4 with L1 on reachable hops2-3 within each checkpoint. For oracle writing, D falls from 34.750 pp at E12 to 7.492 pp at E36, with DeltaD=−27.258 pp [−28.800, −25.816]; for factorized writing it falls from 24.042 to 4.642 pp, with DeltaD=−19.400 pp [−20.542, −18.241]. Both changes are negative in every tested seed (seeds17/29/43: oracle_random −28.725, −25.900, −27.150 pp; joint_fact −20.525, −19.200, −18.475 pp); their 97.5% intervals use independent load bootstrap streams rather than cross-

layout episode pairing. This supports the registered operational load/depth prediction without identifying crosstalk as its sole cause or validating Proposition 4's operator bound as tight.

IVD asks whether key interference explains this shrinkage, on the same frozen r32 M1 checkpoints and a new set C999004 (evaluation seed999004, 2,000 episodes per checkpoint). It keeps the twelve-fact chains, queries, answers and the 14-entity candidate set fixed, adds up to 24 extra facts whose heads lie outside the episode and whose tails lie inside it, and varies the overlap of the extras' keys with the relevant keys at equal load. Its endpoint is IC=G(low)−G(high) at load 24, where G is L4−L1 of M1 on hops2-3; a writer passes if IC≥5 pp and the lower bound of the 97.5% interval is above 0. For oracle_random, with constructed key codes, IC=+13.975 pp [12.642, 15.242] (seeds +14.850, +13.250, +13.825 pp) and the achieved overlap gap is 0.500, so the shrinkage is consistent with key interference, identified by IVD for oracle_random. For joint_fact, whose overlap is selected among candidate heads, IC=+3.208 pp [2.175, 4.208] is below the 5 pp threshold (overlap gap 0.285): the paper does not offer interference as the identified explanation of the IS shrinkage for joint_fact, and states that load, graph layout and candidate distribution were changed together in IS. At load 24 G is below the unmodified base cell in every cell, including low overlap (oracle_random +25.542 pp at low overlap against +33.092; joint_fact +10.833 against +23.383), and a decay-only control leaves joint_fact's G unchanged at +23.383 pp. The extras bring the stored facts to 18 or 36, outside the readers' training distribution, and the constructed codes can themselves be out of distribution; the result narrows the IS wording for one writer and does not make interference the sole cause for either.

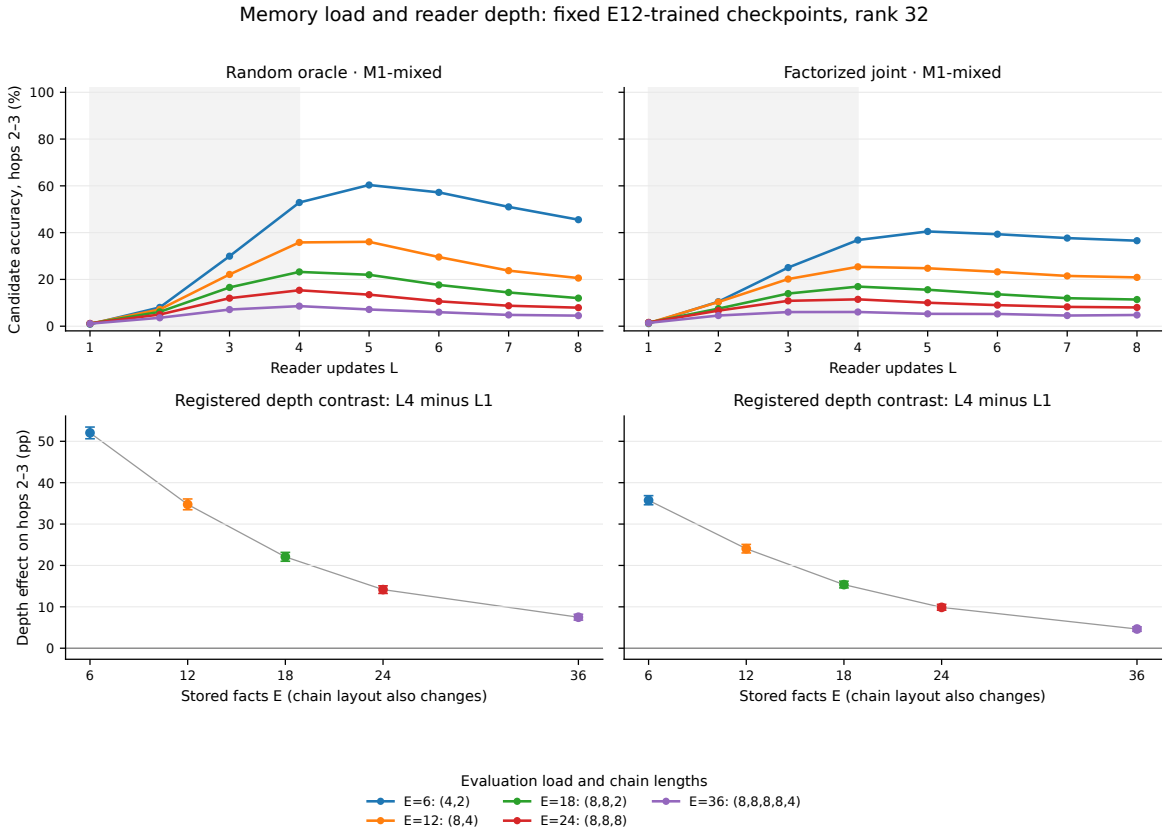


Figure 6: Plan §9.3 item4: fixed E12-trained shared M1 checkpoints across load/layout conditions, averaging seeds17/29/43 and hops2-3 on C999. Top panels show absolute accuracy means without supplied CIs; bottom panels reuse the released 97.5% within-load L4−L1 intervals. The same checkpoint per seed is used at every load, while chain layouts and candidate counts covary.

For models trained at their evaluation load, all tested E12/24/36 M1@4 and B8@4 contrasts against B4@1 meet the registered point≥3 pp and CI lower>0 rule, and all 36 tested-seed effects are positive. B8 adds 9,216 parameters, so it is not parameter-matched to B4 or M1; the percentage depends on the writer's parameter count. At the largest tested load, E36, their reachable-hop margins are:

Writer	M1-mixed@4−B4@1, pp [97.5% CI]	B8@4−B4@1 (B8 +9,216 parameters), pp [97.5% CI]
oracle_random	+4.483 [3.950, 5.033]	+5.942 [5.225, 6.675]
joint_fact	+3.433 [2.983, 3.867]	+5.008 [4.500, 5.500]

The factorized M1 interval need not have its lower endpoint above 3 pp to qualify; the threshold applies to the point estimate. These are separately trained systems with different depth/sharing conditions, not a recheck of hop1 preservation or the practical latency gate; the capability failure boundary beyond E36 was not located. Under the inherited learning rate and fixed-step protocol, all eight load-specific E24/E36 training contrasts against E12-trained models evaluated at the same load are negative, ranging from −6.283 to −2.042 pp, with negative intervals and negative effects in all 24 seed-level contrasts. That result does not show that independently tuned or convergence-matched load adaptation is generally harmful.

The update-task check uses unchanged E12-trained readers with 14 stored facts: 12 current edges plus two obsolete records; its pooled queries cover hops1-4. For B8@4 (+9,216 parameters, not parameter-matched to the shared readers; FR1), the absolute current/stale rates are:

Writer	Current, affected paths, % [97.5% CI]	Current, untouched paths, % [97.5% CI]	Obsolete terminal, affected paths, % [97.5% CI]
oracle_random	41.750 [40.235, 43.300]	37.606 [36.051, 39.211]	0.510 [0.332, 0.712]
joint_fact	28.619 [27.452, 29.735]	30.468 [29.357, 31.648]	6.035 [5.486, 6.591]

Current terminals are selected more often than obsolete ones in these primary B8 conditions, with partial overall success. These are absolute-rate intervals, not paired intervals for current−stale or affected−untouched effects; untouched-path stale rates are undefined. They establish neither a causal benefit from updating nor a guarantee that every reader follows updated paths.

7. Stopping and overshoot: post-hoc selection and training-time regularization

Earlier screening policies

The earlier label-free policies select thresholds using mean validation accuracy, freeze them, and evaluate on screening seed4242 without confidence intervals; no confirmation result is supplied for those pilots. Relative-change stopping on random-oracle r128 M1, from the 20,000-step rank-follow-up checkpoints where rank and training steps are confounded, increases long-hop accuracy from 45.747% at L8 to 70.690%, using 8.301 mean updates, but its hop1 accuracy remains below L1: 87.883% versus 99.983%. The corresponding linear reader also improves long-hop screening accuracy while retaining a hop1 loss. This is a selected empirical tradeoff, not a certificate that the chain terminal has been reached.

The tested r32 factorized stopping policies do not repair the hop1 preservation failure. All 12 factorized family/rule choices fail hop1 preservation; even each family's descriptively least-loss screening rule loses between 24.900 and 41.733 pp relative to L1. The unfactorized joint/M1 screening flag supplies no adaptive benefit either: all three rules use exactly 16 mean updates and reproduce fixed-L16 endpoints, with a negative specificity effect in one training seed (minimum seed W −0.125 pp). These outcomes restrict the tested rules, rather than ruling out every possible halting policy.

Fewer reader updates also do not imply lower runtime. For oracle_random/r128/M1 at batch64, fixed L16 takes 2.912 ms and relative-change stopping takes 11.059 ms; all 48 adaptive family/rule/batch cells are slower than L16 in both CUDA and wall means. This result includes the tested implementation's active-query bookkeeping and candidate decoding, and is not a claim about optimized future implementations or end-to-end latency.

Finally, none of the 96,000 validation-plus-screening episode evaluations has a sufficient factorized contraction bound below one. This establishes the absence of a certificate from that test, not actual noncontraction or divergence. The label-informed true-hop reference is not an accuracy upper bound for these learned readers, reinforcing that update count cannot simply be identified with graph distance.

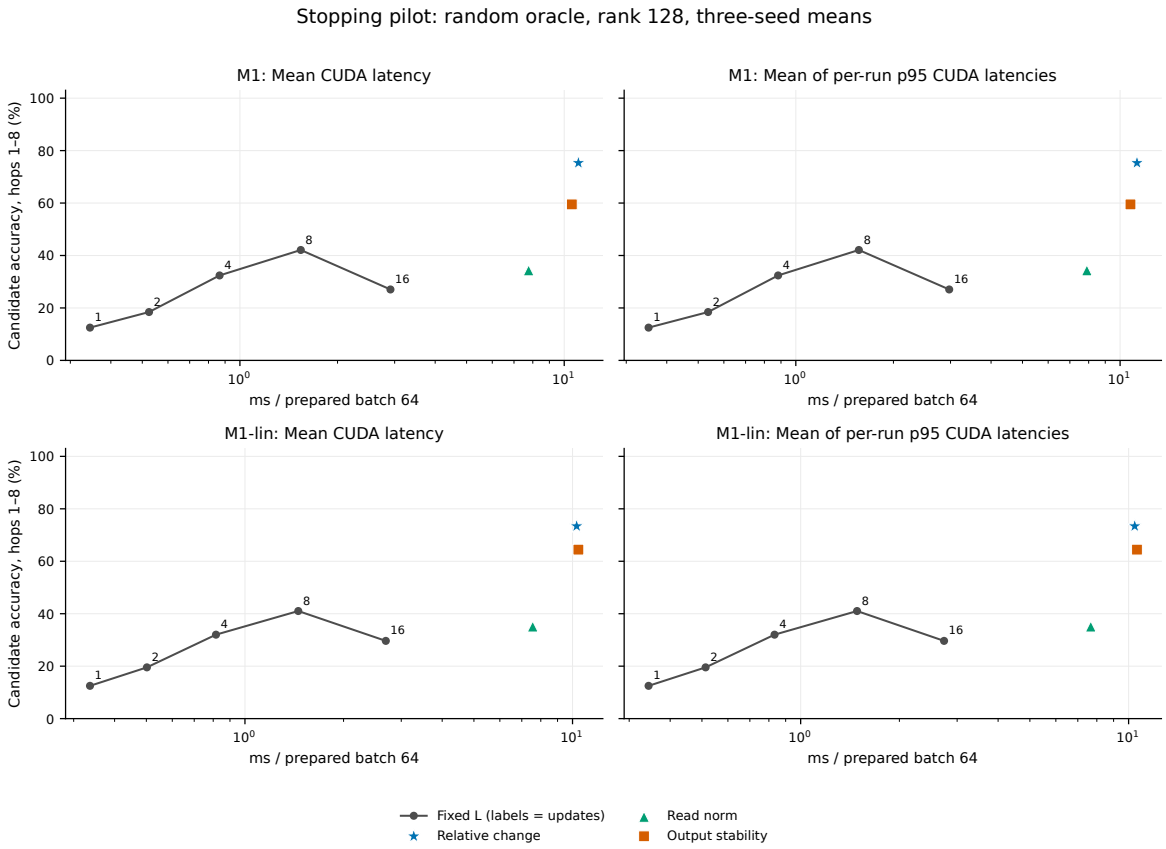


Figure 7: Screening stopping accuracy versus prepared-reader latency, with mean and exported tail-timing views identified separately. Thresholds are selected on validation; plotted screening points have no confidence intervals. Tail columns aggregate timing summaries and must not be read as a pooled request-latency distribution. Fewer steps do not establish a speedup.

Learned post-hoc halting on frozen readers

P4-S-L trains a separate 3,393-parameter halting head for each frozen reader, using state and step features with supervised correctness targets. Inputs exclude answer labels, graph distance, correctness, and decoder-confidence features; the targets still make head training supervised. Reader and writer parameters remain frozen, and one threshold per family is selected on V4242 and locked before fresh confirmation C999002 (test seed999002, 2,000 episodes). The six primary rank32 families are the two activations crossed with factorized joint, frozen factorized, and random-oracle writing.

None qualifies: all pass the reach and specificity requirements but fail the hop1 preservation point threshold H≥−1 pp. Here H/D compare the learned policy with L1 on hop1/hops2-3, and W subtracts the far-hop effect on hops6-8. The following intervals are the registered **99.166667%** paired episode intervals, adjusted for six primary families, conditional on seeds17/29/43.

Writer / reader	H, pp (pooled; seeds17/29/43)	D, pp	W, pp [99.166667% CI]
joint_fact / M1	−33.900; seeds −33.850/−34.950/−32.900	+21.583	+16.517 [15.263, 17.666]
joint_fact / M1-lin	−49.150; seeds −52.050/−46.200/−49.200	+24.733	+15.439 [14.026, 16.886]
oracle_random / M1	−15.700; seeds −12.350/−19.250/−15.500	+32.508	+27.975 [26.379, 29.575]
oracle_random / M1-lin	−20.850; seeds −17.800/−20.350/−24.400	+37.500	+18.839 [16.947, 20.678]
t1_frozen_fact / M1	−21.900; seeds −22.300/−20.450/−22.950	+9.442	+6.892 [5.996, 7.774]
t1_frozen_fact / M1-lin	−26.167; seeds −26.350/−25.500/−26.650	+10.217	+7.550 [6.661, 8.447]

All eighteen primary per-seed H point estimates also fail preservation. The descriptive unfactorized joint/M1 row satisfies the numerical inequalities, but is outside the primary set and is not a formal qualifying repair. Learned halting exceeds each family's validation-selected label-free reference in overall accuracy by +0.883 to +1.615 pp in the four joint_fact/oracle_random rows, with positive adjusted intervals; the two frozen-factorized intervals include zero. These gains coexist with the preservation failure. The label-informed any-step oracle remains 10.250-36.338 pp above the learned policy in primary families; it upper-bounds selection among the tested frozen candidates, unlike the distinct true-hop reference. These results reject this head-training and selection scheme as a repair, without ruling out all learned halting.

Selected depth is also different from executed work. On confirmation, primary-family means span 5.306-10.038 selected updates but 12.740-15.995 executed updates. If no state crosses the threshold, all 16 updates execute before the earliest maximum-probability state is selected; the post-state feature adds one memory read beyond the executed updates. In the separate prepared batch64 timing population, primary learned-policy CUDA means are 20.533-25.253 ms versus 3.072-3.308 ms for fixed L16, with no timing CIs. Those timings include the halting head and bookkeeping, exclude writer/backbone preparation, and do not establish a speedup or end-to-end request cost.

Hop-1-constrained thresholds and continuation-value heads

CVH reuses the nine frozen r32 M1 readers of P4-S-L (random-oracle, joint_fact and t1_frozen_fact writers, seeds17/29/43) on a new confirmation set C999005 (test seed999005, 2,000 episodes per reader). Arm A1 keeps the P4-S-L head, whose label is that the candidate argmax at step s is correct, and chooses its threshold on validation under a hop-1 constraint; arm A2 trains a new head of the same size to predict whether a later read would still lower the answer's loss by a 0.25-nat margin. An arm passes if its hops2-3 gain over L1 is at least 3 pp with the interval above 0, its hop-1 change is at least −1 pp, and its mean executed updates are at most 8; the intervals are 99.166667% (six comparisons) and condition on the three seeds. Every validation threshold preserved hop 1, so the confirmation set was opened for all three families.

Exactly one arm passes: A1 for oracle_random, with θ=0.10, a hops2-3 gain of +9.950 pp [9.153, 10.845] over L1, a hop-1 change of +2.883 pp and 1.968 mean executed updates. A halting head with the correctness label and a hop-1-constrained validation threshold repairs overshoot for oracle_random on these frozen readers. The pass is a registered pattern, not a match to fixed-depth reading: the arm's gain is −23.275 pp [−24.622, −21.942] against fixed L4 and −2.908 pp [−4.122, −1.608] against fixed L16, and the P4-S-L head at its original threshold (A0) still loses 14.800 pp on hop 1. The A2 head for oracle_random gains only +0.492 pp [0.336, 0.675], and both heads for joint_fact and t1_frozen_fact stop at the first read

for every query (θ=0.05, gain +0.000 pp, 1.000 executed updates), so no head repairs overshoot for the factorized families. No latency claim is made for any arm. The passing arm takes 7.829 ms against 3.108 ms for the fixed 16-update read on a prepared batch of 64, and the four arms whose wall time is below fixed L16 (2.395-2.420 ms against 3.081-3.114 ms) stop at step 1 and do not pass. The earlier P4-S-L failures stay reported. No gate depends on CVH.

Random-depth training with a final-state Jacobian penalty

JS changes training instead of fitting an exit to frozen trajectories. Its RS baseline samples one shared-reader depth uniformly from 4-8 per batch; RS+J adds $\max(\mathrm{rad}(J(z_L)) - .9, 0)^2$ with weight 1, averaged over episode/query states at the final training read. Here rad denotes spectral radius, distinct from the reader scale rho. The ReLU transition Jacobian uses the next-update gate at reached z_L, with derivative zero at a zero preactivation; full numerical eigenvalues are computed in float32, rather than estimated by a power method. This is a regularizer, not certified eigenvalue arithmetic or an assertion that the reached state is fixed. The two conditions share initialization and data/depth RNG streams within seed, then optimize separately; learned writers may therefore diverge. Both use M1/ReLU, rank32, 15,000 steps, batch64, and learning rate 0.003.

Fresh confirmation C999003 uses test seed999003 and 2,000 episodes, with thresholds locked on V4242 before test. In this study **short** means hops1-3, rather than the earlier reach endpoint's hops2-3; long remains hops4-8. The registered overshoot statistic is S=Acc16−Acc4 on short hops, and DeltaS=S(RS+J)−S(RS). All following JS contrast intervals are 97.5% paired episode intervals conditional on the three fixed training seeds; the per-seed point effects (seeds17/29/43) follow each pooled value.

Writer	S(RS), pp [97.5% CI]	S(RS+J), pp [97.5% CI]	DeltaS, pp [97.5% CI]
oracle_random	−11.267 [−12.356, −10.050]; seeds −11.900/−11.017/−10.883	−2.833 [−3.800, −1.806]; seeds −3.533/−2.400/−2.567	+8.433 [7.483, 9.418]; seeds +8.367/ +8.617/+8.317
joint_fact	−3.017 [−3.789, −2.272]; seeds −2.800/−1.950/−4.300	−1.772 [−2.484, −1.061]; seeds −1.983/−0.900/−2.433	+1.244 [0.222, 2.200]; seeds +0.817/ +1.050/+1.867

Both writers reduce overshoot, with positive DeltaS in every seed, but neither meets the registered removal criterion: in addition to positive DeltaS and CI, S(RS+J) must be at least −1 pp. Other endpoints expose the cost:

RS+J−RS endpoint	oracle_random, pp [97.5% CI]	joint_fact, pp [97.5% CI]
E2: short hops at L4	+0.789 [0.261, 1.295]; seeds +1.200/+0.883/ +0.283	−0.728 [−1.567, 0.067]; seeds −0.233/−0.733/ −1.217
E3: long hops at L16	−5.547 [−6.200, −4.890]; seeds −5.670/ −4.540/−6.430	−1.443 [−1.990, −0.933]; seeds −0.630/ −1.250/−2.450
E4: stopped, all hops	−1.040 [−1.538, −0.544]; seeds −0.694/ −0.513/−1.913	−0.677 [−1.227, −0.115]; seeds −0.163/ −0.606/−1.263

Thus longer-walk and stopped accuracy decline for both writers, while joint_fact's L4 short-hop interval crosses zero. RS+J's stopped accuracy is 18.704%/15.879% for oracle_random/joint_fact, with 15.938/ 15.028 mean reads under separately validation-selected relative-change thresholds and cap16; absolute-rate and read-count CIs are unavailable. This supplies no measured latency advantage.

The references distinguish random-depth training from the additional penalty. At L4, random-oracle RS scores below both references on short hops; factorized RS scores below M1-L4, while its +0.483 pp contrast with M1-mixed has an interval crossing zero. At L16, RS exceeds both on short hops for both writers. On long hops at L16, RS exceeds both references for oracle_random but scores below both for joint_fact. These contrasts change training history; RS+J−RS is the comparison that tests adding the penalty.

The descriptive L16 mean radius falls from 1.307 to 1.053 for oracle_random and from 0.861 to 0.834 for joint_fact; only 22.238% and 92.698% of RS+J reached states, respectively, have radius below 1. These summaries pool all hops and three seeds and have no radius CIs. A pointwise radius below 1 at a finite state does not certify global contraction, convergence, terminal annihilation, or a correct decoded answer. Lower radii coexist here with the longer-walk and stopped-accuracy losses.

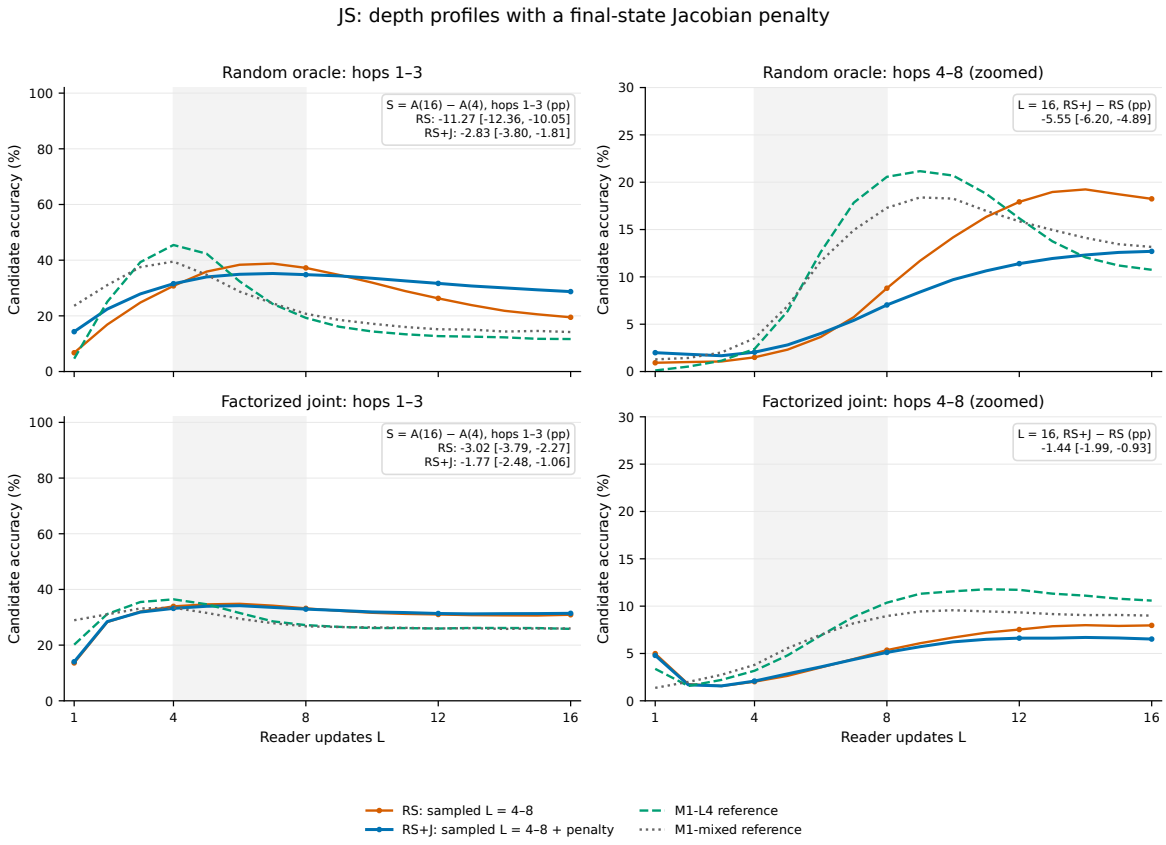


Figure 8: JS C999003: three-seed mean candidate accuracy versus depth, short hops1-3 and long hops4-8, for RS, RS+J, and unchanged M1-L4/M1-mixed references. Shading marks RS training depths4-8; absolute curves have no supplied CIs, while callouts reuse the released 97.5% endpoint intervals. Reduced overshoot retains negative S and costs longer-walk accuracy. Y scales differ by column; the long-hop panels are labeled as zoomed views.

Stay training and a sticky gate

TSM trains eighteen new r32 M1 readers (oracle_random and joint_fact writers, seeds17/29/43, 15,000 steps, the released learning rate 0.003 with no search): T0 is M1-L4 retrained under the same code, T1 trains at depth 8 with an answer loss at every step s≥hop, and T2 adds a content-gated sticky update with two extra parameters. Evaluation uses a new set C999007 (seed999007, 2,000 episodes) at depths 1, 4, 8 and 16, with the released B4 at L1 as the same-rank reference, and 98.75% intervals for the four arm-writer comparisons. An arm passes only if, for both writers, accuracy on hops1-3 differs between L16 and L4 by at most 1 pp (point), hops4-8 at L8 fall at most 1 pp below T0 (point), and the hops2-3 gain of L4 over B4 at L1 is at least 3 pp with the interval above 0. No arm passes for either writer. The gain condition holds in all four arm-writer cells (+16.575 and +16.567 pp for oracle_random T1 and T2, +16.950 and +21.533 pp for joint_fact) and the stay condition in one (joint_fact T1, −0.794 pp), but the far condition fails in all four (−11.667, −11.780, −5.733 and −4.553 pp). Neither stay training nor the sticky gate removed the overshoot under this protocol; fixed-depth training (M1-L4, B8) remains the remedy found, and the result does not generalise from one learning rate. Descriptively, hop 1 at L16 falls by 22.5-44.0 pp relative to each arm's own L1 in all four cells.

F Language-model studies in full

Record section 8: the access protocol, every writer screen, the registered oracle-reader confirmation at two backbones, DPFP and DPFP-X, the per-step readouts, and the summary subsection.

8. A single language-model module: writer screens and oracle-reader confirmation

8.1 Access protocol and scoring

The LLM experiments use frozen `Qwen/Qwen3-0.6B-Base`, revision `da87bfb608c14b7cf20ba1ce41287e8de496c0cd`, with one adapter at the zero-based layer-14 attention `o_proj` (2048→1024); the second-scale confirmation (Section 8.5) uses `Qwen/Qwen3-1.7B-Base`, revision `ea980cb0a6c2ae4b936e82123acc929f1cec04c1` (28 layers, `o_proj` 2048→2048). The backbone uses bf16 and SDPA; the adapter and memory state use fp32. The primary protocol is memory-only: support facts populate the adapter memory, then their backbone KV is discarded before questions are answered. Templated relations use single-token names, and the primary metric is unrestricted full-vocabulary greedy exact match at the one-token answer position. Secondary candidate-restricted accuracy and auxiliary name-probe accuracy measure different objects; neither substitutes for full-vocabulary EM.

The completed writer studies use T1 single-hop facts, seed17, and validation seed4242 with 1,000 episodes and one scored question per episode. Diagnosis and repair use E16, four relations, and head sharing 0.3; ordinary writer fits use M1 at L1, with ranks identified below. Their training batches contain four questions per episode, but that does not make the validation sample 4,000 independent answers. These are sequential validation screens with reused questions and no confirmation intervals or new-training-seed replication. The backbone is frozen during T1 fitting, while eligible adapter parameters, including recurrence matrices, can train; the term writer stage does not imply a frozen reader recurrence.

8.2 Information and output controls, then writer repair

Pilot-1 found no admissible writer capacity under its registered 1,500-step budget: the best full-vocabulary validation EM was 0.3%, and the selected fact-final policy at learning rate 0.0003 failed the capacity criteria at E8/12/16/24. This is a budgeted negative result, rather than an estimate of the architecture's maximum capacity. Separate rank32 frozen-feature probes recover tail identity at the tail-token input with 97.619% accuracy, and question-start identity/relation at the final question position with 99.4%/100.0%. These are independently fitted classifiers over seen name identities, not proof that one shared A jointly preserves all required information.

A supplied-answer-code control trains only a 1024×32 code table and B and reaches 98.7% full-vocabulary EM after 1,500 steps at learning rate 0.003. It bypasses the reader and its rho scale, establishing that this output path can express the answer when its identity is supplied; it does not establish learned retrieval or accessible writer codes. Extending unassisted writer training to 6,400 steps and moving A's input from

`o_proj` to the pre-layer-14 residual did not rescue EM. At the final checkpoints, source-fact key/query cosine top1 is 6.0-7.5%, compared with the 1/16=6.25% ranking baseline, with negative mean true-key margins. This supports a failed-matching diagnosis for those fits, without identifying a unique cause or showing that additional training can never help.

The next screen adds input standardization (W1), normalized address supervision (W2), clean-value auxiliary CE (W3), a pre-tail-key/tail-token-value architecture (W4), and their W1+W2+W3 combination W5. W5 does not include W4; W4 also has more deployed parameters than the other arms. Auxiliary address and value labels exceed answer-only supervision, and the value head is discarded for inference. At 4,800 steps, W2 has 78.9% key top1 but only 0.5% EM; W5 reaches 7.8% EM. Thus improved normalized matching need not imply useful deployed retrieval.

Fresh 32,900-step reruns give W5 14.4% EM, despite 82.4% key top1 and 39.3% candidate accuracy. The diagnostic OA arm, the value/output ceiling, reaches 70.3% EM and 95.7% candidate accuracy. OA substitutes the true learned fact value at every valid question token, bypassing the entire memory read, including addressing, attenuation and interference; it bounds what the learned value encoding and output path can achieve, and is not an addressing control. It is also different from the supplied answer-state control and from P1-L's fact-written random oracle. Neither W5 nor any other eligible learned writer meets the selection threshold at this stage.

8.3 Actual-read distortion and the final negative screen

WP-L4 audits the preceding W5 checkpoint without further training. Its mean query, true-value, and deployed-read norms are 19.012, 30.750, and 159.350; mean normalized squared read error is 26.635. The training-only value head decodes clean values at 85.4% but actual reads at 10.5%. These measurements make Proposition 11's distinction concrete: normalized key rankings omit raw query scale, later-write erasure, and value-mixture geometry. They do not identify which component uniquely causes the final answer failures. Zeroing or shuffling the deployed memory reduces W5 EM from 14.4% to 0.1%/0.2% while question inputs stay fixed; this establishes sensitivity to the tested memory interventions, not successful path following.

The final screen tests R, which supervises the raw read with normalized squared error and a stopped target/denominator; Rρ raises rho to 0.5, and RK adds a distinct-address key-cosine penalty. OAρ is a diagnostic gain variant of OA, the value/output ceiling. The final checkpoints are:

Rank	Arm	Training steps	Full-vocabulary validation EM, %
32	R	23,590	12.0
32	Rρ	24,220	12.2
32	RK	21,980	12.0
32	OAρ, value/output ceiling (diagnostic)	27,040	74.7
128	W5	24,635	29.9
128	R	24,645	33.5
128	OA, value/output ceiling (diagnostic)	31,175	84.0

For the three read-supervised r32 arms, normalized read error remains 0.663-0.669 alongside 12.0-12.2% EM. All are single-seed validation points without CIs; per-arm budgets were fixed by timing, so final comparisons are not matched-step causal effects of rank or gain. The predeclared matched-step OAρ diagnostic at 6,580 steps has higher observed EM than the earlier OA: 43.2% versus 33.7%. This does not make either diagnostic eligible as a learned writer. R at rank128 has clean/read head accuracy 93.3%/19.7%, consistent with remaining read distortion, while a decoder trained on clean values does not measure all information present in a read.

The best eligible final arm is R/r128 at 33.5%, below both the 80% writer gate and the 50% E8-retraining trigger; no E8 retraining occurred. The later DPFP screen did retrain once at E=8 (Section 8.5). The completed selection therefore yields no learned P1-L condition and no P3; the separately registered oracle-reader study is reported in Sections 8.4 and 8.5. The negative result concerns these single-module tasks, super-

vision choices, learning rates, and unequal fixed budgets. It establishes neither impossibility of learned writing in LLMs nor a universal output-path bottleneck.

RFX then separates storage, query and decoder at three frozen checkpoints (R at r128, R at r32 and the W5 rerun at r32; one seed, validation episodes 200,000-203,999, 4,000 queries). It replaces only the read at the final question position, keeps the rest of the LM path, and trains nothing. A replacement rescues if it halves the normalized read error and gains at least 10 pp EM over the better of the deployed read and its gain-fitted version (33.8%, 11.6% and 14.9%). Only an exact ridge store read with the true key rescues, at both R checkpoints (EM 84.2% and 49.05%, against clean controls of 84.4% and 48.8%). The exact store with the raw or unit query reaches 38.75-38.80% at r128 and 13.48-13.65% at r32, and the deployed store with the true key 39.9% and 13.95%, so no single factor rescues. The registered verdict is store+query at R/r128 and R/r32 and none at W5/r32, where the clean control itself reaches only 17.6%. A refit linear decoder on the deployed reads stays below the entity-restricted baseline (18.4%, 6.8% and 8.1%). Read descriptively with the preregistered guide, the deployed store with the true key sits near each checkpoint's own EM rather than near the OA level, which fits write interference in the delta store more than a query bottleneck; this is one seed and not an identification of the cause. The verdicts trigger no LSQ arm, so the separately registered DPFP screen was the only one left at that point; its result, and that of the later DPFP-X, is in Section 8.5. The true-key cells use the label and are not deployable, and a substituted read is a distribution shift for the layers after it.

8.4 Registered P1-L oracle-reader comparison

The symbolic reader selection motivates primary B8 trained at fixed L4, with shared M1-L4 as a secondary comparator. P1-L registers two `oracle_random` conditions, r=r_mem=32 and r=r_mem=128, with seeds17/29/43. The writer constructs memory directly from the fact list using frozen random unit key codes for (entity,relation) and fixed entity value codes, oracle seed17, and lambda=beta=1. Support text is not rendered or forwarded through the LM in this oracle protocol; question encoding and the reader/output path still train. Random codes do not imply orthogonality, lossless memory, or the exact Proposition 3 construction.

This oracle is not the symbolic oracle of Section 5.1. The symbolic oracle takes the question code, the keys, the values and the decoder from one code table, so a read returns exactly the code that addresses the next fact and every transition is identity-like. The language-model oracle keys facts by independent random (entity,relation) pair codes and stores entity value codes from a different table, so every read after the first needs the reader to learn a map from a value code to an unrelated key, and the question encoding and output path must be learned as well. Symbolic oracle results are therefore not carried over to this condition.

The task is T2 E12, chains(8,4), four relations, head sharing 0.3, trained on hops1-3 and evaluated on hops1-8. Per rank, B8@4 is compared with B4@1, fixed-query B5-L4@4, and memory-free B3-L4@4; M1-L4 supplies the shared variant. B8's primary comparison holds rank fixed but is not parameter-matched: its step matrices add 9,216 adapter parameters at rank32 (103,426 to 112,642, +8.9%) and 147,456 at rank128 (475,138 to 622,594, +31.0%), and only the matched shared M1-L4/B4 architecture has equal parameter counts. Adapter parameters and inference time per question are descriptive costs. B0-B2 provide descriptive baselines in both access protocols; B1/B2 train with retained context and are not repeated at r128.

The registered target was 17,000 training steps at r32, with r128 using that target if the timing rule permitted. Pilot-2 selected learning rate 0.001 or 0.003 at $\lfloor S/3 \rfloor$ steps, using training seed17 and validation generator seed4242, before main training, and the committed run locks record S=17,000 at both ranks: at rank32 lr 0.003 for M1-L4 and 0.001 for the other readers, at rank128 lr 0.001 for B3-L4 and 0.003 for the others. Over both learning rates, the selecting validation EM (pooled hops1-3, 3,000 answers) of the memory readers is 0.033-0.133% at rank32 (1-4 answers), so those choices carry almost no information, and 3.1-8.1% at rank128. Confirmation uses split `test`, episode indices [100000,102000), after all required training; this range is not an evaluation RNG seed. With two gated rank conditions, k=2, each gate uses 97.5% pooled paired episode-bootstrap intervals from 2,000 draws and reports seed-specific effects.

On hops2-3, G1-L requires B8@4−B4@1≥3 pp as a point estimate, CI lower>0, and positive effects in every seed. Its hop1 point contrast must be at least −1 pp, and the same checkpoints must have a positive interval on a typed replication with three same-head distractors. G2-L requires positive intervals against both fixed-query B5-L4 and memory-free B3-L4. M1-L4 comparisons and its same-checkpoint L1-8 scan are secondary; B8 has no shared-weight depth scan. No primary gate compares a lower-rank multi-read system with a larger-rank single read.

Two amendments, registered before any confirmation score was read, bound how the confirmation may be interpreted. An equal-parameter claim that more reads beat one read is made only from the secondary contrast M1-L4@4−B4@1 (hops2-3, per rank), and only if it meets the G1-L criterion (point ≥3 pp, 97.5% CI>0, positive in every seed); otherwise B8 is reported with its parameter difference and no equal-parameter claim is made for the language model (Amendment B6). At each rank a memory-use check precedes interpretation: hop1 full-vocabulary EM of B4@1, B8@4 and M1-L4@4 must each exceed B3-L4@4 with a 97.5% CI above 0, and otherwise that rank's G1-L, G2-L and equal-parameter outcomes are reported but not interpreted as evidence for or against multiple reads. No claim about multiple reads in the language model is made at rank32 whatever the gates show. At rank128 the value-to-key transition is a learned lookup over the 768 seen names, so a pass would show that the reader learns it and composes reads inside the language model; it would not establish equivalence with the symbolic oracle, and no claim about unseen names is made (Amendment B7).

Before any confirmation score was read, the training logs and Pilot-2 validation of the main runs were inspected. This is training and validation evidence, not a result of the registered test; the logs are local files, and the committed record is Amendment B7. At rank32, after 17,000 steps, the memory readers (B4, B5-L4, B8, M1-L4) had batch accuracies of 0.391-1.641% against 0.312-0.391% for the memory-free B3-L4, while the context-retained B1/B2 reached 99.375-100.000%; Pilot-2 validation at rank32 gave full-vocabulary EM of at most 0.3% and candidate accuracy of 7.2-9.1% against a 7.1% chance level. The rank32 readers were therefore at or near the memory-free level. At rank128 the batch accuracies were 10.000-10.781% (B4), 10.312-13.516% (B5-L4), 11.250-12.812% (B8) and 28.125-41.250% (M1-L4), against 0.312-0.625% for B3-L4. An in-sample capacity check of the oracle's independent key and value tables puts hop2 addressing at 48.3% at rank32 and 99.85% at rank128 (rotation keys: 99.2% and 100%); it is an upper bound and does not explain the rank32 failure at hop1. The registered prediction was that rank32 fails the memory-use check and rank128 passes it; it was made before any confirmation score, and Section 8.5 reports the check.

WP-L6 asked whether a rank-32 M1-L4 reader can learn the oracle memory when the oracle's keys are tied to its values. O-rot writes the key normalize(Π_ρ g_h) with one fixed orthogonal matrix per relation and the value g_t; O-shared trains a code table jointly with the reader. Both use the P1-L recipe (lr 0.003, seed17, 5,666 steps, validation seed4242), and an arm passes if its final-checkpoint hop-1 full-vocabulary EM at L=4 is at least 30%. Neither passes: O-rot reaches 0.1% and O-shared 11.8%, the registered decision is stop, and the study P1L-S that would have used the selected code does not run. At rank 32 the LM path did not learn these oracle codes within this budget (one seed, one learning rate, 5,666 steps); this is not a proof of impossibility, O-shared above O-rot is descriptive and not a selection, and it adds no statement about multiple reads at rank 32 (Amendment B7).

8.5 Confirmation results and pending items

Memory-use check. The confirmation (Qwen3-0.6B-Base) used split `test`, episode indices [100000,102000), 2,000 episodes, training seeds 17/29/43 and 97.5% paired episode-bootstrap intervals (2,000 draws) in the memory-only protocol, with one oracle memory (oracle seed17) shared by every reader; the intervals condition on the three seeds. Only the `oracle_random` condition ran, because the writer screens selected no learned writer. Hop1 full-vocabulary EM, and each reader minus the memory-free B3-L4@4 in percentage points:

Rank	B4@1	B8@4	M1-L4@4	B3-L4@4	Memory-use check
32	0.067% (+0.033 [0.000, 0.100])	0.067% (+0.033 [−0.050, 0.133])	0.150% (+0.117 [0.017, 0.233])	0.033%	fails
128	19.633% (+19.633 [18.417, 20.933])	19.033% (+19.033 [17.867, 20.367])	47.850% (+47.850 [46.016, 49.667])	0.000%	passes, every seed

The check fails at rank32, so the readers did not learn to use the oracle memory, and by Amendment B7 no claim about multiple reads in the language model is made at rank32: its gate outcomes (G1-L and G2-L fail; B8@4−B4@1 on hops2-3 −0.008 pp [−0.108, 0.100]; M1-L4@4−B4@1 +0.058 pp [−0.050, 0.167]) are reported and not interpreted. The check passes at rank128 in every seed. It is a floor at hop1 and does not show that composition uses the memory.

Rank128 (Qwen3-0.6B-Base). On hops2-3, the gated reader B8 (four unshared reads) exceeds B4@1 by +0.983 pp [0.267, 1.717] (6.342% against 5.358%; seeds +0.925, +0.050, +1.975), below the registered 3

pp bar, so G1-L fails although its hop1 (−0.600 pp [−1.984, 0.767]) and typed-replication (+1.550 pp [0.767, 2.367]) conditions hold; this is a negative language-model result for B8 in this condition, which adds 147,456 parameters (+31.0%) over B4. G2-L passes (+0.833 pp [0.058, 1.550] over the fixed-query B5-L4@4 and +6.342 pp [5.758, 6.933] over the memory-free B3-L4@4), but a G2-L pass without G1-L is not a multi-read claim for B8, and the memory-free reader is at 0.000%. The shared reader does compose: M1-L4@4 exceeds B4@1 on hops2-3 by +21.550 pp [20.383, 22.758] (26.908% against 5.358%; seeds +24.450, +15.100, +25.100), meeting the Amendment-B6 criterion, with equal adapter parameter counts (475,138 each); it also exceeds B5-L4@4 by +21.400 pp [20.217, 22.550] and B3-L4@4 by +26.908 pp [25.758, 27.975]. This is the only equal-parameter claim, and it is bounded by Amendment B7: the reader learns the value-to-key lookup over the 768 seen names and composes reads inside the language model for this oracle memory; it is not equivalent to the symbolic oracle condition, and nothing is claimed for unseen names or entity codes, a learned writer or end-to-end use. Descriptively, M1-L4@4's depth effect (L4 against L1) is +25.925 pp [24.816, 27.050] on hops2-3 and +3.678 pp [3.206, 4.183] on hops6-8, its accuracy on hops1-4 is 47.85%, 35.35%, 18.47% and 7.25% (at most 5% on hops5-8) at the trained depth L=4 (Section 8.6 gives every depth), and B8@4 is 20.567 pp [19.358, 21.783] below it on hops2-3 (a secondary, ungated contrast). The difference between the two readers is one of these two architectures on one fixed oracle memory in 17,000 steps, not a statement about unshared reads in general. The added reads cost time: per question, M1-L4 takes about 1.42 and B8 about 1.57 times B4's 0.392 ms at rank128, and no efficiency claim is made.

Second scale (P5L). P5L repeated the rank128 oracle condition on Qwen3-1.7B-Base (same family; layer-14 `o_proj`, 2048→2048; 17,000 steps; the rank128 learning rates carried over without a pilot) on a range no earlier study used, `test` [104000,106000), with 2,000 episodes, seeds 17/29/43, 97.5% intervals, the same oracle memory and the same 768 seen names. The memory-use check passes in every seed: hop1 EM exceeds the memory-free B3-L4@4 (0.033%) by +30.617 pp [29.167, 32.234] for B4@1, +28.900 pp [27.517, 30.433] for B8@4 and +58.667 pp [56.983, 60.400] for M1-L4@4. The Amendment-B6 contrast M1-L4@4−B4@1 on hops2-3 is +25.783 pp [24.533, 27.026] (31.958% against 6.175%; seeds +24.925, +26.500, +25.925), with 606,210 adapter parameters in each, so the registered decision is that the equal-parameter result holds at a second scale of the same family. It has the limits of the 0.6B result (oracle memory, 768 seen names, trained hops, Amendment B7 rule 3), and in addition one oracle seed, three training seeds and learning rates carried without a pilot; it covers the two sizes tested and no other. Reported only: B8@4−B4@1 is +1.033 pp [0.292, 1.800] (seeds +0.200, +1.725, +1.175), below the 3 pp bar, so G1-L fails again, and its hop1 condition also fails at this scale (−1.717 pp [−3.317, −0.016]) where it held at 0.6B; G2-L passes (+1.675 pp [0.908, 2.383] over B5-L4@4 and +7.167 pp [6.583, 7.758] over B3-L4@4). B8 adds 147,456 parameters (+24.3%) and is not parameter-matched. Descriptively, M1-L4@4 reaches 58.70%, 41.45%, 22.47% and 6.23% on hops1-4 (at most 3.92% on hops5-8) at L=4, with a depth effect of +31.458 pp [30.267, 32.633] on hops2-3 and a walk specificity of +28.181 pp [26.844, 29.425]. These values sit next to the 0.6B ones (+21.550 pp for the B6 contrast; 47.85%, 35.35%, 18.47% and 7.25% on hops1-4) without any test across scales, and two sizes support no scaling trend. Latency is a cost and is not compared across scales: at 1.7B M1-L4 and B8 take about 1.29 times B4's 0.486 ms per question. P1L-C and the per-step readouts were not run at 1.7B.

P1L-C. P1L-C ran at Qwen3-0.6B-Base after the rank128 check passed, evaluating the nine rank128 checkpoints (M1-L4, B8 and B4, three seeds; nothing retrained) on a second range, `test` [102000,104000), with 95% intervals. Keeping the memory only at the final question position lowers the hops1-3 EM of M1-L4@4 from 33.64% to 3.74% (−29.906 pp [−30.756, −29.094]), of B8@4 from 10.48% to 0.99% (−9.489 pp [−10.078, −8.911]) and of B4@1 from 10.39% to 1.26% (−9.133 pp [−9.711, −8.589]), in every seed and hop, including hop1 of the single read. This is descriptive: the readers were trained with reads at every question position, so the drop is a distribution shift and not evidence of leakage, and it does not locate the bridge. Each question position runs its own walk, so the final position is one of several (Section 8.6). Bridge patching, which replaces the state z_1 after the first read at the final question position by the state from a world whose first edge points to another bridge, shifts the answer by +0.391 pp [−0.130, 0.781] for M1-L4 and 0.000 pp [−0.521, 0.000] for B8, against a registered bar of 10 pp with an interval above 0 and a positive shift in every seed. There is no causal bridge claim for either reader. The M1-L4 answer does follow the other chain when the memory itself carries the other bridge (37.24% against 7.03% in the original world), but not when z_1 alone is replaced at the final or at every question position (+0.911 pp [−0.651, 2.344]). The P1-L contrasts replicate on this range (M1-L4@4−B4@1 +20.808 pp [19.725, 21.850], B8@4−B4@1 +0.925 pp [0.283, 1.533]) without changing any gate.

DPFP and P1L-D. A registered writer screen replaced the keys and queries of the WP-L4 learned writer by DPFP features (three arms: D32-3 at rank32, D128-1 and D128-3 at rank128; one seed, T1, E=16, 1,000 validation episodes per arm; the WP-L4 references R at rank32 and rank128, 12.0% and 33.5%, were not

rerun). No arm reaches the 80% writer gate: D32-3 39.0%, D128-1 58.0% and D128-3 60.8%. The best arm is above the 50% trigger, so D128-3 was retrained once at E=8 under the registered rule and reached 73.0%, again below 0.80. The registered decision is stop: no writer is selected, the paper reports a negative result, and no further writer screen follows; P1L-D, which would have run the P1-L learned condition for a selected writer, was not registered and was not run (the LSQ screen triggered by RFX ran no arm, Section 8.3; the later, separately registered SQ2 and LSQT screens are in Appendix I). No learned writer reaches the WP-L4 gate in this project. DPFP does raise the EM (by 27.0 pp at rank32 and by 24.5 and 27.3 pp at rank128 over the references), lowers the normalized read error (0.39, 0.43 and 0.36 against 0.67 and 0.63) and raises the read-value head accuracy (30.2%, 46.0% and 51.0% against 8.8% and 19.7%), but the clean/read gap narrows without closing, and the key overlap of the learned projections in the original space is not lower than R's. Because DPFP changes the feature geometry and the query normalization together (Amendment A1), neither these gains nor the 12.2 pp gain of the E=8 retrain (which also halves the memory load) is attributed to decorrelation alone. A zero- or shuffled-memory audit leaves at most 0.2% EM for every DPFP arm, which shows that the answers depend on the episode's own memory and not that the read is accurate. The result is one seed, one learning rate and fixed budgets, and it is not a proof that learned writing in a language model is impossible. A follow-up registered after this stop, DPFP-X, trained the best DPFP design at rank256 on Qwen3-0.6B-Base (X256) and at rank128 on Qwen3-1.7B-Base (X17), everything else as D128-3. Neither reaches 0.80 (X256 66.2%, X17 59.3%), so the learned-writer line stays stopped and both arms are descriptive rows next to DPFP. X17 differs from D128-3 (60.8%) by −1.5 pp, which the registered rule calls not distinguished, and it is a diagnostic and not a scaling test; both curves were still rising at the final checkpoint (+3.5 and +2.3 pp over the last interval) but the study has no longer run and nothing is extrapolated, and X256's higher EM is not accompanied by a lower read error or a better read-value head than D128-3's. The zero- and shuffled-memory audit again leaves at most 0.4% EM. P1L-S does not run (Section 8.4).

Per-step readouts (descriptive). For M1-L4 and B8 the state z_s after each read s=1...4 was decoded at the answer position on the confirmation episodes (three seeds pooled) through the trained output path and the frozen upper layers (logit lens), by projection onto name-gradient directions (a contrastive and a raw-logit variant) and by a random direction; chance for the top-1 rate of an entity is about 0.1%. At rank128 the logit lens recovers earlier-hop entities from M1-L4's early states, and best at a step s at or above the entity's hop e rather than at s=e: on hop-1 questions the hop-1 entity is the top-1 name in 38.6%, 64.3%, 61.1% and 47.9% of episodes at s=1...4, and on hop-2 questions the hop-2 entity in 1.1%, 11.0%, 30.2% and 35.4%. The rates at s=4 for the question's own hop equal the deployed EM within 0.1 pp, and the hop-1 rates at s=2 and s=3 are close to the EM of the same checkpoint at depth 2 and 3 (64.15% and 61.28%); the corrected record reads this as the answer the reader would give if it stopped at step s. B8's lens decodes names only at s=4 (the hop-1 entity in 19.1% of hop-1 questions) and at most 0.23% at s=1...3. The name-gradient projections and the random direction are at or near chance for both readers (maximum cell 0.28%), and at rank32 every method is at chance (maximum cell 0.27%), which adds nothing because the rank32 readers did not use the memory. These are decoding results: not gate criteria, not an ordered entity trace, not path following, and not evidence for the composition claim above (LP4); bridge patching gives no causal claim. A gradient g_j at the actual injection residual induces local reader sensitivity alpha B^T g_j; a gradient at another site requires the intervening transport Jacobian. This J-lens-inspired proxy differs from published averaged residual-transport lenses and from JS's recurrent transition Jacobian, and its chance result limits what these directions decode without showing that names are absent from z_s. Per-step upper-layer forwards incur diagnostic cost separate from the deployed reader loop. Ordered readability would not alone establish causal path following. The primary-source audit states those limits; the separate output-alignment intervention is not pursued after the writer-screen decision.

TODO , plan §9.3 item6: the symbolic/LLM correspondence figure is not produced; the P1-L confirmation is reported above in tables. The completed writer-screen negatives supply no multihop oracle-reader outcome or general language-reasoning claim.

8.6 The oracle condition at a glance: same-rank table, depth by hop, and seed-level statistics

This subsection restates the language-model results of Sections 8.4 and 8.5 in one place. Every value is a seed mean, or an effect over the three training seeds 17/29/43, at rank128 with one oracle memory (oracle seed17) in the memory-only protocol. The tables were produced after the registered confirmation by a standard-library script (paper_tables_g1.py) from the committed accuracy, timing and endpoint files; the depth matrix and the seed-level statistics are descriptive, add no gate and change no registered decision.

Same-rank table. Exact-match accuracy by hop (%), primary setting; B4@1 is the single read, M1-L4@4 and B8@4 read four times, all at rank128 and at the depth they were trained for.

Backbone	Reader	hop1	hop2	hop3	hop4	hop5	hop6	hop7	hop8
0.6B	B4@1	19.63	5.35	5.37	5.82	5.32	5.00	5.40	5.17
0.6B	M1-L4@4	47.85	35.35	18.47	7.25	3.90	4.50	4.72	4.97
0.6B	B8@4	19.03	6.65	6.03	6.10	6.17	5.38	6.25	6.37
1.7B	B4@1	30.65	6.17	6.18	5.53	5.53	5.62	5.55	5.72
1.7B	M1-L4@4	58.70	41.45	22.47	6.23	3.10	3.78	3.63	3.92
1.7B	B8@4	28.93	8.17	6.25	6.28	5.97	5.73	5.57	6.20

Costs of the same readers. M1-L4 and B4 have equal parameter counts; B8 adds 147,456 parameters (+31.0% at 0.6B, +24.3% at 1.7B). The state bytes are the persistent FP32 delta memory (r_mem × r_mem × 4 bytes per episode; derived from the rank, the same for the three readers because they read the same memory, and excluding the backbone cache). Time per question is a measured cost, compared only within one backbone; no efficiency claim is made.

Backbone	Reader	Adapter parameters	State bytes per episode	Time per question (ms)	Time relative to B4@1
0.6B	B4@1	475,138	65,536	0.392	1.00
0.6B	M1-L4@4	475,138	65,536	0.556	1.42
0.6B	B8@4	622,594	65,536	0.617	1.57
1.7B	B4@1	606,210	65,536	0.486	1.00
1.7B	M1-L4@4	606,210	65,536	0.625	1.29
1.7B	B8@4	753,666	65,536	0.625	1.29

Three kinds of evidence. The results fall into three kinds that answer different questions and are never pooled.

Kind of evidence	Question	Result	Rows
Oracle composition	Does a trained reader compose reads over a privileged memory?	M1-L4@4 exceeds B4@1 at equal parameters on hops2-3 by +21.550 pp (0.6B) and +25.783 pp (1.7B); B8 fails G1-L at both backbones	LP7-LP20
Learned-writer access	Can a learned writer build that memory?	No writer reaches the 0.80 gate in any screen (best final WP-L4 arm 33.5%; DPFP 60.8% and 73.0% at E=8; DPFP-X 66.2% and 59.3%); each registered decision is stop	LL9, DP1, DP4, PD1
Mechanism evidence	Where and how does the walk happen?	Per-step readouts are decoding only; bridge patching at the final question position shifts the answer by +0.391 pp against a 10 pp bar; final-position-only reads lower M1-L4's hops1-3 EM by 29.906 pp (a distribution shift)	LP5, PC1-PC3

Negative interventions and the exact gate each did not meet. The matrix lists each repair or test that failed its registered criterion. Each row is one study under its own conditions (one training seed or three, one optimizer, a fixed budget). The rows record which criterion was not met; they do not show that learned writing, an exact store, unshared reads or composition at rank32 is impossible. Not triggered: the LSQ arms of the RFX rule. Not run: P1L-D and P1L-S. A warm start of B8 from M1-L4 kept the shared solution (Appendix I).

Intervention	Conditions	Result	Registered criterion not met	Rows
Learned writers of the WP-L4 screen (R, W5 and read-supervision variants)	Qwen3-0.6B-Base, ranks 32 and 128, one seed, 1,000 validation episodes	Best eligible arm 33.5% (R, rank128)	Validation EM ≥0.80 (and the 0.50 trigger of the E=8 retraining)	LL9
Read-failure factorial (RFX)	Frozen R/r128, R/r32 and W5/r32 checkpoints, evaluation only	Only an exact store together with the true key (a label) rescues the read; the exact store with the deployed or unit query gives 38.8% and 38.75% against 33.8% at R/r128	Rescue: normalized read error ≤0.5 of the baseline and EM gain ≥10 pp	RX1
Oracle codes at rank32 (WP-L6)	M1-L4, r=r_mem=32, one seed, 5,666 steps	Hop-1 EM 0.1% (O-rot) and 11.8% (O-shared)	Hop-1 full-vocabulary EM ≥30% at L=4	WL1
DPFP writer	Ranks 32 and 128, E=16, E=8 retrain of the best arm, one seed	39.0%, 58.0% and 60.8% at E=16; 73.0% at E=8	Validation EM ≥0.80	DP1
DPFP-X writer	X256: Qwen3-0.6B-Base, r=256; X17: Qwen3-1.7B-Base, r=128; one seed	66.2% and 59.3%	Validation EM ≥0.80	DP4
Oracle readers at rank32	B4, B8 and M1-L4 on the oracle memory, three seeds	Hop-1 EM at most 0.15%; the memory-use check fails	Amendment B7 rule 1: each reader's hop-1 EM above B3-L4@4 with a 97.5% interval above 0 (lower bounds 0.000 for B4, −0.050 for B8)	LP7
B8 (four unshared reads) at 0.6B	Rank128, three seeds	B8@4−B4@1 on hops2-3 +0.983 pp [0.267, 1.717]	G1-L point estimate ≥3 pp (its hop-1 and typed conditions held)	LP9
B8 at 1.7B	Rank128, three seeds	+1.033 pp [0.292, 1.800]; hop 1 −1.717 pp	G1-L point estimate ≥3 pp; hop-1 point estimate ≥−1 pp	LP16
Bridge patching	Rank128, 0.6B, L=4, state after the first read at the final question position	+0.391 pp [−0.130, 0.781] (M1-L4), 0.000 pp (B8)	Shift ≥10 pp, 95% interval above 0, positive in every seed	PC1

Depth by hop (descriptive). Seed-mean EM (%) of the M1-L4 checkpoint, trained at L=4 on hops1-3, evaluated at depths L=1...8 on hops1-8 at Qwen3-0.6B-Base. The best depth of each hop is in bold. The last row, the oracle stop, takes the best depth per hop with the label; it is an upper bound and a maximum over eight noisy cells, not a stopping rule.

L	hop1	hop2	hop3	hop4	hop5	hop6	hop7	hop8
1	37.8	1.1	0.9	1.1	0.8	0.9	1.1	1.1
2	**64.2**	10.9	1.4	1.8	1.6	1.7	1.7	1.8
3	61.3	30.0	7.5	2.9	2.6	2.5	2.9	3.2
4 (trained)	47.9	**35.4**	18.5	7.3	3.9	4.5	4.7	5.0
5	37.2	29.3	**24.2**	14.1	7.2	6.1	6.7	6.7
6	28.1	22.8	20.6	**16.6**	11.6	8.8	8.5	8.9
7	22.6	18.8	17.7	15.4	13.5	10.9	9.5	9.7
8	19.0	16.4	15.0	13.8	**13.5**	**12.4**	**10.6**	**9.9**
oracle stop (best L, label used)	64.2 (L2)	35.4 (L4)	24.2 (L5)	16.6 (L6)	13.5 (L8)	12.4 (L8)	10.6 (L8)	9.9 (L8)

The same at Qwen3-1.7B-Base.

L	hop1	hop2	hop3	hop4	hop5	hop6	hop7	hop8
1	34.1	0.5	0.5	0.5	0.5	0.5	0.5	0.5

L	hop1	hop2	hop3	hop4	hop5	hop6	hop7	hop8
2	**77.6**	10.9	1.2	1.3	1.1	1.2	1.2	1.1
3	70.6	34.6	6.6	1.8	2.0	2.3	2.2	1.9
4 (trained)	58.7	**41.5**	22.5	6.2	3.1	3.8	3.6	3.9
5	44.9	36.5	**29.3**	15.3	5.9	5.2	5.8	6.0
6	34.9	30.2	27.0	**20.9**	12.0	7.5	6.9	7.7
7	29.0	24.6	24.3	20.6	16.9	11.2	8.7	8.8
8	24.1	20.9	21.4	18.4	**17.8**	**14.7**	**10.7**	**9.8**
oracle stop (best L, label used)	77.6 (L2)	41.5 (L4)	29.3 (L5)	20.9 (L6)	17.8 (L8)	14.7 (L8)	10.7 (L8)	9.8 (L8)

At the trained depth the reader reaches hop 3: at L=4 its EM on hops1-3 is 47.9/35.4/18.5% at 0.6B and 58.7/41.5/22.5% at 1.7B, and on hops5-8 it is 3.9-5.0% and 3.1-3.9%, not above the single read B4@1 (5.0-5.8% and 5.5-5.7% on hops4-8). The same checkpoint reaches hops4-8 at 9.9-16.6% (0.6B) and 9.8-20.9% (1.7B) when run deeper (hop4 at L=6, hops5-8 at L=8, the largest depth evaluated, so their best depth is not located), while losing hop 1: at L=8 hop 1 is 19.0% and 24.1%, against 47.9% and 58.7% at L=4 and 64.2% and 77.6% at L=2. These cells are at hops and depths never trained, carry no interval and rest on one oracle seed and three training seeds; the paper does not say that the reader cannot go beyond hop 3, nor that it composes up to hop 8, and no stopping rule for this reader is evaluated.

Seed-level statistics. The registered intervals resample episodes with the three fitted seeds held fixed. Beside each, the tables print the three seed effects and a seed-level interval: the mean of the seed effects $\pm$ t(0.9875; 2 degrees of freedom) = 6.205 times their sample standard error, the two-sided 97.5% level of the registered intervals. Both intervals are conditional on three fitted seeds and one fixed oracle code table (oracle seed17, one layout): neither covers future training seeds or other code tables, and the t-interval treats the three seed effects as exchangeable draws, which three values cannot test.

Qwen3-0.6B-Base.

Contrast (pp)	Seed 17	Seed 29	Seed 43	Seed mean	Seed-level 97.5% interval	Episode bootstrap 97.5% (registered)
B6 contrast M1-L4@4 - B4@1, hops 2-3	+24.450	+15.100	+25.100	+21.550	[+1.50, +41.60]	[+20.383, +22.758]
G1-L reach B8@4 - B4@1, hops 2-3	+0.925	+0.050	+1.975	+0.983	[-2.47, +4.44]	[+0.267, +1.717]
G1-L hop 1 B8@4 - B4@1	-2.300	+1.200	-0.700	-0.600	[-6.88, +5.68]	[-1.984, +0.767]
G1-L typed replication B8@4 - B4@1, hops 2-3	+1.575	+0.600	+2.475	+1.550	[-1.81, +4.91]	[+0.767, +2.367]
G2-L B8@4 - B5-L4@4 (fixed query), hops 2-3	+0.875	+0.100	+1.525	+0.833	[-1.72, +3.39]	[+0.058, +1.550]
G2-L B8@4 - B3-L4@4 (memory-free), hops 2-3	+6.525	+5.675	+6.825	+6.342	[+4.20, +8.48]	[+5.758, +6.933]

Qwen3-1.7B-Base.

Contrast (pp)	Seed 17	Seed 29	Seed 43	Seed mean	Seed-level 97.5% interval	Episode bootstrap 97.5% (registered)
B6 contrast M1-L4@4 - B4@1, hops 2-3	+24.925	+26.500	+25.925	+25.783	[+22.93, +28.64]	[+24.533, +27.026]
G1-L reach B8@4 - B4@1, hops 2-3	+0.200	+1.725	+1.175	+1.033	[-1.73, +3.80]	[+0.292, +1.800]
G1-L hop 1 B8@4 - B4@1	-1.800	-0.300	-3.050	-1.717	[-6.65, +3.22]	[-3.317, -0.016]
G1-L typed replication B8@4 - B4@1, hops 2-3	+0.375	+3.300	-0.300	+1.125	[-5.73, +7.98]	[+0.275, +1.942]
G2-L B8@4 - B5-L4@4 (fixed query), hops 2-3	+0.550	+3.600	+0.875	+1.675	[-4.33, +7.68]	[+0.908, +2.383]

Contrast (pp)	Seed 17	Seed 29	Seed 43	Seed mean	Seed-level 97.5% interval	Episode bootstrap 97.5% (registered)
G2-L B8@4 - B3-L4@4 (memory-free), hops 2-3	+6.225	+8.825	+6.450	+7.167	[+2.01, +12.33]	[+6.583, +7.758]

The seed-level interval excludes 0 for the B6 contrast (M1-L4@4−B4@1) and for G2-L against the memory-free reader at both backbones. It includes 0 for the G1-L contrasts and for G2-L against the fixed-query reader at both backbones, so the G2-L pass over B5-L4@4 is a statement about the fitted seeds. The registered decisions use the episode intervals and are unchanged.

Mechanism wording. The adapter acts at every question position. Each position performs its own vertical walk: its branch output enters that position's residual after the layer-14 attention and is not an input of any later position's adapter, and the frozen upper layers aggregate across positions. Which position's walk carries the answer is untested; "composes reads" in this paper means the iterated reads inside one position's loop (PC1-PC3 give no causal bridge claim and a distribution shift for final-position-only reads).

G Limitations and reproducibility in full

Record section 9.

9. Limitations, reproducibility, and conclusion

The central limitation is conditionality. Writer form, learned representation, reader training depth, and static budget all change the measured composition benefit. Candidate-restricted accuracy is not full-vocabulary answer accuracy. These metrics should not be exchanged when moving from a controlled terminal task to generated language answers.

Five statements are excluded. The language-model oracle is not the symbolic oracle: it needs a learned value-to-key lookup for every read after the first, so symbolic oracle results do not transfer to it. No conclusion about multiple reads in the language model is drawn at rank32. No claim is made that composition holds for unseen entities, beyond the closed 1,024-name vocabulary or for new entity codes: every evaluation draws on the seen pool except the descriptive UNS scope row below. OA is the value/output ceiling, not an addressing control. B8 is not parameter-matched to the single read or to M1: it adds 9,216 parameters, +8.9% at rank32, +31.0% at rank128 and +69.2% in the symbolic unfactorized joint model. Appendix C repeats these and the remaining scope statements beside each principal claim.

The rank128 language-model result is bounded further. It concerns the `oracle_random` memory over the 768 seen names, one oracle seed and layout and three training seeds; its equal-parameter claim is M1-L4 against B4 only, and the unshared reader B8 fails its gate while adding 31.0% parameters (24.3% at the second scale). The second scale (Qwen3-1.7B-Base) carries learning rates from the smaller model without a pilot, and covers two sizes of one family only: no test across scales was registered, so the paper claims no scaling trend, nothing for any other size, and no learned-writer or end-to-end result at either size. P1L-C does not show where the bridge is carried: its read-position drop is a distribution shift, and bridge patching gives no causal claim.

Section 8.6 adds its own limits. Every language-model composition number rests on one random oracle code table (one oracle seed) and three training seeds, so the episode intervals and the seed-level intervals are conditional on those and describe no other code table, layout or future training run; with three seeds the seed-level interval is wide, and it includes 0 for the G1-L contrasts and for G2-L against the fixed-query reader at both backbones. The depth matrix evaluates one L4-trained checkpoint at depths and hops that were never trained, without intervals; its oracle-stop row uses the label and is an upper bound, and hops5-8 peak at the largest depth evaluated. The unshared reader B8 contains the shared one and was trained with one optimizer and no learning-rate pilot of its own, so its failure is not a statement about unshared reads in general; a warm start from M1-L4 keeps the shared solution (Appendix I) and says nothing about why B8 fails from a random start. An exact store was tested only by evaluation on frozen checkpoints (RFX), and the trained least-squares study LSQT left that line open (Appendix I).

UNS evaluates the 256 unseen-pool names of the same 1,024-name vocabulary, which carry fixed random features and were never used in a training episode, with descriptive 95% intervals and no multiplicity

correction. Part A reproduces the 18 committed unseen-pool M1 depth-contrast rows (three writers, hop1, hops2-3 and hops6-8, two confidence levels) exactly from the saved samples of 18 audited runs; the other committed unseen rows stay unaudited. A (writer, reader) result is described as "the depth gain is also present on unseen-pool entities" only if its unseen-pool M1 depth gain on hops2-3 is at least 3 pp with the interval above 0. All six results meet it, labelled by source and never pooled:

Writer	Part A (released mixed-depth M1, S4242): unseen-pool gain, pp [95% CI]	Part B (M1 trained at L4, new episodes C999006): unseen-pool gain, pp [95% CI]
joint	+7.800 [7.067, 8.508]	+47.008 [45.983, 48.075]
joint_fact	+19.958 [19.108, 20.808]	+21.025 [20.083, 21.975]
oracle_random (control)	+31.592 [30.433, 32.750]	+41.267 [39.966, 42.534]

The unseen-minus-seen difference excludes 0 for joint_fact in both parts and for joint in Part B, always with the smaller unseen-pool gain (-3.942, -4.525 and -7.075 pp). The wording is limited to M1's depth gain on hops2-3 and to these readers; the readers were not trained to be tested on new codes, so no sentence that composition holds for unseen entities or beyond the closed vocabulary is justified, and Parts A and B differ in reader and episodes.

The uncertainty also has a narrow estimand. The bootstrap conditions on the fixed training seeds, and the follow-ups reuse confirmation samples rather than supplying independent confirmation of every successive claim. Training-seed failures, longer rank-follow-up training, and the disclosed F2 decision deviation remain relevant even when a pooled interval is positive. The original screening stopping results do not inherit parent-study intervals; P4-S-L and JS instead have their own fresh confirmation sets and scoped uncertainty. So do IVD, CVH, TSM and UNS Part B (C999004-C999007), with 97.5%, 99.166667%, 98.75% and descriptive 95% intervals. The P1-L confirmation (test [100000,102000), 97.5%, k=2) and P1L-C (test [102000,104000), 95%) condition on three training seeds and one oracle seed and layout.

IS now supplies a same-checkpoint load/layout sweep and a bounded update-task diagnostic. Its fixed training protocol and varying layouts limit interpretation: neither operator-bound tightness, a pure crosstalk effect, nor a general benefit from load-specific retraining has been established. IVD identifies key interference for oracle_random only; for joint_fact that reading is withdrawn. The largest tested capability load is not an identified failure boundary, and low stale rates do not imply complete current-answer success. JS supplies reached-state Jacobian measurements and an accuracy tradeoff, not proof of a stable terminal fixed point; verbalizable intermediate directions remain a separate diagnostic question.

Reproduction separates data access from rendering. The figures are produced by paper_figures.py from committed CSV/JSON exports; the figure manifest identifies revisions, selectors, aggregation, and claim IDs. No checkpoints are reevaluated and no bootstrap or timing measurement is rerun for this manuscript. The evidence ledgers retain exact stored values and artifact commits; Appendix B records sets, seeds, and comparisons, and Appendix C the claim contract. Preparation cost and training cost per model remain outside the prepared-query timings. The paper's primary same-rank comparison reports parameters and extra latency as costs, and makes no larger-rank efficiency claim.

Repeated low-dimensional queries are useful under particular writing and access conditions, with the orthogonal-oracle and factorized capability comparisons providing the clearest evidence. The current claim compares several reads with a single read at the same rank and reports their costs. Historical larger-rank controls and the original practical-gate failure remain in the record; neither is rewritten as a system-efficiency success. Post-hoc halting with the P4-S-L protocol does not repair hop preservation in the primary families; a hop-1-constrained threshold gives one registered pass (random-oracle writing, about two executed updates, no latency gain), stay training and a sticky gate do not remove overshoot, and training-time regularization reduces overshoot at a longer-walk cost. The appropriate conclusion is therefore a bounded mechanism result and a set of reproducible failures, with no efficiency claim over increasing rank. The single-module LLM screens, DPFP included, select no learned writer (P1L-D is not run); input probes and oracle controls do not overcome the deployed-read/answer gap, a read-failure factorial finds that only an exact store together with the true key rescues the read, and the rank-32 oracle-code pilot stops. In the oracle-reader confirmation, a shared reader exceeds a single read of equal parameter count at rank128 on two backbone sizes of one family while the registered four-unshared-reads reader fails its gate, bridge patching (0.6B only) gives no causal claim, and at rank32 the readers did not learn to

use the oracle memory; the per-step readouts are descriptive decoding and not evidence for composition. At its trained depth the shared reader reaches hop 3; run deeper, the same checkpoint reaches hops4-8 at 9.9-16.6% (0.6B) and 9.8-20.9% (1.7B) with the depth chosen by the label, while losing hop 1, and each question position runs its own walk (Section 8.6; descriptive).

The additional load study narrows this result further: depth benefits decrease under the registered load/layout changes, while multi-read capability remains positive at the tested training loads through E36. Existing latent-recurrence and interpretability research motivates the next diagnostics; it supplies no missing efficiency, convergence, or end-to-end result for this adapter.

H Related work in full

Record section 2.

2. Related work and boundaries

LoRA supplies the starting parameterization: a frozen base projection plus a trained low-rank branch. Our question concerns context-conditioned computation inside that branch. It does not depend on presenting recurrence itself as a new addition to low-rank adaptation. The dated literature audit is the scope record for the comparisons below; we retain its access limitations rather than turning an incomplete search into a priority claim.

Associative fast weights are the closest antecedent to the memory access pattern. Ba et al. already use an evolving inner state to read fixed fast weights repeatedly, with sustained input during the inner loop. Their construction makes clear that repeated access to a fixed matrix predates this adapter study. Schlag et al. connect linear attention to fast-weight programming and motivate delta-style updates that correct stored key-value associations. We use that lineage to separate the slow parameters that learn a representation from the online matrix that stores the current context.

δ-mem combines associative online state with low-rank corrections to a frozen language model. Its old-state read provides a useful comparison with our separation between writing the context and spending several reader updates on the resulting snapshot. The relevant distinction is the read schedule and query evolution, not a claim that associative memory or context-dependent low-rank corrections are new. Matching these schedules also requires specifying the writer's decay/update order, because nominally similar delta rules can retain different states.

Looped models investigate where repeated computation should occur. MixerLoop repeats a delta-network mixer while applying its dense feed-forward component once; the literature audit distinguishes its repeated mixer updates from the fixed-memory reader considered here. MemReasoner is a close memory-and-composition neighbor, with an end-to-end trained memory-augmented language model. These comparisons motivate describing the exact object being iterated: a low-dimensional query, a whole hidden state, or an update that also rewrites memory. They do not justify transferring a result between those objects without an equivalence argument.

Deep equilibrium models motivate another distinct question: finding and differentiating through a fixed point. Our finite-depth reader need not be a contraction to be useful, and a fixed-point error certificate would not by itself establish correct task predictions. RecLoRA remains a related-work item whose full-text architectural comparison is **TODO**: the committed audit could not verify its page-level claims. We consequently make no claim about absent nonlinearities, routing, or token mixing in that work, and do not use its unverified publication status in the argument. The contribution sought here is a reproducible composition-and-budget study with explicit writer controls and conditional analysis.

Latent recurrence, sharing, and stopping

The latent-reasoning survey distinguishes vertical activation recurrence at fixed token/time from evolution across sequence history. Our reader uses the former axis while the writer updates memory across tokens. Huginn supplies precedents for shared Transformer recurrence, reinjected input, heavy-tailed training-depth sampling, and heuristic prediction-KL exits. Coconut instead adds latent sequence positions by reusing hidden states as input embeddings; its default fixed latent length should not be confused with

guaranteed adaptive halting. These are different placements of latent computation, with different caches and training objectives.

Ouro learns exit allocation for shared Transformer loops and reports degradation beyond trained depth. That is a precedent for studying nonmonotone depth, not a diagnosis of this reader's failure mechanism. Relaxed Recursive Transformers use depth-specific LoRA to relax full-block sharing and train all model parameters; their oracle-exit batching simulations do not establish this adapter's measured efficiency. Mixture-of-Recursions adds learned token-level depth routing, including a variant that reuses first-recursion KV for later queries. Consequently fixed context alone is not a unique distinction. Our specified compact delta state, low-dimensional reader, and separate write schedule define the comparison.

Degradation from extra iterations is a known failure of recurrent depth, separate from this adapter. Bansal et al. report that recurrent networks can lose accuracy when iterated beyond their training depth ("overthinking"), and reduce it with a recall path that reinjects the input together with progressive training. The reader's reinjected u is analogous to a recall path, and no progressive or stay objective is trained here. Think-at-Hard reports that extra loop iterations can turn correct tokens into errors in looped language models, and Kohli et al. report that recurrent-depth transformers can compose knowledge, need more recurrence for deeper compositions, and overthink under excessive recurrence. We therefore read the loss of short-hop accuracy at larger L (Sections 5.1 and 7) as an instance of a known failure in a new setting, not as a defect specific to this reader, and we do not test the remedies those works propose.

STARS already combines sampled depth with Jacobian-vector regularization of looped dynamics. Its practical sampled-state proxy is not a certified spectral radius or global contraction constant. ReLIT combines a frozen-backbone anchor with recursive full-sequence answer/scratchpad updates and ACT halting. These precedents narrow any claim about reinjection, frozen-backbone latent recurrence, or stabilization. They motivate the JS protocol; its separate adapter results are reported in Section 7.

Multi-hop composition, regression memories, and retrieval in weights

The works in this subsection were added after two external literature reviews (rv-fable and rv-astra) opened their primary pages. We cite only what their abstracts and the review notes support, re-read no full text for this manuscript, and draw no priority claim from them.

Latent multi-hop composition in language models has a substantial literature. Yang et al. (2024) look for and intervene on the bridge entity of two-hop queries, and Yang et al. (2025) remove shortcuts and find relation-dependent composition. Balesni et al. report both failures and successes of latent two-hop reasoning across synthetic and natural settings. Biran et al. report that the bridge is resolved early and the second hop starts too late in the layer stack, Yu et al. let early layers read later-layer states, and Liu et al. report that later-hop answers can be decodable before the bridge. Guo et al. find in-context two-hop reasoning with distractors near guessing before fine-tuning. For out-of-distribution composition, Zhang et al. report failure when upper layers map facts to outputs instead of reasoning, and that recurrent-style training that reuses circuitry helped there; Lin et al. report that identity supervision on bridges enables unseen compositions, and Wang et al. that composition does not generalize out of distribution without grokking. Our setting differs: facts reach the reader only through a fixed-size adapter memory, and hops are spent in a reader loop rather than in layer depth. These studies therefore bound what we may claim without predicting our results. In particular, Zhang et al. make it unsafe to read B8's in-distribution advantage over the shared reader (Section 6; B8 is also not parameter-matched) as evidence for the unshared design, because that advantage is measured on trained hops and names; and the paper reports only a descriptive scope row on unseen-pool names of its closed vocabulary (Section 9), not a test of composition over new names or codes.

Associative memory can also be viewed as regression at test time. Wang et al. treat a delta-rule write as one gradient step on a recall loss, with least squares as the exact minimizer. Yang et al. parallelize delta-rule linear transformers over sequence length, and Titans, ATLAS, and TTT layers train the memory itself at test time, ATLAS over a window of past tokens rather than only the latest one. Our writer is the one-step delta update, which need not reach the regression solution when keys are correlated (cf. Proposition 11). Whether that gap contributes to the deployed-read distortion in Section 8.3 is untested, because the audit identifies no unique cause, and we study none of these alternative writers.

Slot and lookup memories give each stored item its own address. End-to-end memory networks learn several memory hops over stored items, and key-value memory networks separate the addressing repre-

sentation from the returned one, so multi-hop reading of a memory and key-value separation are long established and we claim neither. Hopfield networks retrieve stored patterns in a single softmax step, memory layers scale trainable key-value lookup inside language models, and Memory Mosaics compose networks of associative memories. Their state grows with the number of stored items, whereas the reader here uses a fixed-size delta-rule state in which facts superpose. No slot-memory baseline is run, and no result here bears on how a slot memory would behave.

Facts can also live in, or be written into, parameters. Geva et al. interpret feed-forward layers as key-value memories; ROME edits a factual association with a rank-one update and MEMIT scales such edits to many facts, a linear associative write into persistent weights rather than into a resettable per-episode state. Larimar attaches an episodic memory with one-shot writes to a language model; Cartridges and StreamAdapter compress a context into a trained key-value cache and a parameter update, respectively, Cartridges by training on each context, whereas the writers studied here are single-pass delta updates. Three 2026 reports on memories for frozen language models locate the bottleneck in reading or extraction rather than in storage (Frozen Memory Is Not Enough, Prometheus Mind, and a workshop paper on a hidden-state memory layer). That is compatible with, but does not confirm, the read-distortion measurements of Section 8.3, whose cause the audit leaves unidentified.

Two further external reviews (rv-fable and rv-astra, 2026-10-29) added the works below. We cite their abstract pages as the reviewers recorded them, did not re-open the pages for this manuscript, and claim no priority from any of them. We study none of the writers, objectives or architectures named here; our writers are the one-step delta writer, its DPFP-feature variant and the oracle writers. Zahn et al. report that semantic keys overlap and recall falls near chance after a few related facts, while identifier keys do not, a setting close to the key-overlap questions of Section 8.3. Several works read the delta rule as online regression or filtering: Pandey and Singh use online ridge by Sherman-Morrison as approximate key orthogonalization; Gated KalmaNet shows a full-covariance Kalman update to be online ridge regression with fading memory; Kalman Linear Attention filters in information form, scan-parallel; Kalman Delta Networks use a per-channel Kalman gain whose isotropic case is the delta rule; Preconditioned DeltaNet reads the delta rule as a first-order regression step and adds preconditioning; Longhorn derives a state-space update as closed-form online recall; MesaNet optimizes a local regression at test time; and Q-Delta studies keys transformed by subsequent updates. Other works change the objective, the retention or the update: It's All Connected (attentional-bias objectives and retention gates), Nested Learning (a self-modifying model with continuum memory), End-to-End Test-Time Training (a meta-learned initialization), Test-Time Training Done Right (whole-chunk updates), Infini-attention (compressive memory with a delta-style update), Gated DeltaNet-2 (separate erase and write gates), Gated Slot Attention-2 (two-sided association correction through slots), DeltaProduct and Sparse Delta Memory (richer updates or larger states). CAMELoT attaches a training-free associative memory to a frozen language model, as a neighbor of the frozen-backbone setting here. These are different writers from those tested, and the learned-writer stop of Section 8.5 applies only to the writers it tested.

For looped and latent reasoning, DiscoLoop reports that a bridge can be decodable yet misaligned for the next loop and mixes decoded embeddings with continuous states; Fan, Svete and Lee (called LOTUS in rv-astra) supervise looped transformers per position; Chen trains with final-only supervision, a near-identity gate and depth extrapolation, where intermediate supervision hurt; Kuo et al. report that a randomized loop count stabilizes extrapolation; and CODI and Recurrent Self-Improvement distill explicit reasoning into latent reasoning or use deeper teachers. Sanford et al. show that logarithmic depth suffices for k-hop problems, so one hop per read is a design choice of this adapter and not a bound. Liang and Pan warn that decodable intermediate states do not establish a complete serial algorithm, which is the boundary the per-step readouts of Section 8.5 respect: they are decoding only. The orthogonalized read as a training scaffold is reported as a removable training aid and not an inference rescue; whether sharing acts as such a scaffold for B8 is untested here. None of these training objectives, stopping rules or distillation methods is used in this paper.

Verbalizable directions and the readout boundary

Verbalizable Representations Form a Global Workspace in Language Models defines J-lens using averaged transport from intermediate residuals toward final-layer representations and vocabulary readout. Its sparse J-space construction is generally a union of cones from an overcomplete frame, not the adapter's fixed col(B) or a single rank-r subspace. It motivates asking whether intermediate reader states expose the expected entities, while separating readability from causal use. The downstream transport Jacobian

and the reader's recurrent transition Jacobian are different objects. The expanded primary-source audit records versions, access limits, and these distinctions; it supports no new priority claim.

I Follow-up studies G2 to G5: details

This appendix holds the tables behind Sections 3.4 and 4.4, the B8 warm-start paragraph of Section 3.2 and the SQ2 and LSQT paragraphs of Section 4. LSR, B8W and CTC are oracle-memory studies at rank 128 on the 768 seen names with one oracle seed; SQ2 and LSQT are validation-only learned-writer screens. The values of the first three are produced by `scripts/paper_tables_g2.py` from the committed files of each study (`reports/lsr`, `reports/b8w`, `reports/ctc`), and those of the last two by `scripts/paper_tables_g5.py` (`reports/p1l-sq2`, `reports/p1l-lsqt`); each script checks its key values against the figures the study reports state. Rounding is half up on the stored value, so a cell that ends in an exact 5 (for example 64.55) prints 0.1 higher than in the study reports. The evidence ledger holds the exact values (rows LS1 to LS4, BW1 to BW3, CT1 to CT3, SQ1 to SQ3, LQ1 to LQ3) and wins over any shortened wording here.

I.1 LSR: label-free stop rules (G2)

The test range is [106000, 108000), 2,000 episodes at eight hops per checkpoint, and the six checkpoints are the registered rank-128 M1-L4 runs at Qwen3-0.6B-Base (P1-L) and Qwen3-1.7B-Base (P5L), seeds 17, 29, 43. Thresholds were chosen on validation episodes [1000, 3000) and nothing was retrained. The registered intervals are 99.17% paired episode-bootstrap intervals (six comparisons, 10,000 draws shared across seeds, hops and rules); descriptive intervals are 95%. Table 8 in Section 3.4 gives the EM by hop of the fixed-depth rows, R2 and the two depth-per-hop references. Table 34 gives R2 against fixed L=4.

Table 34: R2 against fixed L=4 (which equals the best fixed depth L_c under the validation constraint), pp. Intervals are 99.17% for the registered rows (hops 4 to 8, hop 1) and 95% for the descriptive rows.

	Qwen3-0.6B-Base	Qwen3-1.7B-Base
Hops 4 to 8 gain	+3.90 [3.35, 4.43]	+5.68 [5.09, 6.25]
Hops 4 to 8 gain per seed (17/29/43)	+4.58/+3.25/+3.87	+5.46/+5.71/+5.86
Hop 1 change	+6.28 [5.09, 7.51]	+4.70 [3.48, 5.92]
Hop 1 change per seed	+8.90/+3.95/+6.00	+7.35/+4.40/+2.35
Hop 2 change (descriptive)	−10.83 [−11.85, −9.87]	−12.27 [−13.32, −11.23]
Hop 3 change	−3.28	−3.65
Hops 1 to 3 pooled change	−2.61 [−3.18, −2.06]	−3.74 [−4.32, −3.16]
R2 minus L=8, hops 4 to 8	−3.54 [−3.93, −3.15]	−5.05 [−5.47, −4.62]
R2 minus L=8, hop 1	+34.1	+39.4
R2 minus L=4 by hop 1 to 8	+6.28, −10.83, −3.28, +4.48, +4.98, +4.27, +3.08, +2.68	+4.70, −12.27, −3.65, +7.35, +7.85, +5.32, +4.33, +3.53
Mean reads per question	5.60	5.66
Upper-layer passes per question ($s^* + 1$)	6.60	6.66
Mean reads by hop 1 to 8	4.97, 5.48, 5.67, 5.76, 5.67, 5.70, 5.73, 5.86	4.87, 5.46, 5.72, 5.87, 5.80, 5.77, 5.84, 5.93

The registered conditions (gain of at least 3 pp with an interval above 0, hop 1 not below −10 pp, every seed) are all met at both backbones. R1 and R3 are infeasible at both backbones: none of their 25 validation grid values kept the hop-1 loss against fixed L=4 within 8 pp pooled and 10 pp per seed, so their test statistics were never computed. A prediction recorded before registration (R3: hops 4 to 8 at least 12%, hop 1 at least 55% at 0.6B) was not a gate and is not met. Latency is descriptive and was measured within one validation job, in ms per question (0.6B/1.7B): fixed L=4 0.56/0.62, fixed L=8 0.74/0.81, the L=8 trace pass 0.81/0.87. R2 itself was not timed, because its early exit needs a pass through the upper layers for each lens evaluation, and no latency or efficiency statement is made. At hops 4 to 8 R2 (9.1% and 10.2%) lies between fixed L=4 (5.2% and 4.5%) and fixed L=8 (12.6% and 15.3%), and the validation-chosen depth per hop (O1) reaches 13.0% and 15.8%. A fixed L=5 reaches 8.5% on hops 4 to 8 at 0.6B with hop 1 at

36.6%, close to R2's 9.1% with hop 1 at 54.3%; it lies outside the validation constraint (it lost more than the hop-1 margin) and no interval is attached to this comparison. The intervals condition on three training seeds per backbone and one oracle seed and layout, and they do not cover new training seeds.

I.2 B8W: a B8 copy started from M1-L4 (G3)

The continuations start from the registered P1-L M1-L4 checkpoints (Qwen3-0.6B-Base, rank 128, seeds 17, 29, 43) and run 2,000 steps at learning rate 0.003 with a fresh optimizer on the P1-L stream, in two arms: the shared M1-L4 continued, and a B8 copy with four untied step matrices. Evaluation is on test [108000, 110000), 2,000 episodes, L=4 (B4 at L=1), with 95% paired episode-bootstrap intervals (2,000 draws). The equality gate (bf16 logits, predicted tokens, the control and the fp32 read) held in every seed with a maximum absolute difference of 0 at tolerance 0, and the protocol check (shared arm more than 2 pp below the start) was not triggered. Table 35 gives every registered reference contrast, and Table 36 the accuracy by hop.

Table 35: B8W contrasts on hops 2 to 3 unless stated (pp, 95% interval). The first row is the registered decision; the others are descriptive.

Contrast	Difference [95% interval]	Accuracies (%)	Per seed (17/29/43)
untied@4 − shared@4 (decision)	+0.90 [0.09, 1.71]	32.40 vs 31.50	+1.05/+0.90/+0.75
untied − shared, hop 1	+2.77 [1.55, 4.02]	55.47 vs 52.70	+2.85/+1.70/+3.75
untied − start (M1-L4@4)	+5.99 [5.07, 6.88]	32.40 vs 26.41	+5.08/+6.38/+6.53
shared − start	+5.09 [4.18, 6.01]	31.50 vs 26.41	+4.03/+5.48/+5.78
untied − start, hop 1	+7.45 [6.00, 8.95]	55.47 vs 48.02	+6.40/+8.75/+7.20
shared − start, hop 1	+4.68 [3.33, 6.02]	52.70 vs 48.02	+3.55/+7.05/+3.45
untied − original B8@4 (P1-L)	+26.07 [24.98, 27.13]	32.40 vs 6.33	+29.38/+20.03/+28.80
untied − B4@1	+26.98 [25.87, 28.08]	32.40 vs 5.42	+29.88/+21.38/+29.70
shared − B4@1	+26.08 [24.98, 27.19]	31.50 vs 5.42	+28.83/+20.47/+28.95
start M1-L4@4 − B4@1	+20.99 [19.97, 22.06]	26.41 vs 5.42	+24.80/+15.00/+23.18

The decision rule asks that every seed loses at most 2 pp and that the pooled lower bound is above −2 pp; both hold, and no seed has a loss at all. The 97.5% interval of the decision contrast is [−0.04, 1.81], which touches 0. No superiority test was registered, so the untied arm being nominally above the shared arm is exploratory and is not a finding. Both arms improved by about 5 to 6 pp over the start, so the starting reader had not converged, and this says nothing about untying being better.

Table 36: B8W: exact match (%) by hop, seed means, L=4 (B4 at L=1).

Reader	1	2	3	4	5	6	7	8
Start (M1-L4)	48.0	34.7	18.2	6.8	4.6	5.1	4.9	5.1
Shared, continued	52.7	41.9	21.1	5.5	3.6	4.2	3.7	4.1
Untied copy	55.5	40.8	24.0	4.2	3.4	4.0	3.6	3.6
Original B8 (P1-L)	18.8	6.7	6.0	5.8	5.9	5.9	6.2	6.0
B4@1	20.0	5.6	5.3	5.1	5.6	5.3	5.5	5.1

The starting M1-L4, the original B8 and B4@1 on hops 2 to 3 (26.41%, 6.33%, 5.42%) match the values of P1-L on its own range (26.91%, 6.34%, 5.36%) within 0.5 pp. At hops 4 to 8 both continued arms are below the starting reader, which is descriptive. The relative Frobenius change of the continued tensors from their start is, for R, 0.44 to 0.49 (shared) and 0.48 to 0.58 (untied copies); for U, 0.22 to 0.23 and 0.26 to 0.32; for Q, 0.21 to 0.24 and 0.22 to 0.29. The mean pairwise cosine distance between the four untied copies is 0.191 to 0.215 for R, 0.072 to 0.074 for U and 0.051 for Q. So the copies separated from each other, and the untied arm did not stay tied in practice. Not claimed: that sharing is or is not necessary for composition, the cause of B8's failure from a random start (optimization, update allocation and the inductive bias of sharing are not separated; the original B8 stays at 6.33% on hops 2 to 3), a revised G1-L, equal-parameter status for B8, or anything at rank 32, for learned writers, unseen names, other scales or other training lengths. The study used 1.004 node-hours over ten jobs.

I.3 CTC: a classify-then-code writer (G4)

All values are on validation episodes [0, 1000) with evaluation seed 4242, one seed, full-vocabulary EM. The readers are the frozen P1-L rank-128 checkpoints of training seed 17 (B3-L4, B4, M1-L4). Table 37 gives M1-L4 at L=4 for every memory condition. The hard lookup writes the code of the predicted class; the soft lookup writes the class-probability mixture of the codes; the two differ by at most 0.1 pp at every hop for both arms.

Table 37: CTC: M1-L4 at L=4, exact match (%) by hop on validation episodes [0, 1000). CE and CE+ans are the two registered training arms (class cross-entropy only; with an answer loss added).

Memory	1	2	3	4	5	6	7	8
Oracle memory (reference, $p = 1$)	50.6	36.6	21.5	5.9	2.2	4.2	3.5	4.6
CE, hard (selected)	50.7	36.5	21.6	6.0	2.2	4.2	3.6	4.5
CE, soft	50.6	36.6	21.4	5.9	2.2	4.3	3.7	4.7
CE+ans, hard	50.7	36.5	21.7	6.0	2.2	4.2	3.6	4.6
CE+ans, soft	50.8	36.5	21.7	6.1	2.3	4.4	3.6	4.7
Zero memory	0.0	0.0	0.0	0.0	0.1	0.0	0.0	0.1
Shuffled memory (CE)	0.2	0.0	0.2	0.3	0.2	0.1	0.0	0.3

The per-fact joint classification accuracy is 1.00000 for CE (12,000 of 12,000 facts, every episode fully correct) and 0.99992 for CE+ans (one fact wrong), and a perfect-prediction lookup reproduces the oracle memory bit for bit on these episodes. The gain over B4@1 on hops 2 to 3 is +24.05 pp [21.80, 26.30] for CE hard against +23.90 pp with the oracle memory; hop 1 is 50.7% against a threshold of 40%. Neither registered falsifier fires, and the arm choice is a tie (0.0 pp), which the rule gives to CE. B4 at L=1 gives 18.7/5.3/4.7% on hops 1 to 3 (CE hard) against 18.5/5.4/4.9% with the oracle memory, and 4.5 to 6.0% on hops 4 to 8 in both; with zero memory it gives 0.2/0.1/0.2%. The memory-free B3-L4 gives 0.0% at every hop, and its predictions are identical under the oracle and the zero memory. The writer is not a learned writer in the sense of the DPFP rows: both entity heads read their entity at its own token, and the relation is constant in this task, so the relation head scores 100% at every position. The study shows that the reader transfers to near-oracle codes and uses its memory; it does not show that a model can learn its own codes, and no CTC number is placed beside a DPFP number. Arm CE trained 8,000 steps in 533 s (0.0651 s per step) and arm CE+ans in 2,320 s (0.2883 s per step), because it also runs the frozen reader and its backward pass every step; the study used 1.028 node-hours against 1.8 registered.

I.4 SQ2: ridge storage and the coefficient loss (G5)

The four arms continue the final D128-3 checkpoint (adapter, training-only value head and optimizer state) for 6,000 steps on one fresh training stream at learning rate 0.003, batch 32 with 4 questions per episode, E=16, seed 17, and are evaluated on the 1,000 validation episodes (evaluation seed 4242) at step 0 and every 1,200 steps. The ridge store solves a regularized least-squares fit of the stored values on the DPFP features of the keys ($\eta = 10^{-4}$, the lower bound of the registered clip, because the numerics showed the error is flat up to 10^{-3}); the coefficient loss has weight $w_c = 1$ (Clarification C1, a log2 rounding of the gradient-norm ratio). At step 0 the delta arms equal D128-3 exactly (608 correct, all 1,000 predictions identical to the parent's), and the ridge store on the same weights gives 687 correct. Table 38 gives the trajectories.

The gains over the 6,000 steps are +4.6 pp (S-DE), +2.2 pp (S-DC), +3.6 pp (S-RE, from 687 correct) and +0.8 pp (S-RC). The standard error of a difference of two EMs at 1,000 episodes is about 2.2 pp, and about 4.4 pp for the double difference of the interaction (−0.4 pp), so none of the differences between S-DC, S-RC and their controls is resolved, and S-RE against S-DE (+6.9 pp) is about three standard errors in one seed. Decoding stayed stable: every logged loss and gradient is finite, and the clean-value head of S-RC is between 91.6% and 94.4% at the five checkpoints, above the registered floor of 88.7%. The global gradient-norm clip (1.0) is active at 121 of 121 logged steps in every arm, with a pre-clip norm of about 8 to 10, and D128-3's own training log shows the same (449 of 449 logged steps); the norm is that of the total objective and the gradients of the individual loss terms were not logged. Zeroing or shuffling the memory of each final checkpoint leaves 0 to 2 of 1,000 answers correct. Every arm has 475,138 inference parameters and 393,216 bytes of persistent state per episode (a [128, 768] float32 store, against 65,536 bytes for the rank-128 delta store of the oracle readers). All of this is one seed, on the validation set that selected

Table 38: SQ2: validation exact match (%) and normalized read error at step 0 and the five checkpoints (one seed, 1,000 episodes).

Arm	0	1,200	2,400	3,600	4,800	6,000
S-DE, EM	60.8	61.5	62.7	61.5	63.6	65.4
S-DC, EM	60.8	61.1	63.6	62.3	64.4	63.0
S-RE, EM	68.7	64.2	67.1	70.7	69.3	72.3
S-RC, EM	68.7	63.9	66.8	68.5	68.2	69.5
S-DE, read error	0.363	0.390	0.364	0.370	0.361	0.359
S-DC, read error	0.363	0.388	0.385	0.364	0.364	0.353
S-RE, read error	0.312	0.274	0.242	0.202	0.216	0.196
S-RC, read error	0.312	0.304	0.226	0.217	0.215	0.199

D128-3, and no three-seed confirmation is flagged.

I.5 LSQT: trained least-squares stores (G5)

Each of the three arms is the WP-L4 writer R at rank 128 with the unit query, trained from scratch on the WP-L4 data stream for the registered number of steps (17,970 for LT-K4, 28,005 for LT-RG001, 28,245 for LT-RG1), with E=16, seed 17, batch 32 with 4 questions per episode and learning rate 0.003; evaluation is on the same 1,000 validation episodes. The ridge stores use recency weights and a float64 Cholesky solve; LT-K4 uses four Kaczmarz sweeps. Table 39 gives the trajectories at five evenly spaced checkpoints.

Table 39: LSQT: validation exact match (%) and normalized read error at 20%, 40%, 60%, 80% and 100% of each arm's training (one seed, 1,000 episodes).

Arm	20%	40%	60%	80%	100%
LT-K4, EM	14.9	25.9	30.4	37.6	38.3
LT-RG001, EM	23.6	35.7	42.1	46.9	46.1
LT-RG1, EM	23.0	30.1	35.7	36.6	42.6
LT-K4, read error	0.635	0.617	0.597	0.572	0.554
LT-RG001, read error	0.619	0.604	0.418	0.350	0.512
LT-RG1, read error	0.366	0.398	0.507	0.519	0.499

The decision uses the final checkpoint; for LT-RG001 the checkpoint at 80% (46.9%) lies in the same band. The read error does not follow the EM: LT-RG1 has its lowest read error (0.366) at 20% of training, when its EM is 23.0%, and LT-RG001's read error falls to 0.350 at 80% and returns to 0.512 at 100%. With one seed the checkpoint-to-checkpoint movement is not separated from run noise. The Gram reconstruction of the read is exact in every arm (relative error below 10^{-6}), so what the exact coefficients describe is the deployed read: the store does what it is built to do, and the read is still far from the stored value. Ridge is above Kaczmarz at the final checkpoints (+7.8 pp for LT-RG001 and +4.3 pp for LT-RG1); the arms have different step counts, and at the checkpoints nearest K4's final step count LT-RG001 has 42.1% and LT-RG1 35.7% against K4's 38.3%. The gradient clip is active at every logged step in every arm. Zeroing or shuffling the memory leaves 1 to 4 of 1,000 answers correct, and every arm has 475,138 inference parameters and 65,536 bytes of persistent state per episode. R-r128 is the record of WP-L4, not rerun; it is a valid reference because the test that reproduces it with the DPFP and least-squares options off passes at the science commit. The unit query is part of every LSQT arm and of none of R-r128, so a gain over R-r128 cannot yet be split between the store and the unit query; within the three arms the two ridge parameters differ by 3.5 pp, one seed. Nothing more is attributed here.

References and source records

Project source records: the dated literature audit (`docs/reports/lit_check.md`), the symbolic and LLM evidence ledgers, the binding notation (`docs/notation.md`) and the full theory (`docs/theory.md`). The unresolved RecLoRA comparison remains a TODO.

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